

INTRODUCTION TO STATISTICS

1.1 Introduction to Statistics

Statistics is a part of everyday life. It is common to wonder where and when statistics will be useful. Reading a newspaper, watching television, or browsing the Internet often involves seeing statistical information. There are statistics related to crime, sports, education, politics, and real estate. Newspapers and television programs often present information from surveys or studies. This information helps in deciding whether a statement or claim is correct.

Statistical data is everywhere. It is useful when making important decisions like buying a house or managing a budget. It is also helpful in different professions. Statistics can be found while browsing the Internet, playing sports, or checking the top scores in a video game. But what exactly are statistics? Statistics are numbers that help summarize raw facts and figures in a meaningful way. These numbers reveal key ideas that may not be clear when looking at raw data. Data refers to facts and figures from which conclusions can be drawn.

In today's world of computers and technology, statistics is important in every field. Data and conclusions based on data are encountered daily. Statistics is a scientific discipline that provides methods to understand data. It helps in making smart judgments and informed decisions in situations with uncertainty and variation.

Statistics involves scientific methods for collecting, organizing, summarizing, presenting, and analyzing data. It also helps in drawing valid conclusions and making reasonable decisions. It focuses on the systematic collection of numerical data and its interpretation.

The word '**statistics**' is used in two ways.

1. It refers to numerical facts, such as the number of people in a particular area.
2. It also refers to the study of methods for collecting, analyzing, and interpreting data.

Statistical knowledge is important for three main reasons.

1. It helps in staying informed.
2. It allows a better understanding of issues and enables sound decision-making based on data.
3. It also helps in evaluating decisions that impact daily life.

Statistical knowledge helps in solving real-world problems by providing reliable data for decision-making. It plays a crucial role in business, healthcare, economics, and many other fields.

1.1.1 Meaning and Definitions of Statistics

The word "**Statistics**" comes from the Latin word '**Status**', meaning a group of numbers or figures that represent useful information. The meaning of statistics has evolved over time. Initially, it focused on data collection, but later, methods for analysis and interpretation were developed. As a result, the term "statistics" is used in two different ways.

- In the plural sense, it refers to a collection of numerical figures or data.
- In the singular sense, it refers to the scientific methods and principles used to study, analyze, and interpret data.

The statistics can refer to either **numerical data (plural sense)** or the **scientific methods (singular sense)** used to study and analyze data.

Different scholars have defined statistics in various ways. These definitions are broadly classified into two categories. One category considers statistics as data, focusing on the collection of numerical information. The other category views statistics as a scientific method, emphasizing analysis, interpretation, and decision-making based on data. Both perspectives highlight the importance of statistics in understanding and organizing information effectively.

The application of statistics extends to various fields, including healthcare, education, and technology, where data-driven decisions enhance efficiency and outcomes. Its ability to identify patterns, trends, and relationships makes it a powerful tool for solving real-world problems and improving overall understanding.



Definitions of Statistics in Plural Sense (Statistic as Data)

Statistics in the plural sense refers to numerical facts and figures collected for study and analysis.

1. **A.L. Bowley:** "Statistics are numerical facts in any department of inquiry placed in relation to each other."
 - Statistics refers to numerical data related to a specific area of study.
2. **Yule & Kendall:** "By statistics, we mean quantitative data affected to a marked extent by multiplicity of causes."
 - Statistics is influenced by multiple factors, not just one.
3. **Horace Secrist:** "Statistics are aggregates of facts, numerically expressed, collected in a systematic manner for a predetermined purpose."
 - Statistics is a collection of numerical facts arranged for a specific study.
4. **National Statistical Office (NSO):** "Statistics refers to data that represent a particular phenomenon or trend."
 - It provides quantitative data about trends, such as inflation rates, unemployment rates, and economic growth.



Definitions of Statistics in Singular Sense (Statistics as a Science)

Statistics in the singular sense is a scientific method for collecting, analyzing, and interpreting data.

1. **Croxton & Cowden:** "Statistics or statistical methods may be defined as the collection, presentation, analysis, and interpretation of numerical data."

- Statistics is a scientific process that helps in understanding and interpreting data.
2. **Prof. Seligman:** "Statistics is the science which deals with the methods of collecting, classifying, presenting, comparing, and interpreting numerical data collected to throw some light on any sphere of inquiry."
 - Statistics organizes and classifies numerical facts to explain trends and events.
 3. **Lovitt:** "Statistics is a science that deals with the collection, classification, and tabulation of numerical facts as a basis for explanation, description, and comparison of phenomena."
 - It is a method for presenting and analyzing data to gain insights.
 4. **Sir Ronald A. Fisher (Father of Modern Statistics):** "Statistics is a key scientific tool that helps in drawing conclusions from experiments and data analysis."
 - Statistics applies scientific methods to analyze patterns and relationships.
 5. **Indian Statistical Institute (ISI):** "Statistics is the science of data management, covering collection, classification, and interpretation."
 - It is used in business, economics, healthcare, and social sciences.

1.1.2 Characteristics / Features of Statistics

Statistics is the science of collecting, analyzing, organizing, and interpreting data. It is widely used in various fields like business, economics, medicine, and social sciences. The important features of statistics are listed below.

1. Statistics Deals with Numerical Data

Statistics focuses on numbers and measurable data rather than emotions or opinions. Only information that can be counted or measured is considered statistical data. It does not work with feelings, emotions, or unmeasurable qualities unless they are converted into numerical form.

Example: Statistics Deals with Numerical Data

A teacher records students' marks (85, 90, 78) instead of writing "some students scored well." This numerical data allows for precise analysis, comparison, and evaluation of student performance.

2. Statistics Studies a Group, Not Just One Individual

Statistics always works with groups or collections of data, not individual observations. Studying a single object, person, or event does not provide meaningful statistical analysis. A large dataset helps in drawing valid conclusions.

Example: Statistics Studies a Group, Not Just One Individual

To determine the average height of students in a school, measurements must be taken from a large group of students rather than a single student.

3. Statistics are Affected by Multiple Factors

A statistical observation is influenced by different factors. The final result is a combination of many causes, not just one. Identifying these influencing factors helps in better decision-making.

Example: Statistics are Affected by Multiple Factors

The price of petrol is influenced by global oil rates, market demand, government taxation policies, and transportation expenses. These factors should be considered in statistics for accurate analysis and decision-making.

4. Statistics Helps in Comparison

Statistics enables comparisons between different datasets to find similarities or differences. This is helpful in decision-making and analysis. It allows businesses, researchers, and organizations to study trends and improvements over time.

Example: Statistics Helps in Comparison

A company compares the monthly sales of two different products to see which one performs better. This statistical analysis helps in making informed business decisions, optimizing marketing strategies, and improving product performance.

5. Statistics Must be Collected Systematically

For statistical data to be useful, it must be collected in a structured and planned way. Random or incomplete data collection leads to inaccurate conclusions. Well-organized data ensures reliability and correctness.

Example: Statistics Must be Collected Systematically

A national census follows a structured survey method to collect accurate population data, rather than relying on random responses from a few individuals.

6. Statistics Helps in Making Predictions

By studying past data, statistics helps predict future trends. Businesses, governments, and researchers use statistical analysis to forecast upcoming events or behaviors. This makes planning more effective.

Example: Statistics Helps in Making Predictions

A weather department predicts rainfall based on past climate data and seasonal trends.

7. Statistics Involves Estimation and Approximation

In many situations, obtaining exact numbers is not feasible, so statistics enables us to make reasonable estimates. These approximations are essential for decision-making, especially when complete data is unavailable or impractical to collect.

Example: Statistics Involves Estimation and Approximation

A construction company estimates the amount of raw materials needed for a project based on previous usage patterns instead of measuring every brick or bag of cement individually.

8. Statistics Helps in Decision-Making

Businesses, governments, and individuals rely on statistics to make informed choices. Analyzing statistical data enables problem-solving, strategic planning, and efficient resource allocation.

Example: Statistics Helps in Decision-Making

- A healthcare organization analyzes patient records to determine the most common diseases and plan preventive measures.
- A government uses employment data to develop job creation programs in regions with high unemployment rates.

9. Statistics Can be Represented Visually

Data can be displayed in charts, graphs, and tables for easy understanding. Visual representation helps identify trends and patterns quickly and makes complex data more accessible.

Example: Statistics Can be Represented Visually

- A financial analyst uses line graphs to track stock market trends over time.
- A school uses bar charts to show students' performance in different subjects over the years.

10. Statistics Applies to Different Fields

Statistics is used in almost every field, including medicine, business, sports, and government planning. It helps analyze data and draw meaningful conclusions.

Example: Statistics Applies to Different Fields

In sports, a cricket team uses player statistics (batting average, number of wickets) to select the best players for a tournament.

11. Statistics Works with Variability

Data collected from different sources will show variations, and statistics help in handling this variability. It allows researchers to study patterns and understand differences in data points.

Example: Statistics Works with Variability

The weight of students in a class varies, but statistics help find the average weight.

12. Statistics Can Identify Patterns and Trends

Statistics help find meaningful trends that may not be obvious at first glance. Identifying patterns in data is useful for planning and decision-making.

Example: Statistics Can Identify Patterns and Trends

An e-commerce company studies customer shopping habits to determine which products sell best during festivals.

13. Statistics Involves Sampling for Large Populations

When analyzing an entire population is not possible, a small sample is selected to represent the whole. A well-chosen sample provides accurate results with less effort and cost.

Example: Statistics Involves Sampling for Large Populations

A political survey polls a few thousand voters instead of interviewing every citizen to predict election results.

14. Statistics Helps Detect Relationships Between Variables

Statistics can show how one factor influences another. Understanding relationships between variables helps in predicting outcomes.

Example: Statistics Helps Detect Relationships Between Variables

A study finds that students who spend more time studying generally score higher marks in exams.

15. Statistics can be Used for Hypothesis Testing

Statistical methods help test assumptions or theories to determine if they are true or false. It provides evidence-based conclusions.

Example: Statistics can be Used for Hypothesis Testing

A scientist tests whether a new drug improves recovery time by comparing data from patients who used it versus those who did not.

1.1.3 Objectives of Statistics

Statistics plays a vital role in research, decision-making, and problem-solving across various fields. It helps analyze data, predict trends, and make informed choices based on numerical evidence.

Key Objectives of Statistics

- To Support Research and Studies :** Statistics is widely used in scientific, social, and economic research to analyze data and draw conclusions.
- To Estimate Future Trends and Patterns :** By studying past data, statistics helps forecast future outcomes in business, economics, healthcare, and other fields.
- To Interpret and Analyze Data :** Statistical methods allow for proper evaluation and meaningful conclusions from collected data.
- To Determine Unknown Values :** Using statistical techniques like regression and probability, missing or unknown data can be estimated.
- To Solve Economic and Business Problems :** Statistics helps in planning economic policies, business strategies, and financial management.
- To Assist in Quality Control :** Manufacturing industries use statistical sampling and analysis to ensure product quality and minimize defects.
- To Aid in Decision-Making :** Governments, businesses, and organizations rely on statistical analysis for making informed and strategic decisions.
- To Present Data Effectively :** Statistics organizes and represents data through graphs, tables, and charts. This makes complex information easier to understand.
- To Study Relationships Between Variables :** Statistical methods help identify correlations and dependencies between different factors.
- To Test Hypotheses and Theories :** Statistics validates scientific experiments, medical trials, and economic theories through hypothesis testing and data analysis.

Statistics
compari
statistic

1. Pr

m
w
an
un

2. Pr

un
sin
sta
tre

3. Es

fa
bu
im
an

4. Pr

tin
ma
un

A
in
be

1.1.4 Functions of Statistics

Statistics is used to present numerical facts in a way that is easy to understand. It helps in making comparisons, drawing valid conclusions, and finding useful patterns in data. The key functions of statistics are listed below:

- 1. Presents Facts in Numerical Figures:** Statistics helps in expressing facts in numbers, which makes them more precise and easier to understand. Describing facts in words can be vague, while numbers provide a clear and objective representation. Numerical facts ensure clarity and remove confusion in communication. A well-presented numerical report allows better understanding and decision-making.

Example:	Presents Facts in Numerical Figures
-----------------	--

Instead of saying that recent floods caused "huge damage" to crops, a statistical report would state that "500 acres of paddy and 200 acres of pulses were damaged, causing a total loss of ₹1.2 crores."
--

- 2. Presents Complex Facts in a Simplified Form:** Large amounts of data can be difficult to understand. Statistics provides tools like tables, graphs, and averages to summarize and simplify the data to make it more meaningful. Instead of dealing with thousands of numbers, statistics helps in summarizing key information concisely. It enables people to grasp important trends and patterns without going through raw data.

Example:	Presents Complex Facts in a Simplified Form
-----------------	--

A company tracks the monthly sales of its products. Instead of listing thousands of sales records, a simple bar graph can show sales trends over time. This allows the management to easily identify profitable and under performing products.

- 3. Establishes Relationship:** Statistics helps in discovering relationships between different factors. It can reveal connections that may not be immediately visible. Many aspects of business, science, and economics rely on analyzing relationships between variables to make improvements. Identifying such links helps in understanding how one factor influences another.

Example:	Establishes Relationship
-----------------	---------------------------------

A study may find that as income increases, spending also increases. Similarly, businesses use statistics to analyze the relationship between advertising and sales. If a company finds that more advertising leads to higher sales, it can plan better marketing strategies.

- 4. Provides Comparative Analysis:** Statistics allows comparisons between different groups, time periods, or events. It helps in identifying trends, patterns, and differences. Comparing data makes it easier to track progress, identify weaknesses, and plan corrective measures. It helps in understanding whether certain changes or policies are effective.

Example:	Provides Comparative Analysis
-----------------	--------------------------------------

A school compares student performance this year with last year to measure improvements or declines in results. A company compares the sales of two different products to see which one is performing better. This helps in deciding which product should be promoted more.

5. Improves Knowledge and Decision-Making

Statistics helps individuals and organizations make informed decisions by providing accurate information. It allows for logical and scientific thinking instead of guessing. It improves problem-solving abilities by offering a structured way of analyzing information. Reliable data leads to better planning and avoids unnecessary risks.

Example: Improves Knowledge and Decision-Making

A businessman studies market trends before launching a new product. A sports coach analyzes player performance statistics to select the best players for a match. Without statistical analysis, decisions might be made based on personal opinions rather than real data.

6. Helps in Policy Making

Governments and businesses use statistics to create effective policies based on actual data rather than assumptions. Policies related to taxation, pricing, employment, and investments are guided by statistical studies. Without statistical analysis, policies may not address real problems and could lead to ineffective solutions. Data-driven policies help in the proper allocation of resources.

Example: Helps in Policy Making

The government studies unemployment statistics to create new job programs. A mobile company analyzes customer preferences before launching a new phone model. If policymakers have the correct statistics, they can implement better economic and social programs.

7. Helps in Forecasting

Statistics is used to predict future trends by analyzing past data. It is widely used in economics, business, weather forecasting, and planning. Forecasting helps in reducing risks and preparing for future changes. It allows businesses and governments to take proactive steps instead of reacting to unexpected events.

Example: Helps in Forecasting

A bank predicts future inflation rates based on past economic trends. A retail company estimates demand for products during festivals using past sales data. This helps businesses stock their inventory in advance, avoiding shortages or excess stock.

8. Helps in Testing Hypotheses

A hypothesis is an assumption that needs to be tested with data. Statistics helps in checking whether an assumption is true or false. It ensures that decisions are based on factual evidence rather than intuition. Scientific and business experiments rely on statistical testing to validate their findings.

Example: Helps in Testing Hypotheses

A university believes that the average height of students is 5 feet 6 inches. Statistics can be used to collect height data from a sample of students and test whether this assumption is correct. A pharmaceutical company tests whether a new medicine is more effective than an existing one by analyzing patient recovery rates.

9. He

Ma
ris
to
ess

A
cu
pa

1.2 B

Statistics
work with
collection,

1. Dat

Data
num
(qua

Exa

2. Raw

Raw
sortin
direct

Exa

9. Helps in Making Decisions Under Uncertainty

Many decisions are made under uncertain conditions. Statistics provides tools to minimize risks and choose the best possible option. It allows businesses, individuals, and policymakers to analyze risks before making important choices. Decision-making under uncertainty is essential for planning in dynamic environments.

Example: Helps in Making Decisions Under Uncertainty

A company must decide whether to invest in a new business or not. Statistical analysis of past trends, customer demand, and market risks helps in making a confident decision. A farmer studies weather patterns and rainfall statistics before deciding which crop to plant for the next season.

1.2 Basic Concepts of Statistics

Statistics is the science of collecting, organizing, analyzing, and interpreting data. To effectively work with data, it is essential to understand key statistical terms. These terminologies help in data collection, classification, and analysis. The important terminologies in Statistics are listed below.

1. Data

Data refers to meaningful information collected from various sources. It consists of facts, numbers, or observations related to a particular subject or study. Data can be either numerical (quantitative) or descriptive (qualitative) and is the foundation of statistical analysis.

Examples: Data

- **Numerical Data :** A teacher records the marks of students: 90, 75, 85, 60, 92, 88, 78. The average mark of the class is 79.4.
- **Numerical Data :** The total sales of a company in a year is ₹20 Crore.
- **Descriptive Data :** 60% of people in a city prefer cricket over football and tennis.

2. Raw Data

Raw data is the original, unprocessed information collected directly from a source before any sorting, grouping, or calculations. It is also called unarranged data and is difficult to interpret directly. Raw data must be processed and structured before it becomes useful for analysis.

Examples: Raw Data

- **Unorganized exam scores of students:**
Raw Data : 72, 85, 90, 66, 78, 92, 69, 80. This data is not sorted or analyzed.
After processing: It can be arranged in ascending order or used to find the class average.
- **Unprocessed Daily Sales Figures:**
Raw Data : ₹5,000, ₹7,200, ₹4,800, ₹6,100, ₹3,500, ₹5,900.
These numbers are random daily sales values without analysis.
After Processing: The total sales for the week = ₹32,500 → Now, it is useful data.

3. Primary Data

Primary data refers to original, first-hand information collected directly from a source for a specific purpose. It is collected by a researcher, investigator, or organization through surveys, experiments, observations, interviews, or questionnaires. Since it is freshly collected, it is more reliable, accurate, and specific to the study. It is more reliable than secondary data but can be time-consuming and costly.

Examples: Primary Data

- A company conducts a survey to know customer preferences for a new product.
- A government agency collects population details by visiting every household.
- A business collects customer feedback about a new product through direct interviews.
- After an election, a news agency asks voters whom they voted for as they leave polling stations. This data is primary because it is collected directly from voters.

4. Secondary Data

Secondary data refers to information that has already been collected, processed, and published by someone else for a different purpose. It is used for reference, further research, or analysis. Since it is pre-collected, secondary data is readily available, saving time and effort. However, it may not be as specific, up-to-date, or reliable as primary data. Examples include government records, published research, company reports, and online databases.

Examples: Secondary Data

- The national census report is used by researchers to study population growth.
- A business refers to past sales reports to predict future sales trends.
- A student uses published studies on climate change instead of conducting a new study.
- A historian reads old newspapers and historical records to study past events.

5. Quantitative Data

Quantitative data refers to numerical information that can be measured, counted, or expressed in numbers. It deals with quantities, amounts, or values and can be used for mathematical calculations and statistical analysis.

Examples: Quantitative Data

- A teacher records students' marks as 85, 90, 78, 95, 88, 72. This is quantitative because it consists of numerical values that can be counted.
- The height of students in a school is 165.5 cm, 172.3 cm, 160.8 cm. This is quantitative because height is measured in units.
- A company reports monthly sales: ₹50,000, ₹75,000, ₹1,20,000, ₹98,500. This is quantitative because it involves numerical sales figures.

6. Qualitative Data

Qualitative data refers to descriptive information that cannot be measured numerically but can be categorized based on characteristics, attributes, or labels. It is often used to describe opinions, preferences, qualities, or classifications.

Exa

7. Popul

Popul
statist
the re
large t
In res
is sele
consid

Exam

8. Survey

A surve
behavio
observa
the enti
polls, ec

Types o

• Po

Exam

• I

c

c

• S

n

Examples: Qualitative Data

- Customers rate a product as "Good," "Bad," "Excellent," "Poor." This is qualitative because it describes opinions.
- A showroom categorizes vehicles as SUV, Sedan, Hatchback, Truck. This is qualitative because it classifies cars by type.
- A survey records hair colors of students in a class: Black, Brown, Blonde, Red. This is qualitative because it classifies people into categories.
- Participants in a race are ranked as 1st, 2nd, 3rd, 4th. This is qualitative because it shows ranking but has no fixed numerical difference. (e.g., the time difference between 1st and 2nd place may be 0.2 seconds, while the difference between 2nd and 3rd may be 3 seconds). If we measured actual times (e.g., 10.5s, 12.3s, 14.8s), then it would be quantitative because time is measurable. So, rankings are not quantitative.

7. Population

Population refers to the entire group of individuals, objects, or events under study in a statistical investigation. It includes all elements that share a common characteristic related to the research. The population can be finite (having a fixed number of elements) or infinite (too large to count).

In research, studying the whole population is often difficult, so a sample (a smaller group) is selected for analysis. However, when studying small groups, the entire population can be considered.

Examples: Population

- If a school wants to analyze the average height of students, then all students in the school are the population.
- When the government conducts a national census, the entire country's population is considered.
- If a company wants to study employee job satisfaction, then all employees working in the company are the population.

8. Survey

A survey is a method of collecting information from people to study trends, opinions, and behaviors. It gathers data using questionnaires, interviews, phone calls, online forms or observations and helps in analyzing patterns within a population. A survey can study either the entire population (like a census) or a sample. Surveys are used in market research, political polls, education, and healthcare studies.

Types of Surveys:

- **Population Survey (Census)**
- **Sample Survey**

Examples: Survey

- **Population Survey Example :** The National Population Census collects data from every citizen in a country to record population size, employment status, literacy rate, and economic conditions.
- **Sample Survey Example:** A political opinion poll surveys 5,000 voters to predict election results instead of surveying every voter in the country.

9. Census (Population Survey)

A census is a method of data collection where information is gathered from every member of a population. It is a complete enumeration of all individuals, objects, or units under study.

A census provides accurate and comprehensive data because it includes every element of the population. A census is also known as a **Population Survey**.

Governments and organizations conduct censuses for policymaking, economic planning, and resource allocation. However, censuses are time-consuming, costly, and require significant effort. Despite its challenges, a census remains the most reliable source of population data and is essential for long-term planning.

Examples: Census

- The government collects data on the total population, age distribution, employment status, and literacy rates by surveying every household in the country.
- The agricultural department records the number of farmers, types of crops grown, and land usage by surveying all farms and agricultural units in a region.
- The education department collects data on the number of students, teachers, and school facilities by surveying all schools in a district or a state or country.

10. Sample Survey

A sample survey is a specific type of survey that only studies a portion (sample) of the population instead of everyone. It is used when studying the entire group is impractical, costly, or time-consuming.

It estimates results for the whole population based on the sample data. If the sample is chosen correctly, the results are reliable and generalizable.

Examples: Sample Survey

- Instead of surveying all college students about study habits, researchers select 500 students from different universities to represent the entire population.
- A political opinion poll surveys 5,000 voters instead of interviewing every eligible voter in the country.
- A medical study selects 500 patients to test a new drug instead of testing it on every patient.

11. Sampling (The Process of Selecting a Sample)

Sampling is the process of choosing a subset (sample) from a larger population to represent the entire group. It helps in reducing effort, cost, and time while ensuring that the sample accurately reflects the whole population. In simple terms, sampling is the method used to create a sample for research. It determines how the sample is selected for a study or survey.

Examples: Different Sampling Methods

- **Simple Random Sampling:** Each individual has an equal chance of being selected.
Example: A lottery system selects 100 students randomly from a university for a study on study habits.
- **Systematic Sampling:** Every n^{th} member is selected from a list.

Example: A car manufacturing company picks every 10th product from an assembly line for quality testing.

- **Systematic Sampling:** Every n^{th} member is selected from a list.

Example: A car manufacturing company picks every 10th product from an assembly line for quality testing.

- **Stratified Sampling:** Population is divided into groups, and samples are taken from each group.

Example: A school divides students into grades (classes) and randomly selects 10 students from each grade to analyze academic performance.

- **Cluster Sampling:** Population is divided into clusters, and whole clusters are selected.

Example: A researcher randomly selects 5 out of 50 schools and surveys all students in those 5 schools.

12. Parameter

A parameter is a numerical value or numerical summary that describes a characteristic of an entire population. It is a fixed and exact measure that represents the whole group. A parameter does not change unless the entire population changes. Since parameters involve the entire population, they are usually difficult and expensive to calculate.

Examples: Parameter

- A study finds that 85% of all people in a country own a smartphone. This percentage is a parameter because it describes the entire population.
- The average height of all adults in a country is 170 cm. Since it includes every adult, this is a population parameter.
- The average life expectancy of all people in Japan is 84.5 years. Since it represents the entire population, this is a parameter.

13. Unit/Individual

A unit or individual refers to a single member of a population that is being studied. It is the smallest entity on which data is collected in a statistical study. A unit can be a person, object, event, or any other element that contributes to the data. The total collection of units forms the population.

Each unit must be clearly defined so that data collection remains accurate and consistent. The choice of a unit depends on the nature of the study.

Examples: Unit

- **Census Study:** In a population census, each citizen is an individual unit because data is collected about each person.
- **Student Exam Performance:** In a classroom survey, each student is a unit because their test scores are recorded individually.
- **Employee Satisfaction Study:** In a company-wide survey, each employee is a unit, as data is collected about job satisfaction, salary, and working conditions.

14. Variable

A variable is a characteristic or property that can take different values for different individuals or objects in a population. A variable changes from one unit to another. It is an essential part of statistics because it helps in measuring, analyzing, and interpreting data. Variables can be quantitative (numerical) or qualitative (categorical) depending on whether they represent numerical values or descriptive characteristics.

Examples: Variable

- The marks obtained by students in an exam is a variable because each student has a different score.
- The monthly salary of employees in a company is a variable since different employees earn different amounts.
- The blood group (A, B, AB, O, etc.) of patients in a hospital is a variable as different individuals have different blood types.

15. Attribute

An attribute is a qualitative characteristic that describes an individual, object, or event without using numerical values. Unlike numerical variables, attributes cannot be measured or expressed in numbers. Instead, they are classified into different categories based on specific qualities. Attributes are commonly used in categorical classification to analyze non-measurable characteristics.

Examples: Attribute

- The eye color (black, brown, blue, green, etc.) of students is an attribute because it describes a characteristic that cannot be measured numerically.
- The marital status (single, married, divorced, widowed) of people is an attribute as it classifies individuals into groups.
- The car type (sedan, SUV, hatchback, electric, etc.) is an attribute since it categorizes cars into different groups.

16. Frequency

Frequency refers to the number of times a particular value or category appears in a dataset. It represents how often an event, response, or observation occurs. In statistics, frequency is essential for summarizing data and understanding patterns in a dataset.

Examples: Frequency

- In the following set of numbers, the frequency of 9 is 5 because it appears 5 times:
1, 2, 3, 4, 6, 9, 9, 8, 5, 1, 1, 9, 9, 0, 6, 9.
- A teacher records how many students scored different marks in a test.
Marks: 50, 60, 70, 80, 90
Number of Students (Frequency): 3, 5, 7, 10, 4
Here, 7 students scored 70 marks, meaning the frequency of 70 is 7.

17. Seriation

The process of arranging objects or data points in a particular order or sequence is known as Seriation. It helps in organizing data systematically, making it easier to compare, analyze, and interpret. The arrangement can be chronological (time-based), numerical (smallest to largest), alphabetical, or based on importance or ranking. Seriation is used to simplify complex datasets and highlight trends or patterns.

Examples: Seriation

- A teacher arranges students' test scores from the highest to lowest to identify top performers.

Raw Data: 45, 89, 92, 65, 78

Seriated Data: 92, 89, 78, 65, 45

- A tournament ranks teams based on match wins

Raw Data (Unordered Wins): Team D - 4 wins, Team A - 8, Team C - 6, Team B - 7

Seriated Data (Ordered Highest to Lowest Wins): Team A, Team B, Team C, Team D.

18. Quantitative Variable

A quantitative variable is a type of variable that represents numerical values and can be measured or counted. These variables provide meaningful numerical information and allow for mathematical operations like addition, subtraction, averaging, and comparison. Quantitative variables are further divided into **discrete** and **continuous** variables.

Examples: Quantitative Variable

- Each student scores a different number of marks (e.g., 85, 90, 78). This is quantitative because marks are numerical values that can be counted and compared.
- Employees earn different salaries (e.g., ₹25,000, ₹30,500, ₹42,800). This is quantitative because salary is a measurable numerical value used for financial analysis.
- Fruits have different weights (e.g., 1.2 kg, 2.3 kg, 1.8 kg). This is quantitative because weight is measured using a scale and can have decimal values.

19. Qualitative Variable

A qualitative variable (also called a categorical variable) represents descriptive characteristics that cannot be measured numerically. Instead, it classifies data into categories, groups, or labels. Qualitative variables do not allow mathematical operations like addition or averaging.

Examples: Qualitative Variable

- **Eye Color of Students :** Students may have black, brown, blue, or green eyes. This is qualitative because eye color is a descriptive characteristic and cannot be measured or counted numerically.
- **Blood Groups of Patients :** Patients have different blood groups, such as A+, B-, AB, O+ or O-. This is qualitative because blood type classifies individuals into categories rather than providing numerical values.
- **Customer Satisfaction Level in a Survey :** Survey responses include satisfied, neutral, or dissatisfied. This is qualitative because satisfaction levels are descriptive categories rather than numerical values.

20. Discrete Variable

A discrete variable is a type of quantitative variable that can only take specific, countable values and cannot have fractions or decimals. It represents things that are counted rather than measured.

Examples: Discrete Variable

- **Number of Students in a Classroom :** A classroom has 25, 30, or 35 students. This is discrete because the number of students can only be whole numbers and cannot have fractions.
- **Number of Cars in a Parking Lot :** A parking lot has 10, 15, or 20 cars. This is discrete because the number of cars is a countable value and cannot be a decimal or fraction.
- **Number of Goals Scored in a Football Match :** A team scores 2, 3, or 4 goals in a match. This is discrete because goals are whole numbers and cannot be in decimals like 2.5 goals.

21. Continuous Variable

A continuous variable is a type of quantitative variable that can take any value within a given range, including whole numbers, decimals, and fractions. It is obtained through measurement rather than counting. These variables usually represent measurable quantities.

Examples: Continuous Variable

- **Height of Students in a Class :** Heights can be 150.5 cm, 162.3 cm, or 175.8 cm. This is continuous because height can take any fractional value within a range.
- **Time Taken to Complete a Race :** Runners finish in 10.5 seconds, 12.3 seconds, or 14.8 seconds. This is continuous because time can be measured precisely in fractions of a second.
- **Temperature of a City :** The temperature can be 28.4°C, 30.6°C, or 32.9°C. This is continuous because temperature is measured and can have decimal values.

22. Dependent Variable

A dependent variable is a variable that changes based on the influence of another variable. It is the outcome or effect in an experiment. It changes based on the independent variable. This is the variable that researchers measure to see how it responds to changes in another variable.

Examples: Dependent Variable

- A scientist wants to see how different amounts of water affect plant growth.
Dependent Variable: The height of the plant (it depends on how much water the plant gets).
- A teacher wants to know if study time affects students' exam scores.
Dependent Variable: Marks obtained by students (it depends on how much time they studied).
- A company increases its advertising budget and wants to check if sales increase.
Dependent Variable: Total sales revenue (it depends on how much money is spent on advertising).

23. Independent Variable

An independent variable is something that does not depend on other factors. It is the cause in a cause-and-effect relationship. Researchers manipulate the independent variable to observe its impact on the dependent variable. It is controlled or changed in an experiment to see how it affects the dependent variable.

Examples: Independent Variable

- A scientist wants to see how different amounts of water affect plant growth.

Independent Variable: Amount of water given (because it is changed to observe its effect on plant height).

- A teacher wants to know if study time affects students' exam scores.

Independent Variable: Study time (because it is adjusted to see its impact on marks).

- A company increases its advertising budget and wants to check if sales increase.

Independent Variable: Advertising budget (because it is controlled to measure its effect on total sales revenue).

Differences between Primary Data and Secondary Data

Primary Data	Secondary Data
Primary data is first-hand information collected directly from the source for a specific purpose.	Secondary data is pre-collected information that has already been processed and published by someone else.
It is original, fresh, and specific to the researcher's study.	It is already available and used for reference or further analysis.
It is collected through surveys, experiments, observations, or interviews.	It is obtained from books, government reports, research papers, websites, or company records.
It is highly reliable and accurate but time-consuming and costly to collect.	It is easily accessible and saves time, but may not be as specific, reliable, or up-to-date.
Example: A researcher conducts a survey to collect opinions on a new product.	Example: A company uses a government census report to analyze population trends.

Differences between Discrete Variable and Continuous Variable

Discrete Variable	Continuous Variable
A discrete variable takes specific, separate values and cannot have fractions or decimals.	A continuous variable can take any value within a given range, including decimals and fractions.
It represents countable quantities that remain fixed and do not change gradually.	It represents measurable quantities that can change gradually and smoothly.
Values are distinct and have clear gaps between them.	Values form an uninterrupted scale without gaps.
Examples: Number of students in a class, number of cars in a parking lot, number of books on a shelf.	Examples: Height of students, temperature readings, time taken to complete a race.

Differences between Dependent Variable and Independent Variable

Dependent Variable	Independent Variable
A dependent variable is the outcome or effect that is influenced by the independent variable.	An independent variable is the factor that is controlled or changed to observe its effect on the dependent variable.
It depends on the changes made to the independent variable.	It does not depend on other variables in the experiment.
It is measured and analyzed to understand the impact of the independent variable.	It is manipulated by researchers to examine its effect on the dependent variable.
Examples: Plant growth (depends on the amount of water), student test scores (depend on study time), sales revenue (depends on advertising budget).	Examples: Amount of water given to a plant, study time of students, money spent on advertising.

1.3 Qualitative and Quantitative Data

In statistics and research, data can be broadly divided into two main categories — **Quantitative** and **Qualitative**. Understanding the difference between them is important because the type of data determines the methods used for analysis.

1.3.1 Quantitative Data

Quantitative data refers to information that is expressed in numerical form. It represents measurable quantities, amounts, or values that can be counted or measured. This type of data can be subjected to mathematical operations such as addition, subtraction, multiplication, and division.



Key Characteristics of Quantitative Data

- 1. Numerical in Nature :** Quantitative data is always represented by numbers. These numbers may be whole numbers (integers) or decimal values, depending on the type of measurement. For example, the number of books in a library may be 500 (whole number), while the average weight of students may be 54.8 kg (decimal value).
- 2. Measurable with Standard Units :** Quantitative data is measured using standard units such as centimetres for length, kilograms for weight, litres for volume, rupees for money, and seconds for time. For example, height recorded as 172 cm can be compared directly with another height of 165 cm.
- 3. Mathematically Comparable and Analyzable :** Quantitative data supports various mathematical and statistical operations. Calculations such as averages, percentages, growth rates, and correlations can be performed. For example, if sales figures for three months are ₹50,000, ₹60,000, and ₹70,000, the average sales can be calculated, and the percentage increase can be determined.
- 4. Objective and Factual :** Quantitative data is based on factual measurements rather than opinions or feelings. It represents actual counts or measurements that do not change depending on personal views. For example, a recorded temperature of 30°C remains the same regardless of whether someone feels it is hot or pleasant.
- 5. Discrete Quantitative Data :** Discrete quantitative data represents countable values. These values occur in whole numbers and do not take fractional parts. Examples include the number of students in a class (Example, 45 students) or the number of cars sold in a day (Example, 20 cars).

1.3.2

Qualitative
rather than
Instead,



1. I
2. M
3. C

6. **Continuous Quantitative Data** : Continuous quantitative data represents values that can take any measurement within a range, including decimal values. These values are obtained through measurement rather than counting. Examples include a student's height (e.g., 160.5 cm) or the time taken to run a race (Example, 12.38 seconds).
7. **Comparable Across Time and Place** : Since quantitative data is measured in standard units, it can be compared over time periods or across different locations. For example, the rainfall in Bengaluru during July (210 mm) can be compared with the rainfall in Delhi during the same month (150 mm).
8. **Suitable for Graphical Representation** : Quantitative data can be presented visually through graphs such as histograms, bar charts, line graphs, or scatter plots. This helps in identifying trends, patterns, and relationships in the data. For instance, monthly sales figures plotted on a line graph can show growth or decline over time.

Examples: Quantitative Data

- A teacher records students' marks as 85, 90, 78, 95, 88, 72. These are numerical scores that can be averaged to find the class performance.
- The height of students is 165.5 cm, 172.3 cm, 160.8 cm. These values are measured in centimetres and can be compared.
- A company reports monthly sales as ₹50,000, ₹75,000, ₹1,20,000, ₹98,500. These figures can be used to calculate total sales or growth rates.
- The number of books issued in a library per day is 25, 30, 27, 32, 29. These are countable values.
- The daily temperature of a city is 32.5°C, 31.8°C, 33.2°C, 30.7°C. These are measurable values that can be averaged.
- The weights of fruits in a basket are 1.25 kg, 0.95 kg, 1.40 kg, 1.10 kg. These are continuous measurements.
- A shop records the number of customers each day as 150, 180, 175, 200, 190. This is discrete data.
- The time taken by students to complete a test is 35.2 minutes, 40.5 minutes, 38.7 minutes. These are measurable durations.

1.3.2 Qualitative Data

Qualitative data refers to descriptive information that expresses characteristics, attributes, or labels rather than numbers. It describes qualities or categories and cannot be measured directly in numbers. Instead, it is often analysed by grouping or classifying the data.



Key Characteristics of Qualitative Data

1. **Descriptive in Nature** : Qualitative data is expressed in words, categories, or descriptive labels rather than numbers. It focuses on qualities or characteristics. For example, customer reviews like "Excellent", "Good", "Poor" describe opinions instead of measuring quantities.
2. **Non-Numerical Representation** : It is not measured in numerical units like centimetres or kilograms. Instead, it represents classifications, names, or categories. For example, eye colour categories such as Black, Brown, Blue, Hazel represent qualities, not measurements.
3. **Categorical Classification** : Qualitative data groups information into categories based on attributes, traits, or features. These categories can be:
 - **Nominal** – Categories with no specific order (Example, blood group types: A, B, AB, O).
 - **Ordinal** – Categories with a natural order or ranking (Example, First, Second, Third in a competition).

4. **Subjective in Nature** : It often depends on personal opinions, preferences, or perceptions. For example, a person describing a dress as "beautiful" or "average" is expressing a subjective view that may differ for others.
5. **Not Suitable for Direct Mathematical Operations** : Since qualitative data does not represent measurable quantities, direct mathematical calculations like addition or averaging are not meaningful. For example, adding "SUV" and "Sedan" does not produce a valid result.
6. **May Use Codes or Symbols for Analysis** : In some cases, qualitative data is assigned numbers or symbols for easier analysis, but these numbers only act as labels, not actual measurements. For example, in a survey, 1 = Male, 2 = Female, 3 = Other.
7. **Comparable by Categories** : Qualitative data can be compared based on similarities or differences between categories. For example, comparing "Urban" and "Rural" lifestyles is done through category-based characteristics rather than numbers.
8. **Best Represented through Non-Numerical Visuals** : Qualitative data is often displayed using visuals such as pie charts, bar graphs or frequency tables to show how categories are distributed. For example, a pie chart showing the percentage of customers preferring different mobile brands.

Examples: Qualitative Data

- A survey records customer feedback as "Excellent", "Good", "Average", "Poor". These describe opinions and cannot be measured numerically.
- A showroom categorises vehicles as SUV, Sedan, Hatchback, Truck. These are types of vehicles, not numerical values.
- A survey records hair colours of students as Black, Brown, Blonde, Red. These represent categories based on a physical attribute.
- A school divides students into House A, House B, House C, and House D for competitions. These are category names, not measurements.
- In a competition, participants are ranked as 1st, 2nd, 3rd, 4th. These rankings show order but do not measure exact differences in performance.
- A hospital records patient blood groups as A+, A-, B+, B-, AB+, AB-, O+, O-. These are medical classifications.
- A clothing store lists dress sizes as Small, Medium, Large, Extra Large. These indicate categories, not exact measurements.
- A sports team's jersey colours are Red, Blue, White, Black. These are identifiers, not numerical data.

1.3.3 Difference Between Quantitative and Qualitative Data

Basis	Quantitative Data	Qualitative Data
Nature	Quantitative data is numerical in nature. It consists of numbers that represent counts or measurements. These numbers can be whole numbers or decimals. For example, the height of a student recorded as 165.5 cm or the monthly income of a person as ₹45,000.	Qualitative data is descriptive or categorical in nature. It represents qualities, attributes, or characteristics rather than numbers. For example, the eye colour of a person being Brown, or the customer review of a product as "Excellent".

Measurement	Quantitative data is measured in standard units such as centimetres for length, kilograms for weight, rupees for money, or seconds for time. For instance, the weight of an apple being 1.2 kg.	Qualitative data is not measured in numerical units. Instead, it is identified and classified based on categories or labels. For example, classifying vehicles as SUV, Sedan, or Hatchback.
Analysis	Quantitative data allows mathematical and statistical analysis. Operations such as calculating averages, percentages, correlations, and growth rates can be performed. For example, calculating the average marks of students from scores like 85, 90, 78, 95.	Qualitative data is analysed through grouping, categorising, and descriptive interpretation. For example, analysing survey responses that describe favourite cuisines as "Italian", "Indian", or "Chinese".
Objectivity	Quantitative data is objective and factual because it is based on actual counts or measurements that do not depend on personal opinions. For example, recording the rainfall in a city as 250 mm is a fact.	Qualitative data is subjective because it may involve personal opinions, feelings, or perceptions. For example, one person may describe the weather as "pleasant" while another calls it "hot".
Examples	Examples of quantitative data include the height of students in centimetres, the number of books in a library, the temperature of a city in degrees Celsius, or the income of employees in rupees.	Examples of qualitative data include the colour of cars, the brand of a mobile phone, the taste of food described as "spicy" or "mild", or the performance of a student described as "good" or "poor".
Sub-types	Quantitative data is classified into Discrete Data (countable values such as number of students in a class) and Continuous Data (measurable values within a range such as time taken to run a race).	Qualitative data is classified into Nominal Data (categories without order such as blood groups: A, B, AB, O) and Ordinal Data (categories with a specific order or ranking such as ranks in a competition: 1st, 2nd, 3rd).

Exercise 1 Find out the Key Terms

Determine what the key terms refer to in the following study. We want to know the average (mean) amount of money first-year college students spend at ABC College, excluding books. We randomly surveyed 100 first-year students at the college. Three of those students spent ₹15,000, ₹20,000, and ₹ 25,000, respectively.

Answer:

- **Population**: First-year students attending ABC College this term.
- **Sample**: The sample could be all students enrolled in one section of a course at ABC College (although this sample may not represent the entire population).
- **Parameter**: The average (mean) amount of money spent (excluding books) by first-year college students at ABC College this term in the population.
- **Statistic**: The average (mean) amount of money spent (excluding books) by first-year college students in the sample.
- **Variable**: The amount of money spent (excluding books) by one first-year student. Let X = the amount of money spent (excluding books) by one first-year student attending ABC College.
- **Data**: The data are the amounts spent by the first year students. Examples of the data are ₹15000, ₹20000, and ₹25000.

Exercise 2: Find out the Key Terms

Determine what the key terms refer to in the following study.

An insurance company would like to determine the proportion of all medical doctors who have been involved in one or more malpractice lawsuits. The company selects 500 doctors at random from a professional directory and determines the number in the sample who have been involved in a malpractice lawsuit.

Answer:

- **Population** : All medical doctors listed in the professional directory.
- **Parameter** : The proportion of medical doctors who have been involved in one or more malpractice suits in the population.
- **Sample** : 500 doctors selected at random from the professional directory.
- **Statistic** : The proportion of medical doctors who have been involved in one or more malpractice suits in the sample.
- **Statistic** : The proportion of medical doctors who have been involved in one or more malpractice suits in the sample.
- **Variable** : The variable X = the number of medical doctors who have been involved in one or more malpractice suits.
- **Data** : The data are either yes or no.

Exercise 3: Determine which of the following data is Quantitative or Qualitative:

- | | |
|---|-----------------------------------|
| 1. The marks that students get in a test. | 2. The genders of newborn babies. |
| 3. The area codes in phone numbers. | 4. The heights of buildings. |

Answer:

1. **The marks that students get in a test : Quantitative.**
Marks are numerical values that can be measured and compared on a scale.
2. **The genders of newborn babies : Qualitative.**
Gender is a descriptive category and cannot be measured numerically.
3. **The area codes in phone numbers : Qualitative.**
Although area codes are numbers, they serve as labels for specific geographical areas and do not represent measurable quantities. They indicate locations rather than numerical values that can be counted or used in mathematical operations. Therefore, they are considered qualitative data.
4. **The heights of buildings : Quantitative.**
Heights are measurable numerical values that can take any value within a range.

Exercise 4: Determine which of the following data is Quantitative or Qualitative.

A motor magazine provides descriptions of cars that contain the following data:

- Type of car – hatchback/sedan/compact SUV/SUV;
- Number of doors
- Fuel type – petrol/diesel;
- CO₂ emission rate – grams per kilometre.

Qualitative vs. Quantitative Data:

- 1. Qualitative Data:** Type of car and fuel type are qualitative because they describe categories and do not involve numerical measurements.
- 2. Quantitative Data:** Number of doors and CO₂ emission rate are quantitative because they represent measurable numerical values.

Discrete vs. Continuous Data:

- 1. Discrete Data:** Number of doors is discrete because it takes whole-number values (e.g., 2, 4, or 5) and cannot have fractions.
- 2. Continuous Data:** CO₂ emission rate is continuous because it can take any value within a range, including decimals (e.g., 120.5 g/km or 145.8 g/km).

Exercise 5 Determine which of the following data is Continuous or Discrete:

1. The number of customers visiting a store over a weekend.
2. The amount of water consumed by a country over the past 20 years.
3. The outcomes of rolling a 6-sided die five times.
4. The heights of trees in a forest.
5. Students' shoe sizes in a class.

Answer:

- 1. The number of customers visiting a store over a weekend : Discrete**
The number of customers can be counted, and the values are always whole numbers (Example, 50, 75, or 120).
- 2. The amount of water consumed by a country over the past 20 years : Continuous**
Water consumption is measured and can have decimal values (e.g., 1.75 lakh crore liters) and hence it is continuous data.
- 3. The outcomes of rolling a 6-sided die ten times: Discrete**
There are just 6 possible outcomes of rolling a 6-sided die; since the possible outcomes are a finite number that can only be expressed in whole numbers, this is discrete data.
- 4. The heights of trees in a forest: Continuous**
Tree heights can take any value within a range, including decimal measurements (Example, 5.8 meters or 12.3 meters), so this is continuous data.
- 5. Students' shoe sizes in a class: Discrete**
Shoe sizes follow a specific scale and can only take certain values (e.g., 6, 6.5, 7, or 8). Since they do not form a continuous range but follow fixed increments, they are considered discrete data.

1.4 Construction of Frequency Distribution**1.4.1 What is Frequency Distribution?**

The frequency of an observation tells the number of times the observation occurs in the data. Frequency refers to a number of times a given value appears in a distribution. For example, in the following list of numbers, the frequency of the number 9 is 5 (because it occurs 5 times):

1, 2, 3, 4, 6, 9, 9, 8, 5, 1, 1, 9, 9, 0, 6, 9.

A table, in which the frequencies and the associated values of a variable are written side by side, is known as a frequency distribution. A variable refers to the characteristic that varies in magnitude in a frequency distribution. It may be either discrete or continuous. A discrete variable is that which generally takes integer values. For example, the number of students, the number of books, etc. A continuous variable can take integer or fractional values within the range of possibilities, such as the height or weight of individuals.

A well-organized frequency distribution helps in analyzing trends, making comparisons, and drawing meaningful conclusions from data.



Meaning : Frequency Distribution

A frequency distribution is a table where the values of a variable and their corresponding frequencies are listed side by side.

Examples: Frequency Distribution

A teacher records students' marks and organizes them into a frequency distribution table:

Marks Obtained	Number of Students (Frequency)
40	3
50	6
60	8
70	2
80	7

- The marks (40, 50, 60, etc.) are the values being observed.
- The number of students represents the frequency of each mark.
- The frequency distribution helps in easily identifying the number of students scoring within different mark ranges.

1.4.2 Class Interval

A class interval is a range of values used to group data in a table. It helps when there are too many numbers to list individually. Instead of writing each number separately, similar values are grouped together to make data easier to read and analyze.

Examples: Class Interval

A teacher records students' marks using class interval and organizes them into a frequency distribution table:

Marks Range (Class Interval)	Number of Students (Frequency)
40 - 50	5
51 - 60	8
61 - 70	12
71 - 80	10

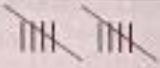
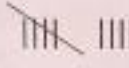
- Instead of listing every student's exact mark, we group them into ranges.
- This makes it easier to analyze trends, like how many students scored between 40-50 or 71-80.

1.4.3 Tally Bar

A tally bar is a simple way to record frequencies using marks or tally marks. It helps in counting occurrences quickly before converting data into numerical values. Each tally mark represents one count, and every fifth mark is drawn diagonally for easy grouping.

Examples: Tally Bar

A teacher records students' marks using class interval and organizes them into a frequency distribution table:

Marks Range (Class Interval)	Tally Marks	Number of Students (Frequency)
40 - 50		10
51 - 60		7
61 - 70		3
71 - 80		9
81 - 90		1
91 - 100		5

- Tally marks group every 5 counts for easy reading.
- The final frequency (total count) is written next to the tally marks.
- Tally bars help in quick counting and organizing raw data.

1.4.4 Types of Frequency Distribution

According to the nature of the variable, the frequency distribution may be of three types.

1. Individual Series (Individual Frequency Distribution)
2. Discrete Series (Discrete (or) Ungrouped Frequency Distribution)
3. Continuous Series (Continuous (or) Grouped Frequency Distribution)

1. Individual Series (or) Individual Frequency Distribution:

An individual series consists of data where each value is recorded separately without grouping or categorization. This type of series is used when data is small, and each observation is listed individually. It is useful when specific observations need to be recorded without forming groups.

Example Marks Obtained by 10 Students in a Test

Student	Marks
Rama	99
Sita	76
Laxman	69
Lava	96
Kusha	89

In this individual series, the marks of each student are recorded separately without any grouping.

2. Discrete Series (or) Discrete Frequency Distribution

A discrete series is a dataset that consists of discrete variables, which can take only specific values and cannot have fractions or decimals. Data is presented in a table showing how often each discrete value appears. It is useful for countable values that do not require grouping. In a discrete series, values do not change gradually, and they remain fixed and exact, unlike continuous variables that can take any value within a given range.

Example

Number of Students Scoring Specific Marks in a Test

Marks Obtained	Number of Students (Frequency)
40	3
50	6
60	8
70	2
80	7

- The marks obtained are whole numbers (40, 50, 60, etc.) and cannot have decimal values.
- The number of students (frequency) is countable and consists of fixed, exact values rather than measurements.
- The values do not change gradually and remain specific and exact. Hence, this is a discrete frequency distribution.

Example:

Frequency distribution of number of children in 25 families is discrete frequency distribution as shown in below table.

Number of children in 25 families : 2, 1, 4, 5, 1, 5, 4, 0, 2, 3, 2, 1, 4, 0, 0, 2, 1, 1, 5, 3, 4, 0, 1, 3, 2

No. of children (X)	Tally Marks	Frequency
0		4
1		6
2		5
3		3
4		4
5		3
Total		N = 25

A bar (|) called tally mark is put against the number when it occurs. After putting this mark four times against the value, a cross tally (||||) is put on these 4 tallies for the fifth mark.

From the sixth mark onwards, we start afresh in the similar manner. This technique facilitates easy counting of the tally marks at the end. The presentation of the data as shown in the above table is known as frequency distribution.

In the example above, the frequency refers to the number of children getting a particular mark in the test. Also note that total frequency must always equal the number of observations. Here, the total we have obtained after tallying the number of children in 25 families which is also the number of observations given.

A frequency distribution such as the one above is called an ungrouped or discrete frequency distribution table. It takes into account ungrouped data and calculates the frequency for each observation singularly.

What happens if we need to consider number of children in 500 families. Would it have been easy to represent such data in the format of ungrouped frequency distribution table? Well, obviously no. To represent a huge amount of information, the data is subdivided into groups of similar sizes known as class or class intervals and the size of each class is known as class width or class size.

3. Continuous Series (or) Continuous Frequency Distribution

A continuous series is a dataset that consists of continuous variables, which can take any value within a given range, including whole numbers and decimals. These variables are obtained through measurement, and they change gradually rather than in fixed steps. Since continuous data can have infinite values within a range, it is often grouped into class intervals to simplify analysis.

Example Heights of Students in a Class (Continuous Frequency Distribution)

- The height of students is a measurable quantity and can take any value within a given range, including decimals (Example, 155.3 cm or 162.7 cm).
- The data is grouped into class intervals (140-149, 150-159, etc.), which helps in organizing a continuous range of values.

Height (cm)	Number of Students (Frequency)
140 - 149	3
150 - 159	6
160 - 169	8
170 - 179	2
180 - 189	7

- The data changes gradually, as a student's height can increase over time, unlike discrete data, which consists of fixed numbers.
- Continuous variables are obtained through measurement, and in this case, height can be measured with accuracy, including decimal values.
- Since the values fall within ranges and can take any possible value within those intervals, this is classified as a continuous frequency distribution.

Example:

When we have a small set of data, like marks of 5 or 10 students, it is easy to list each score and count how many times it appears. But when the data set is large, like marks of 100 students, making a list for every single score becomes difficult and lengthy. The table will be too big and hard to read.

To solve this, we use a grouped frequency distribution table. Instead of listing every score, we divide the data into class intervals (ranges of values) and then count how many students fall in each range.

Take a look at the table below to understand the concept better:

Marks obtained in the test (Class Interval)	Frequency (No of Students)
0-5	4
5-10	14
10-15	35
15-20	33
20-25	10
25-30	4
Total	N = 100

In this table:

- The first column shows marks grouped into class intervals.
- The lowest value in a class interval is called the lower limit.
- The highest value in a class interval is called the upper limit.
- For example, in the class 10–15, the lower limit is 10 and the upper limit is 15.

This is an example of continuous class intervals because the upper limit of one class is the same as the lower limit of the next class. This type of arrangement is known as a **grouped frequency distribution**.

In continuous class intervals, any value that lies exactly on the boundary between two classes is always counted in the class where it is the lower limit. For example, if a student scores 5 marks, it will be included in the 5–10 class interval and not in the 0–5 interval.

A similar concept is disjoint class intervals, where the classes do not touch each other directly. An example of disjoint intervals is: 0–4, 5–9, 10–14, and so on. The frequency distribution for disjoint intervals is prepared in the same way as for continuous intervals.

1.4.5**Important Terms**

Term	Explanation												
Range	It is the difference between highest and lowest value in the data. Range = Highest Value – Lowest Value. Example: The marks of 10 students is shown below 80 65 56 90 75 45 39 61 77 30 Range = 90 – 30 = 60												
Class	'Class' means a group of numbers, in which items are placed such as 0–10, 10–20, 20–30, etc.												
Class Interval	While arranging large amount of data in statistics, they are grouped into different classes to get an idea of the distribution, and the range of such class of data is called the Class Interval. Example : In the table below, marks of 20 students of a class are divided into 5 classes with the size of each class interval being 10 <table border="1"> <thead> <tr> <th>Marks (Class Interval)</th><th>Frequency</th></tr> </thead> <tbody> <tr> <td>0–10</td><td>5</td></tr> <tr> <td>10–20</td><td>13</td></tr> <tr> <td>20–30</td><td>20</td></tr> <tr> <td>30–40</td><td>10</td></tr> <tr> <td>40–50</td><td>6</td></tr> </tbody> </table>	Marks (Class Interval)	Frequency	0–10	5	10–20	13	20–30	20	30–40	10	40–50	6
Marks (Class Interval)	Frequency												
0–10	5												
10–20	13												
20–30	20												
30–40	10												
40–50	6												
Class Limits	The lowest and the highest values which might be taken to define the boundaries of a class are class limits. The lowest value is called lower limit (L.L.) or Lower boundary and the highest value is upper limit (U.L.) or Upper boundary. The lower limit of a class is the value below which there can not be any value in that class. The upper class limit of a class is the value above which no value can belong to that class. Example: 20–30 is a class interval of size 10. 20 is the lower limit and 30 is the upper limit												

Class Frequency	The number of observations corresponding to a particular class is known as the frequency of that class. Example: <table border="1" data-bbox="526 432 1118 508"> <tr> <th>Income</th><th>(₹)</th><th>No. of persons</th></tr> <tr> <td>1000</td><td>2000</td><td>50</td></tr> </table> In the above example, 50 is the class frequency. This means that 50 persons earn an income between ₹1,000 and ₹2,000.	Income	(₹)	No. of persons	1000	2000	50
Income	(₹)	No. of persons					
1000	2000	50					
Inclusive Class	If lower as well as upper limits are included in the same class, such a class is called inclusive class. Here, upper limit of a class is not equal to the lower limit of the next class. In this, the upper limit of a previous class is less by 1 from the lower limit of the next class interval. In short this method allows a class-interval to include both its lower and upper limits within it. Example : 10 - 19, 20 - 29, ... are inclusive classes. In inclusive form, class limits are obtained by subtracting 0.5 from lower limit and adding 0.5 to the upper limit. Thus, class limits of 10 - 20 class interval in the inclusive form are 9.5 - 20.5.						
Exclusive class	If the lower limit is included in the same class and upper limit is excluded from that class but included in the succeeding next class, such a class is called exclusive class. Here, upper limit of a class is equal to lower limit of the next class. For example, a person of age 20 years will not be included in the class-interval (10 - 20) but taken in the next class (20 - 30), since in the class interval (10 - 20) only units ranging from (10 - 19) are included. Example: 10 - 20, 20 - 30, ... are exclusive classes. In exclusive form, the lower and upper limits are known as true lower limit and true upper limit of the class interval. Thus, class limits of 10 - 20 class intervals in the exclusive form are 10 and 20.						
Correction factor	In any problem if the class-intervals are given as the inclusive type, then they should first be converted into the exclusive-type. For this we require a correction factor. It is half of the difference between lower limit of a class and upper limit of the preceding class. Thus, <table border="1" data-bbox="367 1572 1305 1666"> <tr> <td>Correction Factor (C.F.) = $\frac{\text{Lower limit of any class} - \text{upper limit of the preceding class}}{2}$</td> </tr> </table> To get exclusive class intervals from inclusive class intervals, add C.F. to all upper limits and subtract C.F. from all lower limits of the class intervals. Example: Connect the following inclusive class intervals to exclusive class intervals. CI : 20 - 29 30 - 39 40 - 49 50 - 59 Here, $C.F. = \frac{30 - 29}{2} = \frac{1}{2} = 0.5$ By subtracting CF from all lower limits we get lower limits as 19.5, 29.5, 39.5, 49.5 and adding CF to all upper limits we get upper limits as 29.5, 39.5, 49.5, 59.5. The new CI will be 19.5 - 29.5, 29.5 - 39.5, 39.5 - 49.5, 49.5 - 59.5.	Correction Factor (C.F.) = $\frac{\text{Lower limit of any class} - \text{upper limit of the preceding class}}{2}$					
Correction Factor (C.F.) = $\frac{\text{Lower limit of any class} - \text{upper limit of the preceding class}}{2}$							

1.30

Probability and Statistics

Open End Classes

The lower or upper limit is not specified in a class then it is called open-end class. For example: less than (below) or more than (above) a particular class limit. The frequency distribution based on open-end classes is called open-end frequency distribution.

Example:

Class Interval	Frequency
Less than 20	5
20 - 30	13
30 - 40	20
40 - 50	10
More than 40	6

Mid Point or Class Mark

The central value of a class is called mid point or class mark. It is the average of class limits.

i.e.,
$$\text{Mid point} = \frac{\text{Lower limit} + \text{Upper limit}}{2}$$

Example:

Mid point of the class [10 - 20] is
$$\frac{LL + UL}{2} = \frac{10 + 20}{2} = \frac{30}{2} = 15$$

Magnitude or Width of Class Interval

The difference between upper limits and lower limits of a class is called magnitude or width of the class interval. It is denoted by 'c' or 'i'.

Magnitude or Width of class interval is : $UL - LL$

Example: Width of the class interval for exclusive class [20 - 30]

$$= UL - LL = 30 - 20 = 10$$

Marks (Class Interval)	Frequency	Magnitude or Width
0-10	5	$10 - 0 = 10$
10-20	13	$20 - 10 = 10$
20-30	20	$30 - 20 = 10$
30-40	10	$40 - 30 = 10$
40-50	6	$50 - 40 = 10$

In case of inclusive classes, width of the class will be calculate after converting the inclusive class to exclusive form.

Number of Classes

The number of classes can be obtained by using Prof. Sturge's rule of number of classes $(K) = 1 + 3.322 \log N$ where N = Number of observations.

Width of class $= c = \frac{\text{Range}}{\text{Number of classes}} = \frac{\text{Highest Value} - \text{Lowest Value}}{1 + 3.322 \log N}$

Cumulative Frequency

The cumulative frequency of a class is the total of all the frequencies up to and including that class. A cumulative frequency distribution is a frequency distribution which shows the observations 'less than' or 'more than' a specified value of the variable.

There are of two types (i) Less than type

(ii) More than type

Less than cumulative frequency distribution: It is obtained by adding successively the frequencies of all the previous classes including the class against which it is written. The cumulate is started from the lowest to the highest size

More than cumulative frequency distribution: It is obtained by finding the cumulate total of frequencies starting from the highest to the lowest class. The added up frequencies are called cumulative frequencies.

Example: Less than Cumulative Frequency Distribution

Frequency Distribution		Less than Cumulative Frequency Distribution	
Weight	No. of persons	Weight	No. of persons
30 - 40	10	Less than 40	10
40 - 50	15	Less than 50	25
50 - 60	40	Less than 60	65
60 - 70	45	Less than 70	110

Example: More than Cumulative Frequency Distribution

Frequency Distribution		More than Cumulative Frequency Distribution	
Weight	No. of persons	Weight	No. of persons
30 - 40	10	More than 30	110
40 - 50	15	More than 40	100
50 - 60	40	More than 50	85
60 - 70	45	More than 60	45

Frequency density in a class is defined as the number of observations per unit of its width. Frequency density gives the rate of concentration of observations in a class:

Frequency density ($\frac{f}{c}$) is the ratio of class frequency to width of class interval.

$$\text{Frequency density} = \frac{\text{Frequency of the class}}{\text{Width of the class}}$$

Frequency Density

Class interval	Frequency (f)	Width (c)	Frequency Density (f/c)
0 - 5	10	5	10/5 = 2
5 - 20	15	15	15/15 = 1
20 - 35	40	15	40/15 = 2.67
35 - 60	45	25	45/25 = 1.8
60 - 70	20	10	20/10 = 2

It is the ratio of frequency of the value variable to the total frequency.

$$\text{Relative Frequency} = \frac{\text{Frequency of value variable (F)}}{\text{Total Frequency (N)}}$$

Relative frequencies are calculated as

Relative Frequency:

Observation	Frequency (F)	Relative Frequency F/N
3	3	3/10 = 0.3
2	2	2/10 = 0.2
1	5	5/10 = 0.5
Total	N = 10	1

Bivariate Frequency Distribution

In the frequency distributions, the data are classified according to only one characteristic. These distributions are known as **univariate frequency distributions**. There may be a situation where it is necessary to classify data, simultaneously, according to two characteristics. A frequency distribution obtained by the simultaneous classification of data according to two characteristics, is known as a **bivariate frequency distribution**. The frequency distribution which studies two variables for the same populations is referred as **bivariate frequency distribution**.

Bivariate Frequency Distribution

A two-way table is associated with two variables, say X and Y . For each variable, a number of classes can be defined keeping in view the same considerations as in the univariate case. When there are m classes for X and n classes for Y , there will be $m \times n$ cells in the two-way table. The classes of one variable may be arranged horizontally, and the classes of another variable may be arranged vertically in the two way table. By going through the pairs of values of X and Y , we can find the frequency for each cell. The whole set of cell frequencies will then define a bivariate frequency distribution. In other words, a bivariate frequency distribution is the frequency distribution of two variables.

Example: Salary and Expenses of group of individuals, Income and expenditure of a group of individuals, weights of boys and girls in a class etc. It is also called two way table with $m \times n$ cells.

Bivariate Frequency Distribution

Illustration Suppose we have marks of a 15 students in two subjects, say statistics and history and we want to consider the variation of marks in both the subjects simultaneously. Then we have to prepare a table (known as a bivariate table) for the following data:

Marks in Statistics: 60 56 42 33 40 13 27 12 19 50 23 16 32 42 30

Marks in History: 50 45 53 54 49 23 50 24 31 41 47 20 38 21 45

Marks in Statistics	Marks in History				
	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60
10 - 20		= 2	= 1		
20 - 30		= 1		= 1	= 1
30 - 40			= 1	= 1	= 1
40 - 50		= 1		= 1	= 1
50 - 60				= 1	= 2
Total		4	2	4	5
					15

Bivariate Frequency Distribution

We have to consider a pair of data of observations. We mark one stroke (the tally mark) for one pair.

Now consider the case of the 1st student.

Marks in statistics = 60 and Marks in physics = 50

See the row along 50 - 60 $\rightarrow$ 5th row

and the column along 50 - 60 $\rightarrow$ 5th column

Therefore the stroke is marked in the cell (box) where the 5th row and the 5th column intersect each other

In this way we mark the strokes for 15 pairs of marks for 15 students.

Marginal Frequency Distribution	In a bivariate frequency distribution, if one of the variables is neglected and the distribution of the other variable alone is taken into consideration it is known as marginal frequency distribution
Conditional Frequency Distribution	In a bivariate frequency distribution, if distribution of one of the variables subject to a condition imposed on the alternative variable is taken in to consideration, the distribution is known as conditional frequency distribution

1.4.6 Problems

Illustration 1

The following is the distribution for the age of the students in a school.

Age	0 - 5	5 - 10	10 - 15	15 - 20
No. of Students	35	45	50	30

Calculate:

1. The lower limit of the first class interval.
2. The class limits of the third class interval
3. The class mark or mid point for the interval 5-10.
4. The magnitude or width of a class interval

Solution

1. The lower limit of the first class interval i.e. 0-5 is '0'.
2. The class limits of the third class, i.e. 10-15 are 10 (lower limit) and 15 (upper limit).
3. The class mark or mid point is defined as the average of the upper and lower limits of a class. For 5-10, the class mark or mid point is $(5+10)/2 = 7.5$
4. The magnitude or width of a class interval is the difference between the lower and upper class-limits. Here, we have a uniform class size, which is equal to 5 (5 - 0, 10 - 5, 15 - 10, 20 - 15 are all equal to 5).

Illustration 2

Covert the following inclusive Class Interval to Exclusive Class Interval.

Marks	0 - 9	10 - 19	20 - 29	30 - 39	40 - 49	50 - 59	60 - 69	70 - 79
No. of Students	4	16	21	35	28	14	06	03

Solution

Conversion of Inclusive C.I to Exclusive C.I

Marks	No. of Students (f)	E.C.I
0 - 9	4	0.5 - 9.5
10 - 19	16	9.5 - 19.5
20 - 29	21	19.5 - 29.5
30 - 39	35	29.5 - 39.5

40 - 49	28	39.5 - 49.5
50 - 59	14	49.5 - 59.5
60 - 69	06	59.5 - 69.5
70 - 79	03	69.5 - 79.5
Total	N = 127	

$$\text{Correction factor} = \frac{\text{Lower limit of any class} - \text{Upper limit of previous class}}{2} = \frac{10 - 9}{2} = \frac{1}{2} = 0.5$$

∴ This C.F = 0.5 should be subtracted from Lower limit and add back to upper limit of respective class intervals.

Illustration 3

Convert the following less than cumulative sequences distribution to exclusive class interval (E.C.I)

Daily Wages	< 100	< 200	< 300	< 400	< 500	< 600	< 700
No. of Persons	15	33	58	89	109	116	120

Solution

Conversion of less than Cumulative Frequency to E.C.I.

Daily Wages	No. of Persons	Exclusive C.I	Frequency (f)
< 100	15	0 - 100	15
< 200	33	100 - 200	33 - 15 = 18
< 300	58	200 - 300	58 - 33 = 25
< 400	89	300 - 400	89 - 58 = 31
< 500	109	400 - 500	109 - 89 = 20
< 600	116	500 - 600	116 - 109 = 7
< 700	120	600 - 700	120 - 116 = 4
Total			N = 120

Illustration 4

Calculate the class Intervals using the following data

Values (Mid Point):	10	30	50	70	90	110
Frequency:	3	7	10	20	12	08

Solution

Note: When Mid values are given, it has to be corrected through correction factor.

$$\text{Correction factor} = \frac{\text{differences between Mid values}}{2} \quad \text{C.F} = \frac{30 - 10}{2} = 10$$

Conversion of Mid values to exclusive class intervals.

Mid Point Value	Frequency	Exclusive class Interval
10	3	0 - 20
30	7	20 - 40
50	10	40 - 60
70	20	60 - 80
90	12	80 - 100
110	08	100 - 120
Total	N = 60	

Illustration 5

Convert the following more than cumulative sequences distribution to exclusive class interval.

Marks	>10	>20	>30	>40	>50	>60	>70	>80
No. of Students	100	90	77	56	24	12	06	02

Solution

Conversion of more than cumulative frequency distribution to Exclusive C.I.

Marks	No. of Students	Exclusive Class Interval	Frequency (f)
> 10	100	10 - 20	$100 - 90 = 10$
> 20	90	20 - 30	$90 - 77 = 13$
> 30	77	30 - 40	$77 - 56 = 21$
> 40	56	40 - 50	$56 - 24 = 32$
> 50	24	50 - 60	$24 - 12 = 12$
> 60	12	60 - 70	$12 - 06 = 06$
> 70	06	70 - 80	$06 - 02 = 04$
> 80	02	80 - 90	02
Total			N = 100

Problems on Discrete Frequency Distribution**Illustration 1**

Given are the no. of accidents occurred in city. Construct a discrete frequency distribution
0,3,1,2,4,0,0,3,1,3,2,2,4,2,5,5,4,2,0,2,1,3,2,4,3

Solution

Discrete frequency distribution representing number of accidents in a city

No. of Accidents	Tally Marks	Frequency (f)	Relative Frequency f/N
0		4	$4/25 = 0.16$
1		3	$3/25 = 0.12$
2		7	$7/25 = 0.28$
3		5	$5/25 = 0.2$
4		4	$4/25 = 0.16$
5		2	$2/25 = 0.08$
Total		N = 25	1

Problems on Continuous Frequency Distribution

Illustration 1

Given are the weights of 28 students. Calculate frequency distribution with class intervals width 5 using exclusive type and also frequency density for those

103, 87, 90, 97, 104, 111, 101, 79, 96, 99, 104, 96, 81, 92, 64, 79, 122, 109, 119, 67, 84, 91, 100, 99, 96, 77, 96, 79.

Solution

Frequency distribution representing weight of students

Class Intervals	Tally Mark	Frequency (f)	Frequency Density (f/w)
60 - 65		1	$1/5 = 0.2$
65 - 70		1	$1/5 = 0.2$
70 - 75	.	0	$0/5 = 0$
75 - 80		4	$4/5 = 0.8$
80 - 85		2	$2/5 = 0.4$
85 - 90		1	$1/5 = 0.2$
90 - 95		3	$3/5 = 0.6$
95 - 100		7	$7/5 = 1.4$
100 - 105		5	$5/5 = 1$
105 - 110		1	$1/5 = 0.2$
110 - 115		1	$1/5 = 0.2$
115 - 120		1	$1/5 = 0.2$
120 - 125		1	$1/5 = 0.2$
Total		N = 28	

Illustration 2

Prepare a frequency distribution from the following data by using exclusive method taking four as width of the class:

27	15	13	21	14	14	27	11	21	17
10	37	21	10	29	15	11	21	26	19
20	21	25	26	38	13	33	23	37	29
14	11	13	35	10	13	18	17	27	20

Solution

Class Intervals	Tally Marks	Frequency (f)
10 - 14		10
14 - 18		7
18 - 22		9

22 - 26	II	2
26 - 30	IIII II	7
30 - 34	I	1
34 - 38	III	3
38 - 42	I	1
Total		N = 40

Illustration 3

Marks of 50 college students given below. Prepare a frequency distribution with an interval mark 5 each using inclusive class intervals using Sturge's rule.

44	63	47	53	40	39	51	42	31	49
45	52	60	48	55	43	49	35	40	36
50	35	54	59	41	62	40	49	48	48
45	60	44	40	59	47	58	58	38	33
58	36	47	54	48	41	60	44	51	40

Solution

Range = Highest value - Lowest value = 64 - 32 = 32

The number classes as per Sturge's rule are obtained as follows.

Number of class intervals (K) = $1 + 3.322 \log N = 1 + 3.322 \log (50) = 6.64 = 7$ classes approximately

$$\begin{aligned} \text{Size of class intervals} = c &= \frac{\text{Range}}{\text{No. of class intervals}} \\ &= \frac{32}{7} = 5 (\text{Approx}) \end{aligned}$$

The required continuous frequency distribution is prepared as:

Table showing Frequency Distribution of Weight of Students.

Class Interval	Tally Marks	Frequency (f)
30 - 34	II	2
35 - 39	IIII I	6
40 - 44	IIII III I	11
45 - 49	IIII III III	13
50 - 54	IIII II	7
55 - 59	IIII I	6
60 - 64	IIII	5
Total		N = 50

1.5 Classification of Data

Raw data is the unorganized information collected during data gathering. In its original form, raw data is difficult to understand, compare, or analyze. Since it is unstructured and messy, it cannot be used directly for drawing conclusions. To make sense of this data, it must be properly organized.

Organized data allows for easy comparison and meaningful analysis. The best way to structure data is by looking for similarities or grouping similar values together. This process helps transform disordered and mixed data into a clear and meaningful format.

The main goal of organizing data is to make it easier to compare, analyze, and interpret. This is done by dividing the data into categories or groups based on specific characteristics. This method of sorting data into different groups or classes is called **classification**. Each group or category is known as a **class**. In simple terms, **classification** helps arrange data into a structured format to make it easier to compare and analyze.

1.5.1 Meaning and Definition



Meaning : Classification of Data

- ▲ Classification is the process of arranging data into similar groups based on common characteristics. It helps in organizing raw data and makes it easier to understand, compare, and analyze.
- ▲ **Example 1 :** The population of a town can be classified based on different characteristics such as gender, age, marital status, or occupation. This helps in analyzing demographic trends and planning resources for the community.
- ▲ **Example 2 :** A hospital collects patient records and classifies them based on disease type, age, or treatment history. This helps in tracking disease trends and improving healthcare services.
- ▲ **Example 3 :** A company analyzes employee salaries by classifying them into different pay scales based on experience, which helps in studying salary distribution and making wage policy decisions.

Raw data is often unstructured and difficult to interpret. Proper classification allows users to group similar data for better analysis and decision-making.



Definitions : Classification of Data

- ▲ **According to Connor,** "Classification may be defined as the process of arranging things in groups or classes according to their resemblances and affinities".
- ▲ **According to Horace Secrist,** "Classification is the process of arranging data into sequences and groups according to their common characteristics, or separating them into different but related parts".
- ▲ **According to E.L. Thorndike:** "Classification is the systematic grouping of data or objects based on their similarities, which helps in better understanding, comparison, and analysis."

1.5.2 Functions and Importance of Classification

Classification plays an important role in organizing and analyzing data effectively. It helps in structuring raw information, making it easier to interpret, compare, and draw meaningful conclusions.

The key functions of classification are:

1. It controls the data.
2. It facilitates the data.
3. It helps in the data.
4. To help in the data.
5. It provides a deeper understanding of the data.
6. It gives a clear character to the data.
7. To give an element of filtering to the data.
8. To provide a table, etc.

1.5.3 Requirements of Classification

A well-structured classification should be clear and concise. It should be easy to understand and use. It should be a good classification.

1. **Comprehensiveness** : The classification should cover all the data.
2. **Clarity** : The classification should be clear and easy to understand.
3. **Homogeneity** : The classification should be homogeneous.
4. **Suitability** : The classification should be suitable for the purpose.
5. **Stability** : The classification should be stable.
6. **Elasticity** : The classification should be elastic.
7. **Mutual Exclusivity** : The classification should be mutually exclusive.
8. **Simplicity** : The classification should be simple.

The key functions of classification are as follows:

1. **It condenses the Data** : Classification simplifies large volumes of data. This makes it more manageable and easier to study.
2. **It Facilitates Comparison** : Grouping similar data allows for better comparisons between different categories or variables.
3. **It Helps to Study the Relationships between Variables** : Classification makes it easier to analyze how one factor influences another.
4. **To highlight the Effect of one Variable by Eliminating the Effect of Others** : It helps in isolating the impact of a particular variable by controlling other factors.
5. **It Provides Further Statistical Analysis** : Well-classified data serves as a foundation for deeper statistical studies, including correlation and regression analysis.
6. **It gives Importance to Significant Features of the Data** : It helps in focusing on key characteristics.
7. **To give Importance to the Prominent Data Collected while Separating the Optional Elements** : Classification ensures that the most relevant information is highlighted while filtering out less important details.
8. **To Prepare Data for Tabulation and Analysis** : Classified data can be easily structured into tables, charts, and reports for better visualization and interpretation.

1.5.3 Requisites of Good Classification of Data

A well-structured classification system is essential for effective data organization and analysis. It should be clear, logical, and adaptable to different requirements. The following are the key requisites of a good classification of data:

1. **Comprehensiveness** : Classification should include all the items in the dataset so that every piece of data is assigned to a suitable group or class. No data should be left unclassified.
2. **Clarity** : There should be no confusion about where a particular data item belongs. The classification should be well-defined and unambiguous to avoid overlapping or misinterpretation.
3. **Homogeneity** : The items within each group or class should share similar characteristics. A classification system is effective only when data within each category maintains uniformity.
4. **Suitability** : The attribute or characteristic used for classification should align with the objective of the analysis. A classification method is meaningful only when it serves the intended purpose.
5. **Stability** : The classification system should remain consistent over time for a particular type of study. Frequent changes in classification may lead to confusion and incorrect analysis.
6. **Elastic** : The classification should be flexible enough to accommodate changes. As the purpose of classification evolves, the basis of classification should be adjustable to fit new requirements.
7. **Mutual Exclusiveness** : Each data item should belong to only one category at a time. There should be no overlapping between different classes to maintain accuracy and clarity.
8. **Simplicity** : The classification should be simple, logical, and easy to understand. A complex system may lead to errors and inefficiency in data interpretation.

9. **Comparability:** The classification should allow easy comparison of data over different periods or across different groups. This helps in identifying trends and patterns effectively.
10. **Scientific Approach:** The classification should be based on logical and scientific principles rather than arbitrary grouping. A well-defined system ensures accuracy and reliability in analysis.

1.5.4 Types of Classification

The nature of classification depends on the purpose and objective of investigation. Data can be categorized in different ways based on various factors. The following are some of the most commonly used types of classification:

1. Geographical (or Spatial) Classification
2. Chronological (or Temporal) Classification
3. Qualitative Classification
4. Quantitative Classification

1. Geographical Classification

When data is classified based on geographical locations such as countries, states, cities, or districts, it is known as **Geographical Classification** or **Spatial Classification**. This type of classification is helpful in analyzing regional trends, population distribution, and economic conditions in different areas.

Examples Geographical / Spatial Classification

- The population of India can be classified by states such as Maharashtra, Karnataka, and Tamil Nadu.
- COVID-19 cases can be classified by country to compare infection rates across different nations.
- Classification of Production of Food Grains in Different States in India.

State	Production of food grains (in '000 tons)
Tamil Nadu	4500
Karnataka	6600
Andhra Pradesh	2800
Telangana	5200

Geographical classification is widely used in demographics, economics, business planning, and environmental studies.

2. Chronological / Temporal Classification

When data is grouped based on time periods such as years, months, weeks, or days, it is known as **Chronological Classification** or **Temporal Classification**. In this classification, data is arranged in either ascending or descending order according to time. It helps in studying trends, patterns, and changes over time.

Examples Chronological / Temporal Classification

- The rainfall pattern of a city over the last 10 years can be analyzed using chronological classification.
- The monthly sales of a product can be classified to track seasonal demand trends.
- Profits of a Company From the Year 2001 to 2005

Year	2001	2002	2003	2004	2005
Profits in Crores	175	195	250	220	350

Chronological classification is widely used in economics, business forecasting, weather analysis, stock market trends, and historical research. It helps in understanding past trends, predicting future outcomes, and making informed decisions.

3. Qualitative Classification

When data is classified based on non-measurable characteristics, it is called **Qualitative Classification**. These characteristics or attributes cannot be expressed in numbers but can be categorized based on certain qualities such as gender, marital status, color, honesty, intelligence, and literacy.

Examples	Qualitative Classification
----------	----------------------------

- | |
|--|
| <ul style="list-style-type: none"> • The population can be classified based on marital status into married and unmarried individuals. • Fruits can be classified based on sweetness levels into very sweet, mildly sweet, and not sweet. • People can be grouped based on employment status into employed and unemployed. |
|--|

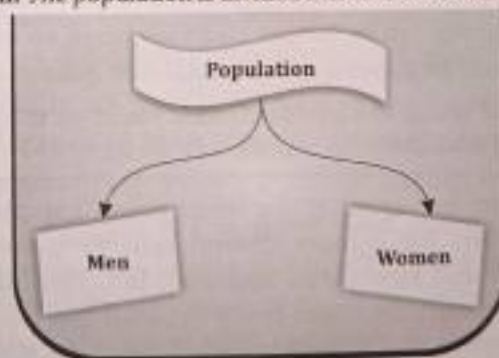
Types of Qualitative Classification:

a. Simple Classification (Dichotomous Classification)

Simple classification, also known as **dichotomous classification**, refers to the process of dividing data into only two distinct categories based on a specific attribute. It is the most basic form of qualitative classification, where each item either belongs to a group or does not belong to it. This type of classification is useful when dealing with characteristics that have only two possible values.

Examples	Simple Qualitative Classification
----------	-----------------------------------

- | |
|---|
| <ul style="list-style-type: none"> • Employment Status: The working population can be classified into "employed" and "unemployed" based on the attribute "employment." • Education Level: The population can be categorized as "literate" and "illiterate" based on the attribute "education." • Customer Satisfaction: A company can classify customers as "satisfied" and "unsatisfied" based on the attribute "customer feedback." • Gender Classification: The population is divided into men and women based on the attribute "sex." |
|---|



It is commonly used when the attribute has only two possible values, such as Yes/No, Present/Absent, True/False. It simplifies the classification process and helps in quick decision-making.

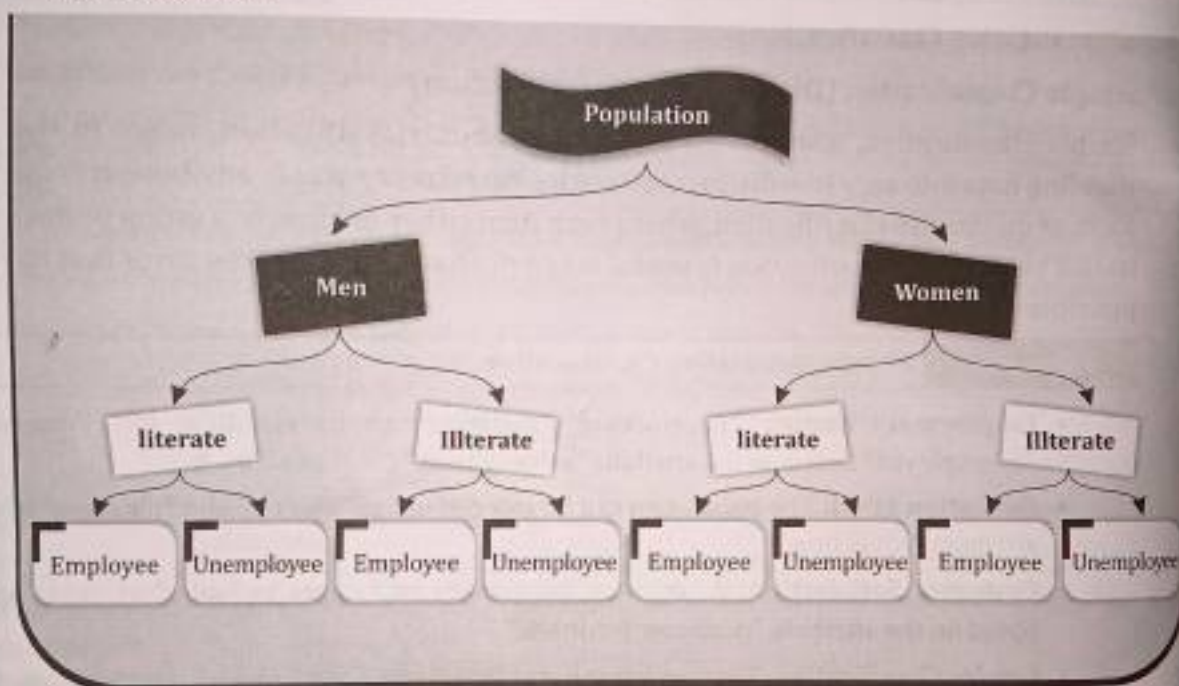
It is useful for quick data organization and analysis, especially when working with binary categories. However, when data involves multiple characteristics, a more complex classification system, such as manifold classification, is required.

b. Manifold Classification

Manifold classification is a type of qualitative classification where data is categorized based on more than one attribute at a time. Unlike simple classification, which divides data into only two groups, manifold classification organizes data into multiple subcategories for better analysis and understanding.

Examples Manifold Qualitative Classification

- The population is first divided into men and women based on the attribute "sex."
- Each group is further subdivided into literate and illiterate based on the attribute "literacy."
- Within each group, a further classification is done based on employment status into employed and unemployed.



- The employed group can be further classified based on the type of job, such as government employees, private employees, and self-employed individuals.
- The unemployed group can also be divided based on the reason for unemployment, such as actively seeking jobs, students, retired individuals, or homemakers.

Manifold classification is essential for organizing complex data that involves multiple characteristics. It allows for a more comprehensive understanding of data relationships. It provides a detailed and structured approach for classification. Since it involves multiple classifications, the structure becomes more detailed and informative. This makes it easier to analyze patterns, compare trends, and make informed decisions.

4. Quantitative

Quantitative classification is based on data that is quantifiable or measured. It is used for analyzing income, expenditure, and profits.

Data classified by quantitative classification helps in analyzing income, expenditure, and profits.

Example:

Weights
No of Students

In Quantitative Classification, weights are classified and recorded. This method helps in analyzing data.

Example:

Age (Years)
No of Patients

In this example, patients are classified by age ranges. In each range, the number of patients is recorded.

Quantitative classification is used in public administration and distributions.

1.6 Diagram

One of the key functions of a diagram is to make data understandable. It helps in visualizing the volume of data and its relationships.

A more effective way to present data is by using geometrical figures. Numerical information can be presented in a more understandable manner.

Presenting data using diagrams helps in understanding information and making decisions. It is a representation of data that is easier to understand.

Diagrams are used to present data in a more understandable manner. They help in visualizing the volume of data and its relationships.

1. Quantitative Classification

Quantitative classification is used when data is measurable in numerical terms and can be counted or measured. It includes variables such as age, height, weight, income, production, sales, expenditure, and profits.

Data classified based on quantitative measurements such as age, height, weight, prices, production, income, expenditure, sales, and profits is termed as **Quantitative Classification**. This classification helps in analyzing numerical trends and distributions in statistical studies.

Example: The weights of the students in a class of 100

Weights	30-40	40-50	50-60	60-70	70-80	80-90
No of Students	5	18	25	28	19	5

In Quantitative Classification, data is grouped based on measurable numerical values. In this example, student weights are classified into specific ranges (class intervals), and the number of students in each range is recorded. This makes it easier to analyze weight distribution in the class.

Example: Age Distribution of Patients in a Hospital

Age (Years)	0-10	11-20	21-30	31-40	41-50	51-60
No of Patients	12	8	35	47	55	99

In this example, patient ages are grouped into specific age ranges (class intervals), and the number of patients in each range is recorded, allowing for easy analysis of the age distribution in the hospital.

Quantitative classification is widely used in education, business, healthcare, manufacturing, and public administration to organize numerical data efficiently. It helps in identifying patterns, trends, and distributions that support better decision-making.

1.6 Diagrammatic Representation of Data

One of the key functions of statistics is to present large and complex data in a simple and easily understandable form. Classification and tabulation help in organizing data systematically, but as the volume of data increases, it becomes difficult to comprehend even after classification and tabulation.

A more effective way to present statistical data is through diagrams and graphs. These visual tools use geometrical figures such as points, lines, bars, squares, rectangles, and circles to simplify complex numerical information. Even a layperson can quickly grasp the data by looking at a diagram.

Presenting data using diagrams and graphs is one of the most effective ways to simplify complex information and make it visually appealing. Unlike textual or tabular presentations, diagrammatic representation transforms numerical data into images to make trends, comparisons, and relationships easier to understand. This method enhances clarity and helps in decision-making by providing a quick visual summary of information.



Meaning : Diagrams and Graphs

1. Diagrams

A diagram is a pictorial representation of statistical data that simplifies complex information by using geometric shapes such as bars, circles, or pictograms. It visually highlights the key facts and relationships in a dataset. Diagrams make data easy to interpret and provide a quick overview of patterns and trends.

Examples of Diagrams Include:

- Bar diagrams (used for comparing different categories)
- Pie charts (used to show proportions of a whole)
- Pictograms (uses images to represent quantities)

Diagrams are commonly used when presenting qualitative data, such as preferences, categories, and comparisons. While they may not provide precise numerical accuracy, they are useful for communicating data in an engaging and visually appealing manner.

2. Graphs

A graph is a mathematical representation of data that displays relationships between numerical values using a coordinate system. Graphs show trends, patterns, and variations in data. It helps in understanding fluctuations over time or comparisons between different variables.

Examples of Graphs Include:

- Line graphs (used to show trends over time)
- Histograms (used for frequency distributions)
- Scatter plots (used to show relationships between two variables)

Graphs are used when dealing with large datasets that require detailed analysis of numerical data and relationships. They help in making statistical inferences and drawing meaningful conclusions from data.

1.6.1

Meaning and Definition of Diagrammatic Presentation

Diagrammatic presentation of data refers to the graphical representation of numerical information using various types of diagrams and graphs. This method converts raw data into a visual format that quickly interprets trends, patterns, and relationships between different data points. Diagrammatic representation is widely used in statistics, economics, business, and scientific studies to present data in an attractive and meaningful manner.



Definition : Diagrammatic Presentation of Data

1. According to Croxton and Cowden:

"A good statistical diagram is an effective device for presenting complex statistical facts in an attractive, simple, and understandable manner."

2. According to Spiegel:

"A graph is a mathematical picture of data, which simplifies statistical analysis and makes it easy to draw conclusions from large datasets."

1.6.2

Diagrammatic interpretation

- To present
- To make
- To facilitate
- To highlight
- To make
- To improve
- To reduce
- To increase
- To provide

1.6.3

Diagrammatic method are:

- Data is
- Simplified
- Facilitates
- Widely
- Helps in
- Makes
- Useful for
- Provides
- Reduces
- Allows for

1.6.4

Advantages



Advantages

- Provides
- Simplifies
- Facilitates
- Enhances
- Helps in

1.6.2 Objectives of Diagrammatic Presentation

Diagrammatic presentation of data is used to simplify numerical information and improve data interpretation. The primary objectives are:

- To present statistical data in a visual and attractive manner.
- To make complex data more understandable and accessible.
- To facilitate quick comparisons between different sets of information.
- To highlight trends, variations, and relationships in data.
- To make numerical data more appealing for reports and presentations.
- To improve decision-making by offering a clearer understanding of the dataset.
- To reduce reliance on lengthy textual explanations.
- To increase the effectiveness of business reports, research findings, and academic studies.
- To provide an engaging and interactive way to communicate statistical insights.

1.6.3 Features of Diagrammatic Presentation

Diagrammatic representation enhances data clarity and improves analysis. The key features of this method are:

- Data is represented visually using graphs, charts, or diagrams.
- Simplifies complex numerical data, making it easier to understand.
- Facilitates quick comparison of different datasets.
- Widely used in business reports, financial analysis, and statistical studies.
- Helps in identifying trends, patterns, and relationships between data points.
- Makes presentations more engaging and visually appealing.
- Useful for both small and large datasets to convey key insights.
- Provides an accurate and precise way to represent numerical information.
- Reduces the effort needed to analyze and interpret raw data.
- Allows for better communication of findings in a simplified manner.

1.6.4 Advantages and Disadvantages of Diagrammatic Presentation



Advantages of Diagrammatic Presentation

- Provides a clear and visual representation of data.
- Simplifies complex numerical information for easy understanding.
- Facilitates quick comparisons between different data sets.
- Enhances readability and engagement in reports and presentations.
- Helps in identifying trends, patterns, and relationships easily.



Disadvantages of Diagrammatic Presentation

- Not suitable for representing highly detailed data with precision.
- May require additional explanation or textual support for context.
- Can sometimes be misleading if not designed correctly.
- Difficult to present very large datasets without making the graph cluttered.
- Requires software or tools to create, which may not be available to all users.

1.6.5

General Rules for Constructing Diagrams and Graphs

1. **Provide a Clear Title** : A suitable and descriptive title should be placed above the diagram or graph to indicate what the data represents.
2. **Select an Appropriate Scale**: The scale should be chosen carefully to ensure accurate representation of data without distortion.
3. **Ensure Neatness and Clarity** : The diagram should be well-drawn, visually appealing, and free from unnecessary elements.
4. **Avoid Overloading with Excessive Information** : The diagram should focus on key data points without cluttering it with too many details.
5. **Use Suitable Colors and Shades** : Different components should be clearly distinguished using shades, colors, or patterns for better understanding.
6. **Provide an Index or Legend** : If different colors, shades, or symbols are used, an index should be included to explain their meanings.
7. **Ensure Completeness** : All necessary elements such as labels, axes, units, and numerical values should be present for a complete representation.
8. **Make it Simple and Self-Explanatory** : The diagram or graph should be easy to interpret without requiring additional explanations.

1.6.6

Types of Diagrams

Diagrams are used to present statistical data in a simple and visually appealing way. They help in understanding large amounts of information quickly. Different types of diagrams are used depending on the type of data and the purpose of the presentation. The main types of diagrams are:

1. **One-Dimensional Diagrams (Bar Diagrams)** : These diagrams use bars to represent data. The length of each bar shows the value of the data. These include different types of bar charts, such as simple bar diagrams, multiple bar diagrams, and sub-divided bar diagrams.
2. **Two-Dimensional Diagrams** : These diagrams use areas of shapes to represent data. The bigger the shape, the larger the value. These include pie charts and rectangular diagrams, where data is represented in proportion to the total area of a shape.
3. **Three-Dimensional Diagrams** : These include cube diagrams and spherical diagrams, where data is represented in three-dimensional space for enhanced visualization.

1.6.7 One Dimensional Diagrams (or) Bar Diagrams

One Dimensional (or) Bar diagrams are one of the most commonly used methods for presenting statistical data. They are shaped like rectangles, where only the length of the bar represents the value of the data, while the width remains uniform throughout the diagram. These diagrams are called one-dimensional diagrams because only one dimension (length) varies, while the width stays constant. If the dataset contains a large number of values, lines may be used instead of bars to save space.

Features of Bar Diagrams

- The rectangular shape in the diagram is known as a bar and represents the value of a variable.
- Bars can be arranged vertically or horizontally depending on the requirement.
- Each bar is placed at an equal distance from the others.
- All bars originate from a common baseline or axis.
- The width of all bars remains the same, while the height changes based on the value of the variable.
- Bars are usually arranged in ascending or descending order, unless a specific order is required.

Advantages of Bar Diagrams

- Easy to understand, even for those unfamiliar with charts.
- Simple and effective for comparing two or more datasets.
- Suitable for presenting a large number of observations clearly.
- Helps in identifying trends and patterns in data at a glance.
- Can be used for both categorical and numerical data representation.

Important Types of One Dimensional Diagrams (or) Bar Diagrams

1. **Simple Bar Diagram** : A Simple Bar Diagram is used to compare different things using bars of the same width but different heights. The taller the bar, the larger the value. For example, it can show the number of students in different schools or sales of a product over the years.
2. **Sub-Divided Bar Diagram (Component Bar Diagram)** : A Sub-Divided Bar Diagram breaks a bar into smaller parts to show how a total is divided. For example, if a company's total expenses are shown in one bar, the bar can be divided into parts like salaries, rent, and electricity costs.
3. **Multiple Bar Diagram (Compound Bar Diagram)** : A Multiple Bar Diagram shows two or more bars next to each other to compare different groups. For example, it can be used to compare the number of boys and girls in different classes or the sales of two products in different months.
4. **Percentage Bar Diagram** : A Percentage Bar Diagram shows data in terms of percentage instead of actual numbers. Each bar represents 100%, and it is divided into sections to show how different parts contribute. For example, it can show what percentage of people use buses, trains, and cars for travel in a city.

1.6.7.1 Simple Bar Diagram

A **Simple Bar Diagram** is a type of bar chart used to represent a single set of numerical data graphically. It consists of rectangular bars of equal width, where the height of each bar corresponds to the value it represents. The bars can be drawn either vertically or horizontally, with equal spacing between them. A Simple Bar Diagram is useful for comparing values of a single variable across different categories, such as time periods, regions, or groups. It helps in visualizing trends and variations in an easy-to-understand format. When data needs to be represented based on one variable, a Simple Bar Diagram is the best choice. It is widely used in business, economics, education, agriculture, and research to present statistical data clearly and effectively.

Structure of a Simple Bar Diagram

- The bars in a Simple Bar Diagram are rectangular and vertical or horizontal, depending on how the data is presented.
- All bars have equal width and are placed at equal distances from each other.
- The height of each bar represents the value of the variable. The taller the bar, the higher the value.
- The X-axis (horizontal axis) represents the different categories, such as years, products, or locations.
- The Y-axis (vertical axis) represents the numerical values, such as production, sales, profits, or population.
- All bars start from a common baseline, ensuring easy comparison.

Characteristics of a Simple Bar Diagram

- Only one variable is represented in the diagram.
- Bars are evenly spaced and separated by equal intervals.
- All bars have the same width, but the height varies depending on the value it represents.
- Easy to understand as the data is visually displayed.
- It can be drawn both vertically and horizontally, depending on preference.

Advantages of Simple Bar Diagram

- Easy to construct and understand, even for those unfamiliar with statistics.
- Clearly presents numerical data and allows for quick comparisons.
- Helps in identifying trends over time.
- Useful for decision-making in business, government, and research.
- Can be used for various fields, such as economics, agriculture, education, and commerce.

Illustration

Represent a bar diagram.

Year
Production

Solution

1. Represent

- T
- T
- T

2. Trend

- V
- T
- I

3. Purpose

- T
- I

Illustration

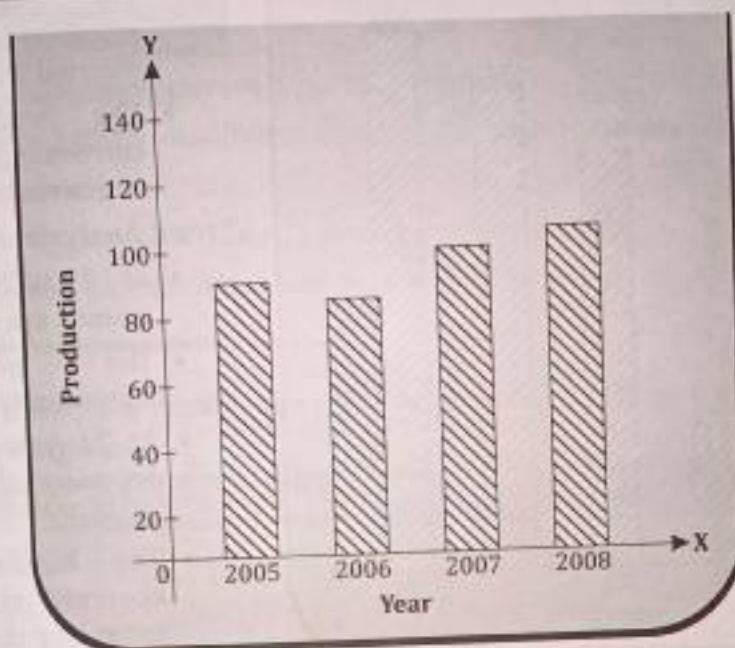
The following table shows the number of married women in various age groups.

Age (in years)
Married women

Illustration 1

Represent the following data regarding the production of wheat (in 000's tons) by simple bar diagram.

Year	2005	2006	2007	2008
Production	90	85	100	105

Solution**1. Representation of Data**

- The X-axis (horizontal axis) represents the years (2005, 2006, 2007, and 2008).
- The Y-axis (vertical axis) represents the production of wheat in 000's tons.
- The height of each bar corresponds to the wheat production for the respective year.

2. Trend Analysis

- Wheat production fluctuates over the years.
- The production was 90,000 tons in 2005 and slightly decreased to 85,000 tons in 2006.
- In 2007, production increased to 100,000 tons and further rose to 105,000 tons in 2008.

3. Purpose of the Diagram

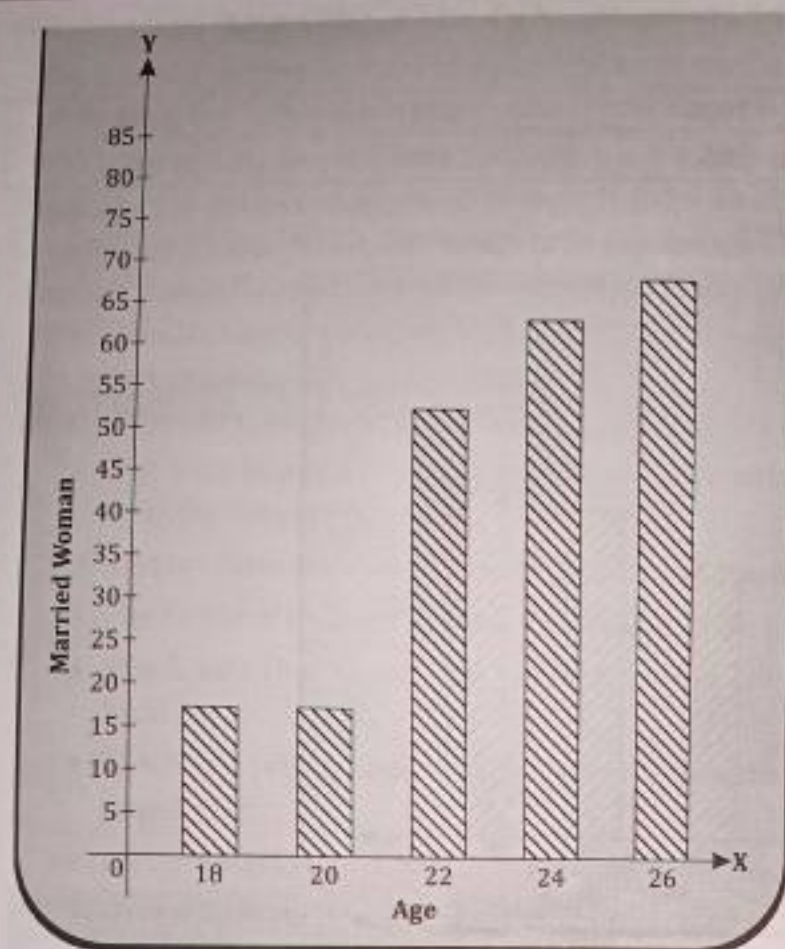
- The diagram provides a visual representation of production trends over the years.
- It helps in comparing the production levels across different years quickly.

Illustration 2

The following table gives the percentages of married women of Karnataka population under various ages. Represent the data by a simple bar diagram.

Age (in years)	18	20	22	24	26
Married women (in %)	17	17	51	62	66

Solution



Representation of Data

- The X-axis (horizontal axis) represents the age groups of women (18, 20, 22, 24, and 26 years).
- The Y-axis (vertical axis) represents the percentage of married women in Karnataka.
- The height of each bar corresponds to the percentage of married women at each age.

Trend Analysis

- At 18 and 20 years, only 17% of women are married.
- The percentage rises significantly at 22 years (51%).
- At 24 years, 62% of women are married, showing a steady increase.
- The highest percentage is observed at 26 years (66%), indicating that most women get married by this age.

1.6.7.2 Multiple Bar Diagram

A **Multiple Bar Diagram** is an extension of the Simple Bar Diagram, where multiple bars are used for each category to compare two or more related variables. These bars are placed side by side to visually represent multiple sets of data for a single attribute. This type of diagram is particularly useful for analyzing trends, comparisons, and relationships between different groups over time or across categories.

Structure of a Multiple Bar Diagram

- Each category has two or more bars placed next to each other for easy comparison.
- The bars are of equal width and separated by uniform spacing.
- Different bars are distinguished using colors, patterns, or shading to represent different variables.
- The X-axis (horizontal axis) represents the categories (Example, years, regions, product types).
- The Y-axis (vertical axis) represents the values (Example, population, sales, profits, production).
- A legend or footnote is included to explain the markings or colors used.

Characteristics of a Multiple Bar Diagram

- Used for comparing multiple variables within a single attribute.
- Bars are placed side by side for each category.
- Different shades, colors, or patterns are used to differentiate between data groups.
- Helpful for tracking trends over time or across different groups.
- Bars remain of the same width, while height represents the magnitude of the value.

Advantages of Multiple Bar Diagram

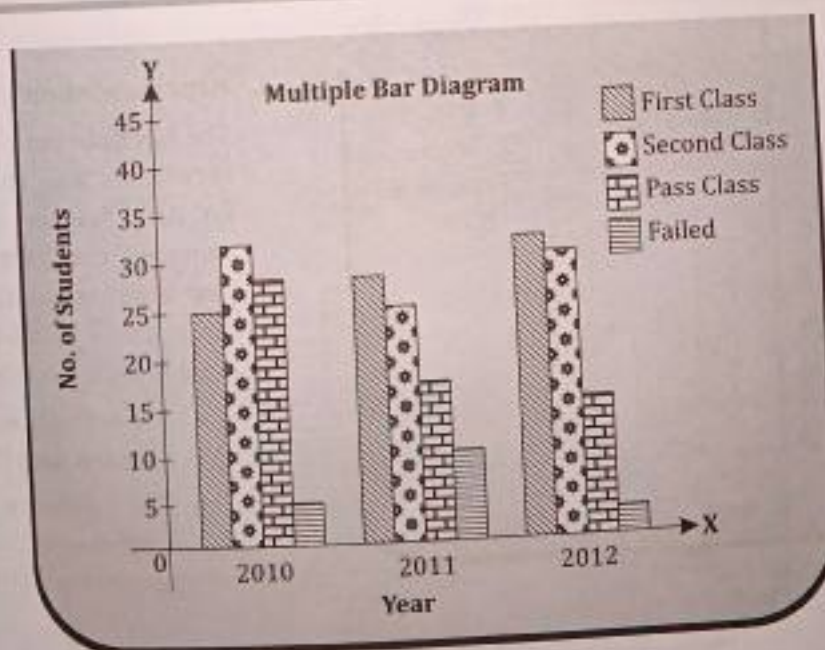
- Enables direct comparison between different variables.
- Simplifies complex data by visually representing multiple attributes together.
- Easy to interpret trends over time.
- Useful for comparing multiple groups across the same category.
- Widely used in research, business, economics, and social studies.

Illustration 3

The following table shows the results of degree students of a college for the last three years. Draw a multiple bar diagram.

Years	First class	Second class	Pass class	Failed
2010	25	32	28	5
2011	28	25	17	10
2012	32	30	15	3

Solution



1. Representation of Data

- X-Axis (Horizontal Axis) represents the years (2010, 2011, 2012).
- Y-Axis (Vertical Axis) represents the number of students.
- **Bars and Legends:**
 - First Class students are represented by diagonally shaded bars.
 - Second Class students are shown with dotted pattern bars.
 - Pass Class students are displayed using horizontal line patterns.
 - Failed students are represented by plain bars.

2. Trend Analysis

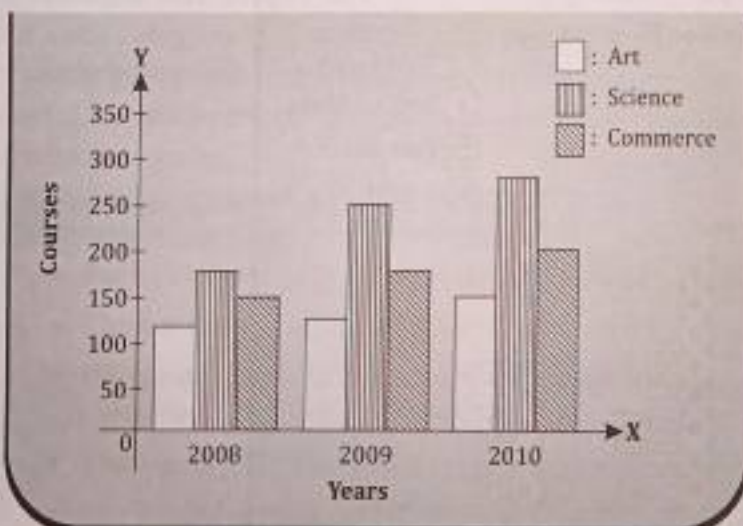
- **First Class:** The number of students achieving First Class increased steadily from 25 in 2010 to 28 in 2011 and further to 32 in 2012.
- **Second Class:** The number of Second Class students declined from 32 in 2010 to 25 in 2011, but rose again to 30 in 2012.
- **Pass Class:** The number of students in Pass Class dropped significantly, from 28 in 2010 to 17 in 2011, and further to 15 in 2012.
- **Failed Students:** The number of failed students was highest in 2011 (10 students) but dropped sharply to 3 in 2012, indicating improved academic performance over time.

Illustration 4

The table below shows the number of students admitted to a college across three disciplines (Arts, Science, and Commerce) over three academic years (2008, 2009, and 2010). Present this data using a Multiple Bar Diagram and analyze the trends.

Years	Arts	Science	Commerce
2008	120	180	150
2009	130	250	175
2010	150	280	200

Solution



1. Representation of Data

The Multiple Bar Diagram visually represents the admission trends for Arts, Science, and Commerce courses from 2008 to 2010. Each year is plotted on the X-axis, while the number of students admitted is represented on the Y-axis.

- The Arts course is represented by blank bars (□).
- The Science course is represented by vertical striped bars (|||).

- The Commerce course is represented by diagonal striped bars (\\).

Each group of bars shows the number of students admitted in that particular year. This makes it easier to compare student enrollment trends across different disciplines.

2. Trend Analysis

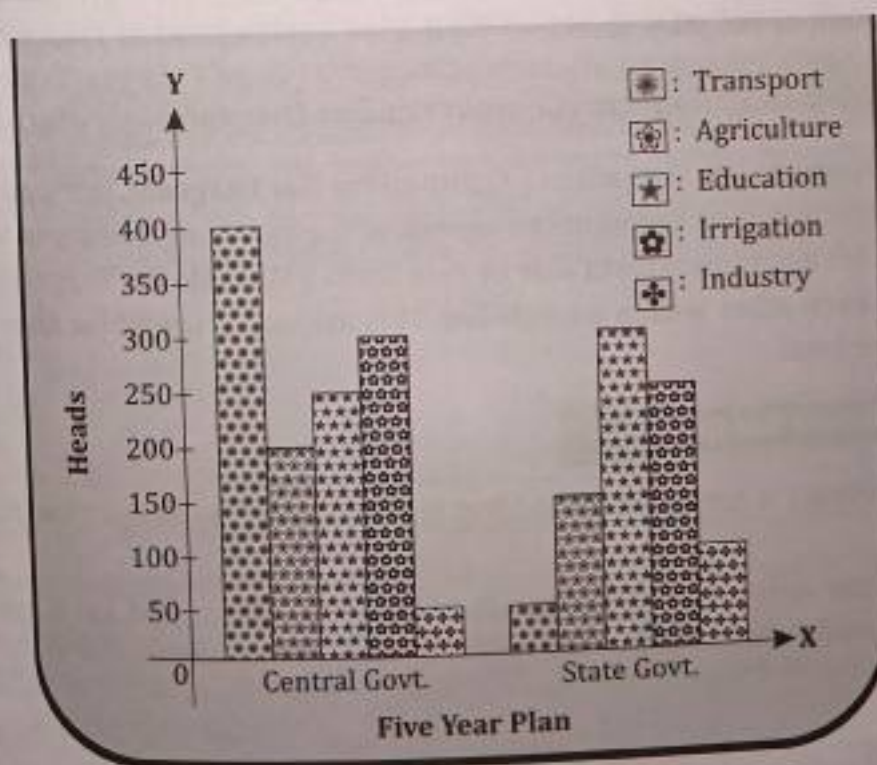
- Arts Admissions:** The number of students admitted to Arts increased steadily from 120 in 2008 to 130 in 2009 and further to 150 in 2010. This indicates a gradual rise in interest in Arts courses over the years.
- Science Admissions:** Science saw the highest number of admissions compared to the other two disciplines. Admissions increased significantly from 180 in 2008 to 250 in 2009, then further to 280 in 2010. This suggests a growing preference for Science courses among students.
- Commerce Admissions:** The number of students in Commerce also showed a consistent upward trend from 150 in 2008 to 175 in 2009 and 200 in 2010. This reflects a rising demand for Commerce-related education during this period.

Illustration 5

The table below represents the amount allocated (in thousand crores of rupees) by the Central Government and State Government for different sectors under the 12th Five-Year Plan. Present this data using a Multiple Bar Diagram and analyze the trends.

Heads	Transport	Agriculture	Education	Irrigation	Industry
Central Govt.	400	200	250	300	50
State Govt.	50	150	300	250	100

Solution



1. Representation of Data

The Multiple Bar Diagram represents the sector-wise allocation of funds in the 12th Five-Year Plan by the Central and State Governments. The X-axis represents the different sectors, while the Y-axis represents the amount allocated in thousand crores of rupees.

- Each sector has two bars representing Central Government and State Government allocations.
- Different patterns and markings distinguish the sectors for easy identification.
- This visualization helps in comparing government spending across different sectors.

2. Trend Analysis

- **Transport:** The Central Government allocated the highest amount (400 thousand crores), while the State Government allocated only 50 thousand crores. This indicates that transport infrastructure development was primarily the responsibility of the Central Government.
- **Agriculture:** The Central Government allocated 200 thousand crores, while the State Government allocated 150 thousand crores. Both governments shared responsibility, but the Central Government invested slightly more in agriculture.
- **Education:** The State Government allocated more funds (300 thousand crores) compared to the Central Government (250 thousand crores). This highlights the State Government's higher involvement in education-related projects and policies.
- **Irrigation:** The Central Government allocated 300 thousand crores, while the State Government allocated 250 thousand crores. Both governments made substantial investments, emphasizing the importance of irrigation projects in economic growth.
- **Industry:** The State Government allocated twice as much (100 thousand crores) as the Central Government (50 thousand crores). This suggests a greater focus on industrial development at the state level.

1.6.7.3

Sub-Divided Bar Diagram (Component Bar Diagram)

A **Sub-Divided Bar Diagram**, also known as a **Component Bar Diagram**, is a type of bar chart where each bar is divided into different segments to represent various components of a whole. Instead of placing separate bars for each component side by side (as in a Multiple Bar Diagram), the components are stacked on top of each other within a single bar. This method is useful for showing how different parts contribute to the total.

Structure of a Sub-Divided Bar Diagram

- Each bar represents a total value, divided into multiple segments that represent different components.
- The height of the entire bar corresponds to the total value, and each component's height represents its proportion of the whole.

- Correlation
- to
- Diff
- The
- The
- A le

Characteristics

- Rep
- Com
- Help
- Each
- Uses

Advantages

- Clear
- Effect
- Easy
- Avoid
- Wide

Illustration

The table below shows the different years of the 12th Five-Year Plan, and the component

Category
Cost of Raw Materials
Labour
Power
Debt
Total

Solution

Using a Sub-Divided Bar Diagram, the data for each year can be represented. This method is easier to compare

- Components within each bar are stacked from the largest at the base to the smallest at the top to maintain a consistent order across all bars.
- Different colors, patterns, or shading distinguish the various components within a single bar.
- The X-axis represents the categories (e.g., years, regions, departments).
- The Y-axis represents the total values (e.g., sales, population, production).
- A legend or footnote explains the colors or patterns used for the components.

Characteristics of a Sub-Divided Bar Diagram

- Represents how different parts contribute to a total.
- Components are stacked within a single bar instead of being placed separately.
- Helps in understanding the composition of data in an easy and visual way.
- Each bar maintains the same order of components across categories for consistency.
- Uses colors or shading to differentiate between various components.

Advantages of Sub-Divided Bar Diagram

- Clearly shows the composition of different groups within a total.
- Effective for comparing categories where each category has multiple sub-categories.
- Easier to analyze contributions of individual components to a whole.
- Avoids overlapping of bars, making the chart more readable.
- Widely used in education, business, finance, and social studies for data representation.

Illustration 6

The table below provides information on the cost components of manufacturing a chair over different years. The cost is categorized into four major components: Raw Material, Labour, Polish, and Delivery Charges. The objective is to visually represent the contribution of each component to the total cost in the years 2010, 2015, and 2020 using a Sub-Divided Bar Diagram.

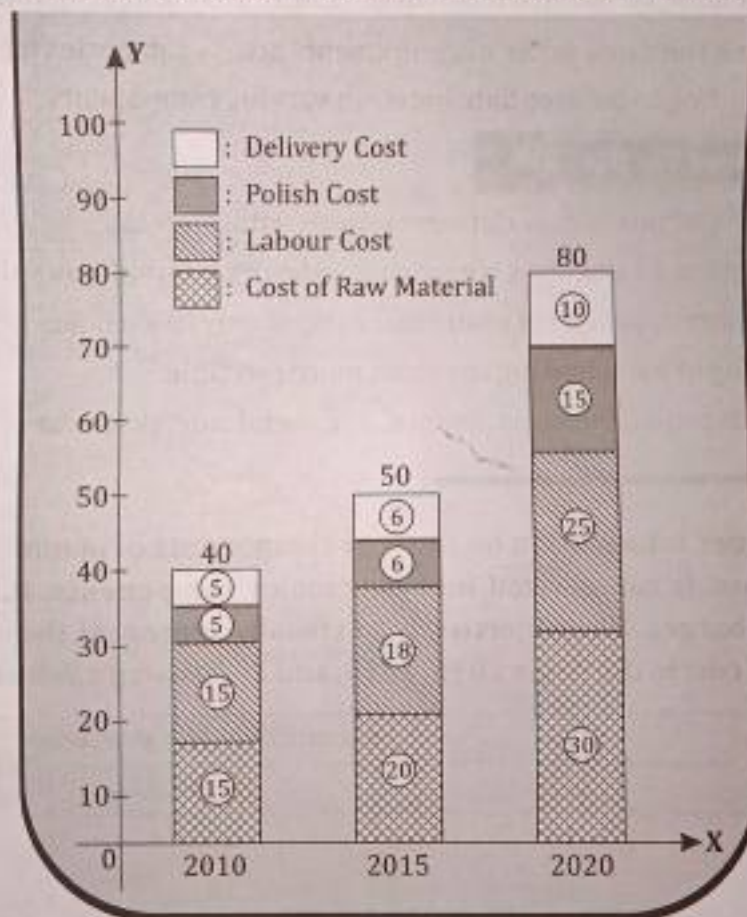
Category	Cost per chair (in ₹) year wise		
	2010	2015	2020
Cost of Raw Material	15	20	30
Labour Cost	15	18	25
Polish	5	6	15
Delivery	5	6	10
Total	40	50	80

Solution

Using a Sub-Divided Bar Diagram, we can stack the different cost components on top of each other for each year. This will help visualize the contribution of each cost category to the total cost and make it easier to compare changes over time.

Category	2010		2015		2020	
	Cost (in ₹)	Cumulative Cost (in ₹)	Cost (in ₹)	Cumulative Cost (in ₹)	Cost (in ₹)	Cumulative Cost (in ₹)
Cost of Raw Material	15	15	20	20	30	30
Labour Cost	15	$(15 + 15) = 30$	18	$(20 + 18) = 38$	25	$(30 + 25) = 55$
Polish	5	$(30 + 5) = 35$	6	$(38 + 6) = 44$	15	$(55 + 15) = 70$
Delivery	5	$(35 + 5) = 40$	6	$(44 + 6) = 50$	10	$(70 + 10) = 80$
Total	40		50		80	

On the basis of above table required subdivided bar diagram is given below:



1. Representation of Data

The given table presents the cost breakdown of manufacturing a chair for the years 2010, 2015, and 2020. The total cost is divided into four key components: **Cost of Raw Material, Labour Cost, Polish Cost, and Delivery Cost**

Each component contributes to the total cost, and the cumulative cost is calculated for each year. A Sub-Divided Bar Diagram (Component Bar Diagram) is used to visually represent each cost component adds up to the total.

Illustration

The following table shows the number of students during the years 2000, 2001, and 2002 in Arts, Science, Commerce, Engineering, and Medical.

Year	Arts	Science	Commerce	Engineering	Medical
2000	10	15	20	25	30
2001	12	18	22	28	32
2002	14	20	24	30	34

Solution

Faculty

Arts

Science

Commerce

Engineering

Medical

- The bars represent the total cost of a chair for each year.
- Each bar is divided into four sections, showing the contribution of each cost category.
- The height of each segment corresponds to the cost of that component.

2. Trend Analysis

- The total cost of manufacturing a chair has increased over the years:

- 2010: ₹40

- 2015: ₹50

- 2020: ₹80

This shows a continuous rise in production costs.

- Cost of Raw Material and Labour Costs have seen significant increases over time.
 - Raw material cost increased from ₹15 in 2010 to ₹30 in 2020, indicating rising material prices.
 - Labour cost increased from ₹15 in 2010 to ₹25 in 2020, showing higher wages.
- Polish and Delivery Costs have also increased, but at a slower rate compared to raw materials and labour.
- The steepest cost increase is observed from 2015 to 2020, where the total cost jumped by ₹30, compared to a ₹10 increase from 2010 to 2015.

Illustration 7

The following data represents the student enrollment in different faculties of a university during the years 2008 to 2010. The enrollment figures are categorized into five faculties: Arts, Science, Commerce, Engineering, and Medical. Construct a Sub-Divided (Component) Bar Diagram to visually represent the distribution of students across these faculties over the three years.

Year	Faculties				
	Arts	Science	Commerce	Engineering	Medical
2008	200	150	50	30	20
2009	250	200	80	50	40
2010	300	250	100	80	50

Solution

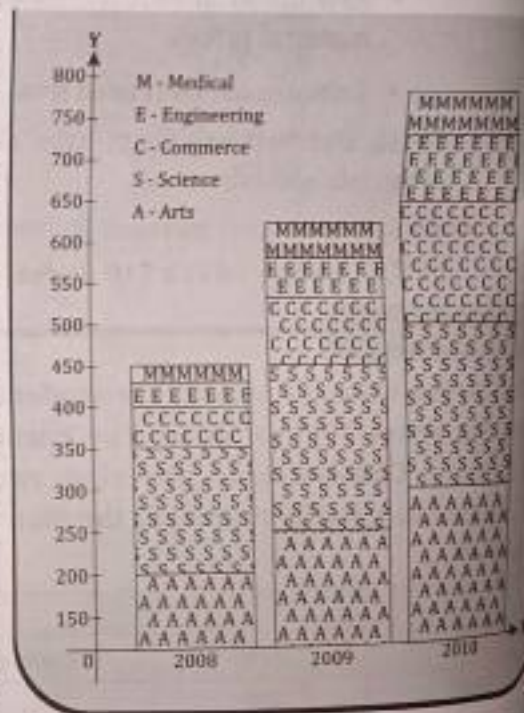
Faculty	2008		2009		2010	
	Actual	Cumulative	Actual	Cumulative	Actual	Cumulative
Arts	200	200	250	250	300	300
Science	150	$(200 + 150) = 350$	200	$(250 + 200) = 450$	250	$(300 + 250) = 550$
Commerce	50	$(350 + 50) = 400$	80	$(450 + 80) = 530$	100	$(550 + 100) = 650$
Engineering	30	$(400 + 30) = 430$	50	$(530 + 50) = 580$	80	$(650 + 80) = 730$
Medical	20	$(430 + 20) = 450$	40	$(580 + 40) = 620$	50	$(730 + 50) = 780$

1. Representation of Data:

- The given data is presented using a Sub-Divided (Component) Bar Diagram, which visually represents the student enrollment in five different faculties (Arts, Science, Commerce, Engineering, and Medical) across the years 2008, 2009, and 2010.
- Each bar represents a year, and within each bar, different segments correspond to the number of students in each faculty.
- The cumulative values are calculated step by step to show the total student enrollment for that year.
- The increasing height of the bars indicates the overall growth in student strength over the three years.

2. Trend Analysis:

- **Overall Growth:** The total number of students has increased from 450 in 2008 to 780 in 2010. This reflects a consistent rise in university enrollment.
- **Arts and Science Growth:** The Arts faculty had the highest number of students in each year, increasing from 200 in 2008 to 300 in 2010. Similarly, the Science faculty also showed significant growth from 150 to 250.
- **Commerce, Engineering, and Medical:** While Commerce and Engineering faculties saw steady increases, the Medical faculty had the least number of students, although it still exhibited gradual growth.
- **Cumulative Increase:** Every segment in the bars has increased over time, showing that all faculties experienced growth, with major contributions from Arts and Science streams.
- **Comparison:** The rate of increase in Engineering and Medical faculties is relatively higher in later years. This indicates a growing interest in professional courses.



1.6.7.4 Percentage Bar Diagram (Percentage Sub-Divided Bar Diagram)

A **Percentage Bar Diagram**, also known as a **Percentage Sub-Divided Bar Diagram**, is a graphical representation used to show the proportion of different components in a total value expressed as a percentage. Unlike a Sub-Divided Bar Diagram, where the total height of the bars varies according to the absolute values, in a Percentage Bar Diagram, all bars are of equal length (100%) and each component is represented as a percentage of the whole.

First of all the total of the given variable is assumed equal to 100. Then the percentage is calculated for each and every component of the variable. After then the cumulative percentage are calculated.

for every component
like sub-divided

Conversion

Percentage

Structure of

- Each bar represents a year, and within each bar, different segments correspond to the number of students in each faculty.
- The length of the bars indicates the overall growth in student strength over the three years.
- Each component is represented by a different color.
- Cumulative values are calculated step by step to show the total student enrollment for that year.
- Different faculties are represented by different colors.
- The X-axis represents the years.
- The Y-axis represents the number of students.
- A legend is provided to identify the faculties.

Characteristics

- Represents the proportion of different components in a total value.
- All bars have an equal length of 100%.
- Components are represented by different colors.
- Uses color coding to distinguish between different components.
- Helps in comparing the relative proportions of different components across different categories.

Advantages of

- Standardized representation.
- Easy to interpret.
- Clearly shows the relative proportions of different components.
- Useful for comparing the relative proportions of different components across different categories.
- Prevents misinterpretation.

Illustration 8

The following table shows the number of students enrolled in a course in the year 2020, categorized by unit. The total number of students has increased by 20% over the previous year. The following table shows the proportional distribution of students across the units.

for every component. Finally the bars are subdivided into the cumulative percentage and presented like subdivided bar diagram.

$$\text{Converting to Percentage} = \frac{\text{Actual Value}}{\text{Total of the Actual Value}} \times 100$$

Structure of a Percentage Bar Diagram

- Each bar represents a total value, with different segments indicating proportions of components in percentage terms.
- The length of all bars is the same (100%) to ensure a uniform scale.
- Each component's height is determined based on its percentage of the total value.
- Cumulative percentages are calculated to determine the segments within each bar.
- Different colors, shades, or patterns differentiate the various components.
- The X-axis represents categories (e.g., years, locations, departments).
- The Y-axis represents percentage values, where every bar reaches 100%.
- A legend or footnote explains the colors or patterns used for the components.

Characteristics of a Percentage Bar Diagram

- Represents data as percentages rather than absolute values.
- All bars have the same height (100%) for easy comparisons.
- Components are stacked within the bar, showing relative proportions.
- Uses colors, shading, or patterns to differentiate components.
- Helps in understanding the composition of a whole in different categories.

Advantages of a Percentage Bar Diagram

- Standardized for comparison, as all bars are of equal length.
- Easy to interpret when comparing proportions across different categories.
- Clearly shows the distribution of components within a total.
- Useful for visualizing relative differences between groups.
- Prevents misinterpretation that can arise due to varying total values in other types of diagrams.

Illustration 8

The following table provides data on the cost per unit of production in the years 2010 and 2020, categorized into Material Cost, Labour Cost, and Delivery Charges. The total cost per unit has increased over the years. Construct a Percentage Bar Diagram to represent the proportional distribution of these costs for both years.

Category	Cost Per Unit in 2010	Cost Per Unit in 2020
Material	20	32
Labour	25	36
Delivery	5	12
Total	50	80

Solution

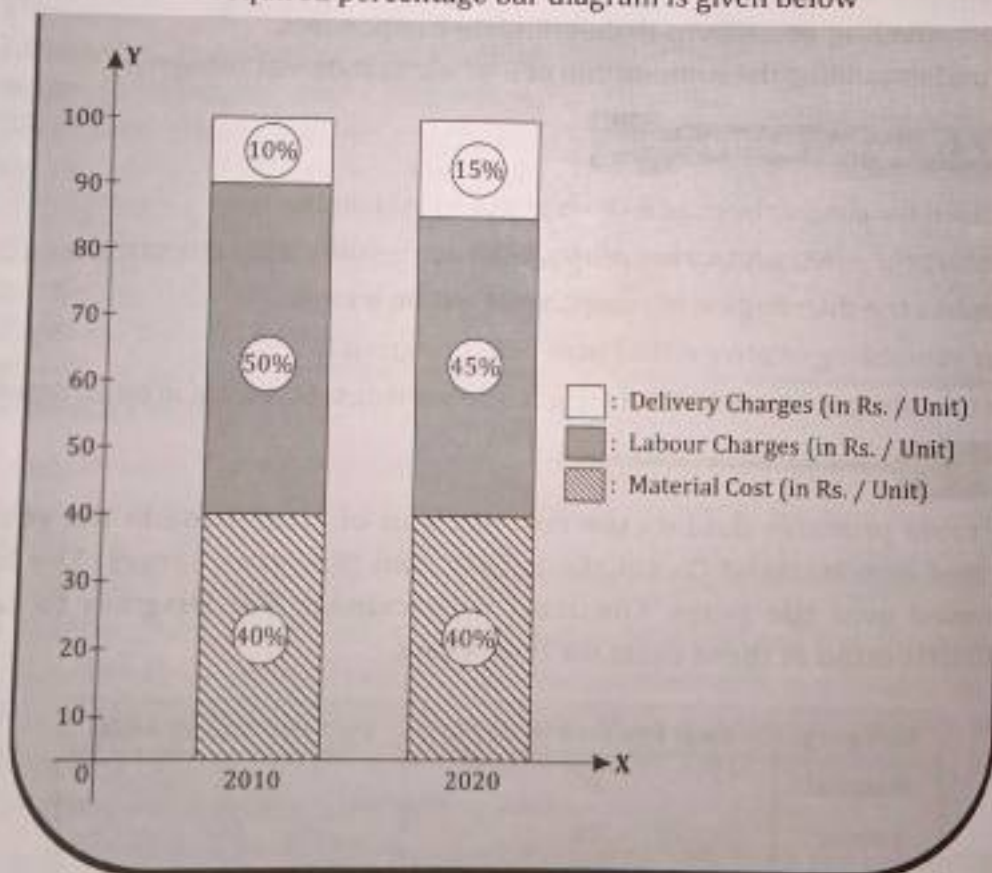
Calculation of Percentage

Category	Year 2010	Year 2020
Material	$\frac{20}{50} \times 100 = 40$	$\frac{32}{80} \times 100 = 40$
Labour	$\frac{25}{50} \times 100 = 50$	$\frac{36}{80} \times 100 = 45$
Delivery	$\frac{5}{50} \times 100 = 10$	$\frac{12}{80} \times 100 = 15$

The percentage and cumulative percentage for both the years in various category is shown below.

Category	Cost Per Unit (2010)			Cost Per Unit (2020)		
	Cost	% Cost	Cumulative % Cost	Cost	% Cost	Cumulative % Cost
Material	20	40	40	32	40	40
Labour	25	50	$(40 + 50) = 90$	36	45	$(40 + 45) = 85$
Delivery	5	10	$(90 + 10) = 100$	12	15	$(85 + 15) = 100$
Total	50	100		80	100	

On the basis of above table required percentage bar diagram is given below



Illustration

The following on their occ Industrial, Ag Diagram to v

1. Representation of Data:

The Percentage Bar Diagram represents the proportion of Material Cost, Labour Cost, and Delivery Charges in total cost per unit for the years 2010 and 2020. The height of both bars is kept equal to 100%, and the segments within each bar represent the percentage share of each cost component.

- X-Axis: Represents the years (2010 and 2020).
- Y-Axis: Represents the percentage of total cost (ranging from 0% to 100%).
- Each bar is divided into three segments to represent the Material Cost, Labour Cost, and Delivery Charges as a proportion of the total cost.
- Different shading patterns are used to differentiate between the three components.
- Cumulative percentages are used to determine the segment heights within each bar.

2. Trend Analysis:

- **Increase in Total Cost:** The total cost per unit has increased from ₹50 in 2010 to ₹80 in 2020. This reflects rising production costs.
- **Labour Costs Have Increased:** Labour costs accounted for 50% of the total cost in 2010 and increased to 45% in 2020. Despite the increase in absolute cost from ₹25 to ₹36, its percentage share has slightly decreased. This indicates that other costs have risen at a faster rate.
- **Significant Increase in Delivery Charges:** Delivery charges have doubled from ₹5 in 2010 to ₹12 in 2020. The share of delivery charges in total cost has increased from 10% to 15%. This suggests that increased logistics and transportation expenses over time.
- **Material Cost Remains Constant in Proportion:** Material cost remains at 40% of the total cost in both years, despite its absolute increase from ₹20 to ₹32. This suggests that while material costs have increased, they have remained proportionally stable relative to the total cost.

Illustration 9

The following table presents the percentage distribution of men and women in India based on their occupational classification. The data categorizes employment into four sectors: Industrial, Agricultural, Household Industries, and Miscellaneous. Construct a Percentage Bar Diagram to visually represent the occupational distribution of men and women in India.

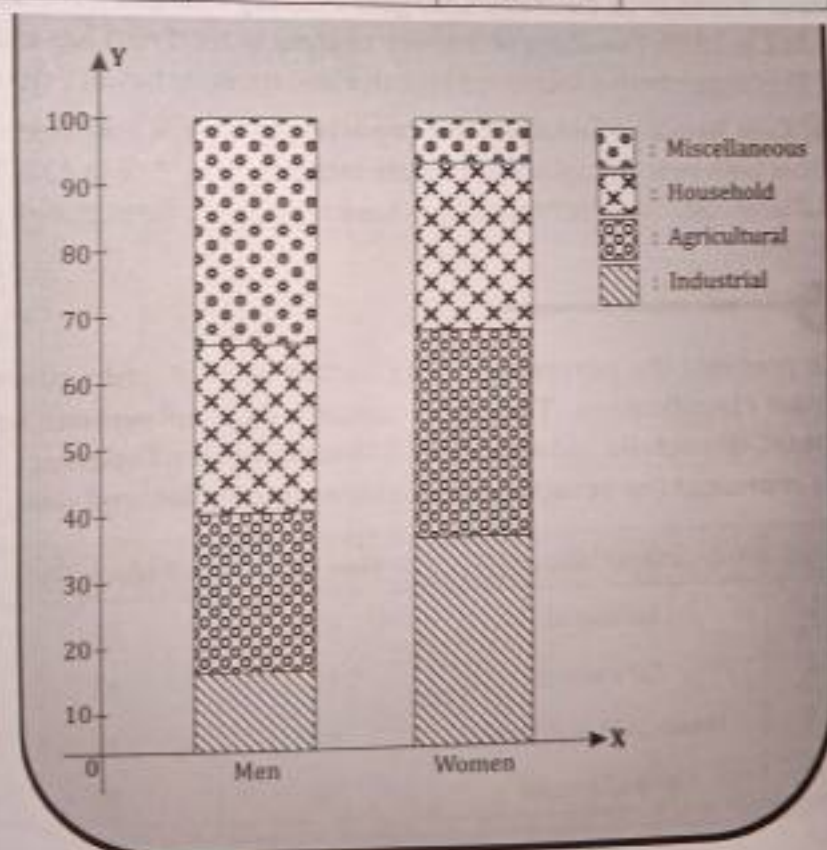
Sl. No.	Occupation/ labour	Men (%)	Women (%)
1	Industrial	45	55
2	Agriculture	55	45
3	House hold industries	61	39
4	Miscellaneous	88	12

Solution

Calculation of Percentage

Occupation/ Labour	Men	Women
Industrial	$\frac{45}{249} \times 100 = 18$	$\frac{55}{151} \times 100 = 36.42$
Agriculture	$\frac{55}{249} \times 100 = 22$	$\frac{45}{151} \times 100 = 29.8$
House hold industries	$\frac{61}{249} \times 100 = 25$	$\frac{39}{151} \times 100 = 25.83$
Miscellaneous	$\frac{88}{249} \times 100 = 35$	$\frac{12}{151} \times 100 = 7.95$

Men	Percentage %	Cumulative %	Women	Percentage %	Cumulative %
45	18	18	55	36.42	36.42
55	22	(18 + 22) = 40	45	29.8	66.22
61	25	(40 + 25) = 65	39	25.83	92.05
88	35	(65 + 35) = 100	12	7.95	100
Total = 249			Total = 151		



1. Represent

The Perce
India by u
segment

- X-A
- Y-Ax
- Each
- Hou
- Diffe
- occu
- Cum
- the b

2. Trend An

- High
- indu
- pres
- Men
- secto
- Indu
- men
- Men
- that
- only
- High
- is sp
- in th
- fema
- uncl

1.6.8 Two

Two-Dimensiona
circles. In these
Diagrams (such
area of a shape
Dimensional Dis

These diagrams
one country has
not clearly show
comparison.

1. Representation of Data

The Percentage Bar Diagram represents the occupational distribution of men and women in India by using a stacked percentage format. Both bars have an equal height of 100%, and each segment within the bars represents the proportion of workers in a particular sector.

- X-Axis: Represents gender categories (Men and Women).
- Y-Axis: Represents percentage of workforce (ranging from 0% to 100%).
- Each bar is divided into four segments, corresponding to Industrial, Agricultural, Household, and Miscellaneous occupations.
- Different patterns/shading styles are used to distinguish between the different occupational categories.
- Cumulative percentages are calculated to determine the height of each segment within the bar.

2. Trend Analysis:

- **Higher Female Participation in Industrial Work :** The percentage of women in the industrial sector (55%) is higher than that of men (45%). This indicates a greater female presence in industrial occupations.
- **Men Are More Involved in Agriculture and Household Industries :** The agriculture sector employs 55% of men, while only 45% of women are engaged in it. Household Industries have 61% male workers compared to 39% female workers, showing that more men are involved in this sector.
- **Men Dominate Miscellaneous Jobs :** The miscellaneous category includes occupations that do not fall under the primary three sectors. Here, 88% of workers are men, whereas only 12% are women.
- **Higher Occupational Diversification Among Men :** Men's workforce distribution is spread across all four sectors, while women's participation is more concentrated in the industrial and agricultural sectors. The miscellaneous category has the lowest female representation, indicating that women are less involved in various informal or unclassified jobs.

1.6.8 Two Dimensional Diagrams

Two-Dimensional Diagrams are a way of presenting data using shapes like rectangles, squares, and circles. In these diagrams, both height and width are used to show values. Unlike One-Dimensional Diagrams (such as Bar Diagrams), which only consider height, Two-Dimensional Diagrams use the area of a shape to represent the data. The larger the area, the bigger the value it represents. **Two-Dimensional Diagrams** are also known as **Surface Diagrams** or **Area Diagrams**.

These diagrams are useful when data values are very different from each other. For example, if one country has a population of 100 million and another has 5 million, a simple bar diagram may not clearly show the difference. In such cases, Two-Dimensional Diagrams provide a better visual comparison.

Types of Two-Dimensional Diagrams

The important types of Two-Dimensional Diagrams include:

1. Rectangular Diagrams
2. Square Diagrams
3. Circular Diagrams (Pie Diagrams)

Among these, Pie Diagrams (Circle Diagrams) are the most commonly used. Circles drawn with area proportional to the magnitude of the observations constitute a Pie Diagram. This method allows for a clear representation of proportions by dividing a circle into sections based on percentage values.

Features of Two-Dimensional Diagrams

- Both height and width are considered to represent data.
- Larger values appear as bigger shapes, making comparisons easy.
- Used when there are big differences in data values.
- Common shapes used include rectangles, squares, and circles.
- Pie Diagrams (Circle Diagrams) are the most popular type.

When to Use Two-Dimensional Diagrams?

Two-Dimensional Diagrams are helpful in situations where one-dimensional bar charts are not enough to show clear differences. Some common uses include:

- Comparing land areas of different countries using rectangular diagrams.
- Showing budget distribution in a company using pie diagrams.
- Representing different crops grown in a region using square diagrams.
- Analyzing the proportion of various age groups in a city using circular diagrams.

Why Use Two-Dimensional Diagrams?

- Better for comparing large and small values together.
- Easy to understand at a glance.
- Shows proportions clearly in percentage-based data.
- Commonly used in business, economics, and population studies.

Advantages of Two-Dimensional Diagrams

- Provides a better visual representation of large numerical differences.
- Ideal for proportion-based comparisons, such as market share, budget allocation, or land use.
- Simplifies complex data and makes it easier to understand.
- Used widely in business, economics, demographics, and agriculture.
- Effective in representing total values with their subdivisions (Example, different types of expenses in a company).

A Pie Diagram is a circular diagram of the whole of a whole that to

Structure

- ...
- ...
- ...
- ...
- ...
- ...

Characteristics

- R...
- A...
- U...
- V...
- C...

Advantages

- Ea...
- Cle...
- Ide...
- Pre...
- Wi...

Steps to

Step 1 :

Step 2 :

Step 3 :

1.6.8.1 Pie Diagrams (or) Circular Diagrams

A Pie Diagram, also known as a **Circular Diagram**, is a graphical representation of data in the form of a circle. The circle is divided into different sections (slices), where each slice represents a proportion of the total. This type of diagram is used when data needs to be shown as percentages or fractions of a whole. The entire circle represents 100% of the data, and each component is shown as a part of that total.

Structure of a Pie Diagram

- The entire circle represents the total value (100%).
- The circle is divided into different slices based on the proportion of each category.
- The size of each slice is proportional to the percentage it represents.
- Each sector (slice) is labeled with the corresponding percentage or value.
- Different colors, patterns, or shades are used to distinguish categories.
- A legend or footnote explains the different sections of the diagram.

Characteristics of a Pie Diagram

- Represents data as percentages of a whole.
- All sections together sum up to 100%, ensuring clarity in data representation.
- Useful for comparing proportions rather than absolute values.
- Visually easy to interpret for non-technical audiences.
- Commonly used in surveys, budgets, and financial reports.

Advantages of a Pie Diagram

- Easy to understand and visually appealing.
- Clearly shows the proportion of each category in a dataset.
- Ideal for displaying percentage-based data in an intuitive way.
- Prevents misinterpretation of large and small values.
- Widely used for business, research, and social studies.

Steps to Construct a Pie Diagram

- Step 1:** **Find the Total Value:** Add up all the given values. This total represents 100% of the data.
- Step 2:** **Calculate the Angle for Each Part:** Find the sector angles (in degrees) of each part keeping in mind that total angle around the centre of a circle is of 360° .
- $\text{Angle} = (\text{part}/\text{total}) \times 360$
- Step 3:** **Convert Each Part into a Percentage:** Find the percentage of each part taking the total obtained in step 1 as 100 percent.
- $\text{Percentage} = (\text{part}/\text{total}) \times 100$
- This helps in labeling the sections.

Step 4 : Draw the Pie Diagram:

Draw a circle and divide it into sectors, where each sector (or area of the sector) of the circle with corresponding angles obtained in step 2 will represent the size of corresponding parts. Diagram thus obtained is nothing but pie chart fitted to the given data.

Illustration 1

A company was founded by four partners A, B, C, and D, who agreed to distribute profits and losses in the ratio 4:3:2:1. In the year 2010, the company earned a total profit of ₹14,000. Represent the individual shares of profit for each partner using a pie chart.

Solution

Given ratio is 4 : 3 : 2 : 1

∴ Sum of ratios = 4 + 3 + 2 + 1 = 10

Calculation of Degrees and Percentages:

Partners	Profits (in ₹)	Sector Angles (in degree)	Percentages
A	$14,000 \times \frac{4}{10} = 5,600$	$\frac{5,600}{14,000} \times 360 = 144$	$\frac{5,600}{14,000} \times 100 = 40$
		or $\frac{4}{10} \times 360 = 144$	
B	$14,000 \times \frac{3}{10} = 4,200$	$\frac{4,200}{14,000} \times 360 = 108$	$\frac{4,200}{14,000} \times 100 = 30$
		or $\frac{3}{10} \times 360 = 108$	
C	$14,000 \times \frac{2}{10} = 2,800$	$\frac{2,800}{14,000} \times 360 = 72$	$\frac{2,800}{14,000} \times 100 = 20$
		or $\frac{2}{10} \times 360 = 72$	
D	$14,000 \times \frac{1}{10} = 1,400$	$\frac{1,400}{14,000} \times 360 = 36$	$\frac{1,400}{14,000} \times 100 = 10$
		or $\frac{1}{10} \times 360 = 36$	
Total	14,000	360	100

On the basis of above calculation, pie chart which shows the shares of profit of the four partners is shown below:

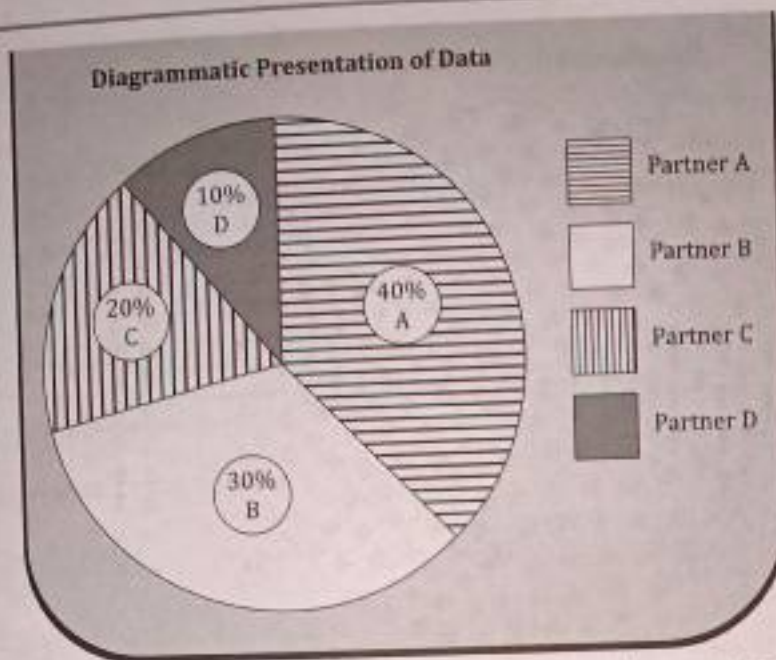
- The p
perce
- Large
lower
- Differ

Illustration

The percent
is given bel
Labour : 20
other exper
Represent t

Solution**Items**

Labour
Bricks
Cement
Steel
Wood
Supervisi
Other exper



- The pie chart divides the total profit into four proportional segments, each representing the percentage of total profit received by each partner.
- Larger sectors indicate higher shares in the total profit, whereas smaller sectors represent lower shares.
- Different shading patterns help distinguish each partner's share in the profit distribution.

Illustration 2

The percentage breakup of the cost of constructing a house in Bangalore (excluding land cost) is given below:

Labour : 20%, Bricks : 12%, Cement : 20%, Steel : 15%, wood : 13%, supervision : 15%, other expenses : 5%.

Represent the construction cost distribution using a pie diagram.

Solution

Items	Percentage	Sector Angles (in degree)	Cumulative Angles
Labour	20	72	72
Bricks	12	43.2	$(72 + 43.2) = 115.2$
Cement	20	72	$(115.2 + 72) = 187.2$
Steel	15	54	$(187.2 + 54) = 241.2$
Wood	13	46.8	$(241.2 + 46.8) = 288$
Supervision	15	54	$(288 + 54) = 342$
Other expenses	5	18	$(342 + 18) = 360$

$$\text{Convert to Degrees} = \frac{\text{Percentage}}{100} \times 360 = \frac{20}{100} \times 360 = 72$$

Diagrammatic Presentation of Data

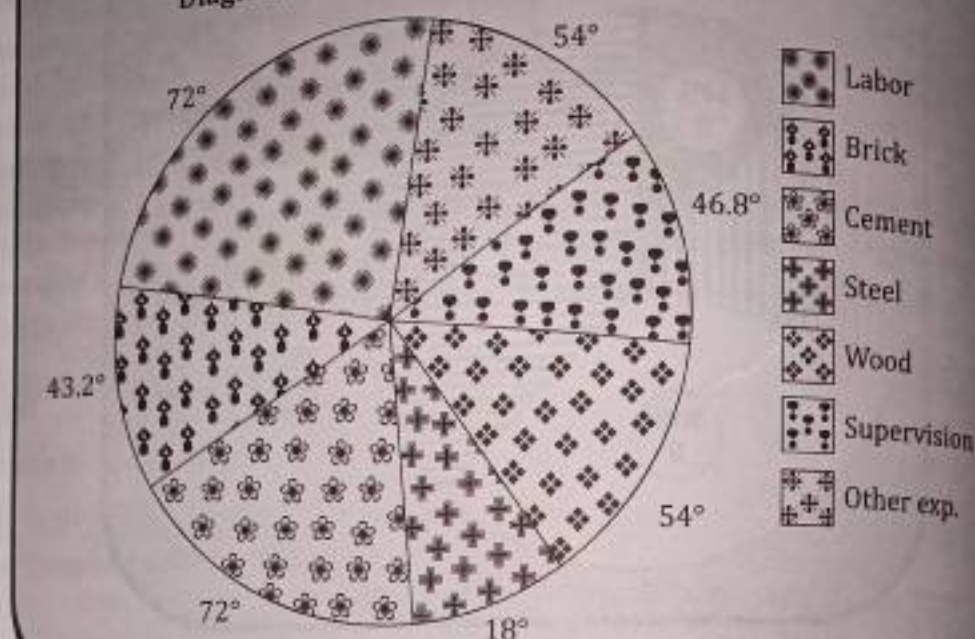


Illustration 3

The income of the government from different sources is given below. Represent the distribution of revenue using a pie diagram.

Source	Revenue (in million ')
Customs	80
Excise	190
Income Tax	160
Corporate Tax	75
Miscellaneous	35
Total	540

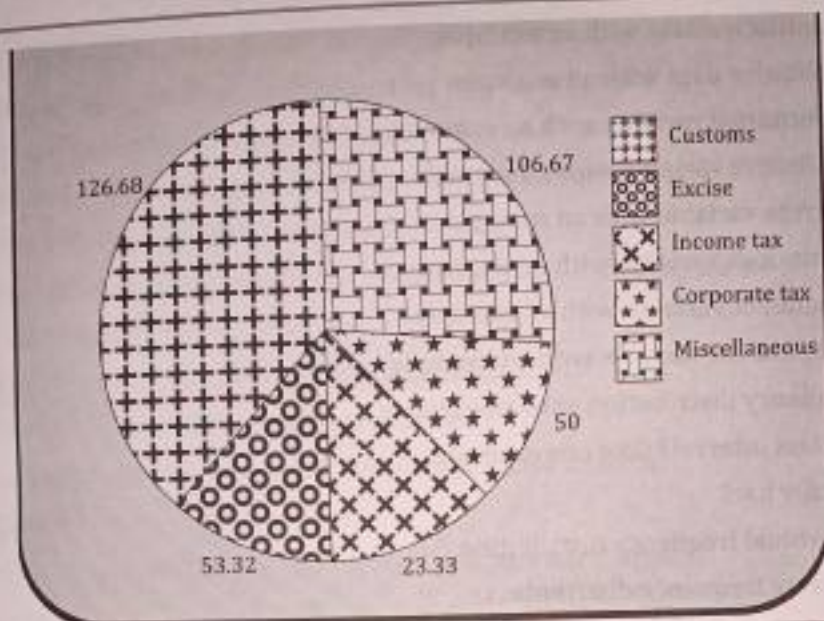
Solution

Source	Revenue (in million ')	Percentage	Angles in Degrees	Cumulative
Customs	80	14.81	53.32	53.32
Excise	190	35.19	126.68	$(53.32 + 126.68) = 180$
Income Tax	160	29.63	106.67	$(180 + 106.67) = 286.67$
Corporate Tax	75	13.89	50	$(286.67 + 50) = 336.67$
Miscellaneous	35	6.48	23.33	$(336.67 + 23.33) = 360$
Total	540	100	360	

1.7 Review

Short Answer

1. Define St
2. Define St
3. What is t
4. Define ra
5. Define pr
6. Define se
7. Define po
8. Define sa
9. Define su
10. Define ce
11. Define sa
12. What is s
13. Define pa
14. Define st
15. Define ur
16. Define va
17. Define at
18. Define fre
19. What is s



1.7 Review Questions

Short Answer Type Questions

1. Define Statistics in the plural sense with an example.
2. Define Statistics in the singular sense with an example.
3. What is the meaning of data in Statistics? Give an example.
4. Define raw data with an example.
5. Define primary data with an example.
6. Define secondary data with an example.
7. Define population in Statistics with an example.
8. Define sample with an example.
9. Define survey with an example.
10. Define census with an example.
11. Define sample survey with an example.
12. What is sampling? Give an example.
13. Define parameter with an example.
14. Define statistic with an example.
15. Define unit or individual with an example.
16. Define variable with an example.
17. Define attribute with an example.
18. Define frequency with an example.
19. What is seriation? Give an example.

20. Define quantitative data with an example.
21. Define qualitative data with an example.
22. Define quantitative variable with an example.
23. Define qualitative variable with an example.
24. Define discrete variable with an example.
25. Define continuous variable with an example.
26. Define dependent variable with an example.
27. Define independent variable with an example.
28. Define frequency distribution with an example.
29. What is a class interval? Give one example.
30. What is a tally bar?
31. Define individual frequency distribution.
32. Define discrete frequency distribution.
33. Define continuous frequency distribution.
34. What is meant by range in statistics?
35. Define class limits with an example.
36. What is inclusive class interval? Give an example.
37. What is exclusive class interval? Give an example.
38. Define class mark (mid-point).
39. Define magnitude or width of class interval.
40. What are open-end classes?
41. What is cumulative frequency distribution?
42. Define frequency density.
43. Define relative frequency.
44. What is bivariate frequency distribution?
45. What is marginal frequency distribution?
46. What is conditional frequency distribution?
47. What do you mean by Classification of Data?
48. Define Classification of Data.
49. Mention the Types of Classification.
50. Mention the Types of Qualitative Classification.
51. What is Qualitative Classification?
52. What is Simple Classification?
53. What is Manifold Classification?
54. What is Quantitative Classification?
55. What is diagrammatic presentation of data?

6. Define diagram.
7. Mention two a
8. How does diagr
9. List the differen
10. What are the ge
11. Differentiate be
12. What is a bar di
13. Why are bar dia
14. What are the ke
15. Mention two ad
16. List the differen
17. How does a sim
18. What is a sub-d
19. Explain the pur
20. What are the ad
21. What is a pie di
22. How does a pie
23. What are the ke
24. Explain the step
25. Mention two ad

Long Answer Type Questions

1. Write the features
2. Explain the obje
3. Explain the func
4. Differentiate be
5. Differentiate be
6. Differentiate be
7. Differentiate be
8. Explain frequen
9. Explain class int
10. Explain differen
11. Explain inclusive
12. Explain the follo

56. Define diagrammatic representation according to Croxton and Cowden.
57. Mention two advantages of using diagrams for data interpretation.
58. How does diagrammatic presentation help in statistical analysis?
59. List the different types of diagrams used for data presentation.
60. What are the general rules for constructing diagrams?
61. Differentiate between one-dimensional and two-dimensional diagrams.
62. What is a bar diagram?
63. Why are bar diagrams called one-dimensional diagrams?
64. What are the key characteristics of a bar diagram?
65. Mention two advantages of using bar diagrams for data analysis.
66. List the different types of bar diagrams.
67. How does a simple bar diagram differ from a multiple bar diagram?
68. What is a sub-divided bar diagram?
69. Explain the purpose of using a percentage bar diagram.
70. What are the advantages of using a percentage bar diagram over a sub-divided bar diagram?
71. What is a pie diagram?
72. How does a pie diagram represent data?
73. What are the key features of a circular diagram?
74. Explain the steps involved in constructing a pie diagram.
75. Mention two advantages of using pie diagrams for data analysis.

Long Answer Type Questions

1. Write the features of Statistics.
2. Explain the objectives of Statistics.
3. Explain the functions of Statistics.
4. Differentiate between primary data and secondary data.
5. Differentiate between discrete and continuous variables.
6. Differentiate between dependent and independent variables.
7. Differentiate between quantitative and qualitative data.
8. Explain frequency distribution with a suitable example.
9. Explain class interval and tally bar with suitable examples.
10. Explain different types of frequency distributions: individual, discrete, and continuous with examples.
11. Explain inclusive and exclusive class intervals with examples.
12. Explain the following terms with examples: range, class, class limits, class mark, magnitude of class interval.

13. Write a note on cumulative frequency distribution (less than and more than type) with examples.
14. Define frequency density and relative frequency. Explain with examples.
15. Explain bivariate frequency distribution with an illustration.
16. What is Classification of Data? Explain the Functions and Importance of Classification
17. Write the Requisites of Good Classification of Data.
18. What is Classification of Data? Explain the types of Classification of Data.
19. What is diagrammatic presentation? Explain its objectives and features.
20. Differentiate between diagrams and graphs with at least five key points.
21. What are the advantages and limitations of diagrammatic presentation? Explain with examples.
22. Write down the general rules for constructing diagrams and graphs.
23. List and explain the different types of diagrams used in statistical analysis.
24. What is a one-dimensional diagram? Explain its features and advantages.
25. What is a simple bar diagram? Write its features and explain its structure.
26. Explain the steps involved in constructing a simple bar diagram. Illustrate with an example.
27. What is a multiple bar diagram? Write its features and explain its structure with an example.
28. How does a multiple bar diagram help in comparing different datasets? Explain with examples.
29. What is a sub-divided bar diagram? Write its characteristics and explain its structure.
30. How does a sub-divided bar diagram represent data? Explain with a real-life example.
31. What is a percentage bar diagram? Explain its structure and advantages with examples.
32. What is a two-dimensional diagram? Explain its meaning, types, and features.
33. How does a two-dimensional diagram differ from a one-dimensional diagram? Explain with examples.
34. What are pie diagrams? Explain their features and the steps involved in their construction.
35. How does a pie diagram represent data proportionally? Explain with an example.
36. What are the advantages of using a pie chart for statistical analysis?
37. Explain the procedure for constructing a pie diagram. Illustrate with an example.
38. The following is the distribution for the age of the students in a school.

Age (Years)	0-5	5-10	10-15	15-20
No. of Students	35	45	50	30

Calculate:

1. The lower limit of the first class interval.
2. The class limits of the third class interval.
3. The class mark or mid-point for the interval 5-10.
4. The magnitude or width of a class interval.

39. Con

Ma
No

40. Co

Da
No

41. Con

Mic
Fre

42. Con

Mar
No43. Give
distr

0, 3,

44. The
distr

103,

99, 9

45. Prepa
meth46. The fo
using

4

4

5

43

58

47. Repre
diagramYe
Produ

39. Convert the following inclusive class intervals to exclusive class intervals:

Marks	0-9	10-19	20-29	30-39	40-49	50-59	60-69	70-79
No. of Students	4	16	21	35	28	14	6	3

40. Convert the following less than cumulative frequency distribution to exclusive class intervals:

Daily Wages	<100	<200	<300	<400	<500	<600	<700
No. of Persons	15	33	58	89	109	116	120

41. Convert the following mid-point values into exclusive class intervals:

Mid Point Value	10	30	50	70	90	110
Frequency	3	7	10	20	12	8

42. Convert the following more than cumulative frequency distribution to exclusive class intervals:

Marks	>10	>20	>30	>40	>50	>60	>70	>80
No. of Students	100	90	77	56	24	12	6	2

43. Given are the number of accidents that occurred in a city. Construct a discrete frequency distribution:

0, 3, 1, 2, 4, 0, 0, 3, 1, 3, 2, 2, 4, 2, 5, 5, 4, 2, 0, 2, 1, 3, 2, 4, 3

44. The following are the weights (in kg) of 28 students. Construct a continuous frequency distribution using exclusive class intervals of width 5 and also calculate frequency density:

103, 87, 90, 97, 104, 111, 101, 79, 96, 99, 104, 96, 81, 92, 64, 79, 122, 109, 119, 67, 84, 91, 101, 99, 96, 77, 96, 79

45. Prepare a frequency distribution from the following marks of 40 students, using exclusive method and class width = 4:

27, 15, 13, 21, 14, 14, 27, 11, 21, 17,

10, 37, 21, 10, 29, 15, 11, 21, 26, 19,

20, 21, 25, 26, 38, 13, 33, 23, 37, 29,

14, 11, 13, 35, 10, 13, 18, 17, 27, 20

46. The following are the marks of 50 students in a college test. Prepare a frequency distribution using inclusive class intervals of width 5 (apply Sturges' rule for deciding number of classes):

44, 63, 47, 53, 40, 39, 51, 42, 31, 49,

45, 52, 60, 48, 55, 43, 49, 35, 40, 36,

50, 35, 54, 59, 41, 62, 40, 49, 48, 48,

45, 60, 44, 40, 59, 47, 58, 58, 38, 33,

58, 36, 47, 54, 48, 41, 60, 44, 51, 40

47. Represent the following data regarding the production of wheat (in '000 tons) by simple bar diagram.

Year	2010	2011	2012	2013
Production	80	95	110	120

39. Convert the following inclusive class intervals to exclusive class intervals:

Marks	0-9	10-19	20-29	30-39	40-49	50-59	60-69	70-79
No. of Students	4	16	21	35	28	14	6	3

40. Convert the following less than cumulative frequency distribution to exclusive class intervals:

Daily Wages	<100	<200	<300	<400	<500	<600	<700
No. of Persons	15	33	58	89	109	116	120

41. Convert the following mid-point values into exclusive class intervals:

Mid Point Value	10	30	50	70	90	110
Frequency	3	7	10	20	12	8

42. Convert the following more than cumulative frequency distribution to exclusive class intervals:

Marks	>10	>20	>30	>40	>50	>60	>70	>80
No. of Students	100	90	77	56	24	12	6	2

43. Given are the number of accidents that occurred in a city. Construct a discrete frequency distribution:

0, 3, 1, 2, 4, 0, 0, 3, 1, 3, 2, 2, 4, 2, 5, 5, 4, 2, 0, 2, 1, 3, 2, 4, 3

44. The following are the weights (in kg) of 28 students. Construct a continuous frequency distribution using exclusive class intervals of width 5 and also calculate frequency density:

103, 87, 90, 97, 104, 111, 101, 79, 96, 99, 104, 96, 81, 92, 64, 79, 122, 109, 119, 67, 84, 91, 101, 99, 96, 77, 96, 79

45. Prepare a frequency distribution from the following marks of 40 students, using exclusive method and class width = 4:

27, 15, 13, 21, 14, 14, 27, 11, 21, 17,

10, 37, 21, 10, 29, 15, 11, 21, 26, 19,

20, 21, 25, 26, 38, 13, 33, 23, 37, 29,

14, 11, 13, 35, 10, 13, 18, 17, 27, 20

46. The following are the marks of 50 students in a college test. Prepare a frequency distribution using inclusive class intervals of width 5 (apply Sturges' rule for deciding number of classes):

44, 63, 47, 53, 40, 39, 51, 42, 31, 49,

45, 52, 60, 48, 55, 43, 49, 35, 40, 36,

50, 35, 54, 59, 41, 62, 40, 49, 48, 48,

45, 60, 44, 40, 59, 47, 58, 58, 38, 33,

58, 36, 47, 54, 48, 41, 60, 44, 51, 40

47. Represent the following data regarding the production of wheat (in '000 tons) by simple bar diagram.

Year	2010	2011	2012	2013
Production	80	95	110	120

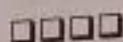
Year	1st class	2nd class	Pass class	Failed
2015	26	33	30	7
2016	31	27	21	12
2017	37	32	20	7

Years		2013	2014	2015	2016	2017	2018
Sales	Sugar	23	28	32	40	46	50
	Wheat	36	45	64	62	69	70

Food Quantity Consumed (In million Tons)			
Rice	Ragi	Wheat	lower
350	375	290	400

Subjects	Marks of Students	
	Student A	Student B
Language	75	82
English	85	94
Statistics	95	95
Economics	90	90
Business studies	90	90
Accountancy	94	98

Construct a pie diagram.



MEASURES OF CENTRAL TENDENCY

2.1 Introduction

In the previous chapter, different methods to arrange and display data were studied. These methods help in understanding the general pattern of data, such as its shape and how it is spread out. However, just organizing data is not enough to fully understand it. To get a clearer and deeper understanding, data needs to be summarized using numbers, which is done through statistics.

Summarizing data is an essential step in statistics. When a large amount of raw data is collected, it may look unorganized and difficult to analyze. To make it easier to handle, the data is first arranged into tables and frequency distributions. These methods help in grouping the data and showing patterns more clearly. However, even after this, the data may still be too large to interpret easily. To further simplify it and highlight its key features, mathematical measures are used.

One of the most important statistical concepts for understanding data is **Central Tendency**. This concept helps in finding a single value that represents the entire dataset. Instead of looking at all the numbers separately, a central value is chosen that gives an idea of the overall trend of the data.

2.1.1 Meaning and Definitions of Central Tendency



Meaning : Central Tendency

Central tendency is a general statistical concept that explains how data values tend to cluster around a central value. It shows that most data points in a dataset are concentrated near the middle rather than being spread out randomly. It helps in simplifying large datasets by finding a single value that represents the overall trend. The concept of central tendency helps in summarizing and comparing data easily instead of analyzing each number individually.



Definitions : Central Tendency

- **Prof. Bowley:** "The central tendency of a dataset is a single value that represents the entire dataset by showing the position around which other values are distributed."
- **W.L. King:** "Central tendency refers to the tendency of data to cluster around a central value, which serves as a representative figure for the dataset."

2.2.2 Measures of Central Tendency (Averages)



Meaning : Average (or) Measures of Central Tendency

Measures of Central Tendency refer to mathematical methods used to find a single value that represents a whole set of numbers. This central value gives an idea of where most of the data points lie in a dataset. It helps in understanding the general trend of the data and makes it easier to compare different datasets.

A measure of central tendency is also known as an **"average"** because it provides a central or typical value around which other data points are distributed. It is widely used in statistics to summarize large amounts of data in a simple way. When a large amount of data is collected, it may be difficult to analyze each number separately. To make the data easier to understand, a single representative value is chosen. This value is called the **"central value"** or **"average"**. It shows the middle or central point of the dataset.

Example : If five students scored 40, 50, 60, 70, and 80 in a test, instead of looking at each score separately, an average score can be calculated to represent all of them. This helps in quickly understanding how well the group performed overall.



Definition: Average (A Measure of Central Tendency)

The concept of an average has been explained by different experts in various ways. Some important definitions are:

1. **According to Clark and Sekkade**, "An average is an attempt to find one single figure to describe the whole of figures". This means an average is a single number that represents a group of numbers.
2. **According to Murray R. Spiegel**, "Average is a value which is typical or representative of a set of data". This means an average is a number that best represents a group of values.
3. **According to Croxton and Cowden**, "An average is a single value within the range of the data that is used to represent all the values in the series. Since an average is somewhere within the range of data, it is sometimes called a *measure of central value*". This means an average is a number that lies within the given data range and represents the entire dataset.
4. **According to Sipson and Kafka**, "A measure of central tendency is a typical value around which other figures *congregate*". This means an average is a central value around which other numbers in the dataset are grouped.

In simple terms, Central tendency is the general concept that data tends to cluster around a central value, while measures of central tendency are the specific methods used to calculate and express the central value such as mean, median, and mode.

Think of central tendency as the idea of determining the average temperature of a city over a year. Measures of central tendency are the different ways to calculate it—like taking the daily average temperature (mean), finding the most common temperature recorded (mode), or identifying the middle value from a list of yearly temperatures (median).

2.1.3

Measures of
The key of

1. To S
coll
the
2. To E
For
the
3. To E
on a
man
4. To A
if th
acad
5. To P
stand
befor
6. To E
elim
7. To A
polic
social

2.1.4

An ideal av
central tend
the key requ

1. Easy
and c
2. Base
repre
3. Math
consi
4. Not A
very l
5. Capab
analys
6. Stable
from d

2.1.3 Objectives of Measures of Central Tendency (Averages)

Measures of central tendency aim to provide a single, representative value that summarizes a dataset. The key objectives are:

1. **To Summarize Large Amounts of Data** : The human mind finds it difficult to process a large collection of numerical figures. A single representative value (average) helps in summarizing the data. This makes it easier to understand and remember.
2. **To Facilitate Comparison** : Averages allow for easy comparison between different datasets. For example, the wage levels of workers in two different factories can be compared by analyzing the average wages of workers in each.
3. **To Help in Decision-Making** : Many decisions in research, planning, and business depend on average values. For example, if a company's average monthly sales are declining, the sales manager may take corrective measures to boost sales.
4. **To Analyze Trends and Patterns** : Averages helps in identifying trends over time. For example, if the average marks of students in a school increase yearly, it suggests an improvement in academic performance.
5. **To Provide a Basis for Further Statistical Analysis** : Many statistical techniques, such as standard deviation and correlation, require an understanding of the central value of data before performing further calculations.
6. **To Ensure Fair Representation of Data** : Averages provide a fair representation of data by eliminating extreme values and fluctuations. This helps to get a clearer picture of the dataset.
7. **To Assist in Policy Formulation** : Governments and businesses use average values to develop policies. For instance, the average income of a population helps in designing tax policies and social welfare programs.

2.1.4 Requisites of an Ideal Average

An ideal average should accurately represent a dataset and provide a meaningful summary of its central tendency. It must be reliable, stable, and applicable across various fields. The following are the key requisites of an ideal average:

1. **Easy to Understand and Calculate** : It should be simple so that anyone can easily interpret and compute it without difficulty.
2. **Based on all Observations** : Every value in the dataset should be considered to ensure a true representation of the data.
3. **Mathematically Defined** : A good average must have a clear mathematical formula to ensure consistency and accuracy.
4. **Not Affected by Extreme Values** : It should not be significantly influenced by very high or very low values to prevent distortions.
5. **Capable of Further Mathematical Treatment** : It should allow for advanced statistical analysis such as variance, standard deviation, and correlation.
6. **Stable and Consistent** : The average should provide similar results even when calculated from different samples of the same population.

7. **Capable of Handling Open-End Distributions** : It should be computable even when dataset has class intervals that are not fully specified.
8. **Unique and Definite** : The same dataset should always produce the same average value to ensure accuracy.
9. **Should be Representative** : The average must reflect the true central tendency of the data and not be misleading.
10. **Widely Applicable** : It should be useful in various fields such as business, economics, science, and social research.
11. **Comparable Across Datasets** : A good average should allow meaningful comparisons between different datasets or time periods.
12. **Resistant to Sampling Variations** : It should remain stable when derived from different samples of the same population to ensure reliability.

2.1.5 Different Measures of Central Tendency (or) Different Types of Averages

There are several ways to determine the central value of a dataset. The most commonly used measures of central tendency are:

1. **Arithmetic Mean** : The sum of all values divided by the total number of values. It is the most widely used measure of central tendency.
2. **Median** : The middle value in an ordered dataset. If the number of observations is even, the median is the average of the two middle values.
3. **Mode** : The most frequently occurring value in a dataset. A dataset may have one mode (unimodal), two modes (bimodal), or more (multimodal).
4. **Geometric Mean** : The geometric mean is used when dealing with growth rates, percentages, or proportional changes over time. It provides a better measure than the arithmetic mean when values have large variations. Multiply all values and take the n^{th} root of the product (where n is the number of values).
5. **Harmonic Mean** : The harmonic mean is best used when averaging values that are rates, speeds, or ratios. It gives a better estimate than the arithmetic mean when numbers vary significantly. It is calculated by dividing the total number of values by the sum of their reciprocals.

2.2 Arithmetic Mean

Before discussing the Arithmetic Mean, it is important to introduce some basic notations. Suppose there are n observations, represented as $x_1, x_2, \dots, x_n$. The sum of these observations is written in compact form using the summation symbol (Σ , called sigma) as follows:

$$\Sigma x_i = x_1 + x_2 + x_3 + \dots + x_n$$

Here, i is a positive integer representing the position of each observation in the dataset. Since there are n observations, i ranges from 1 to n .

The **Arithmetic Mean** is calculated by dividing the sum of all observations by the total number of observations. It is one of the simplest and most commonly used measures of central tendency.

Arithmetic Mean can be Computed in two ways:

1. **Simple Arithmetic Mean** : All observations are treated equally, and each value contributes equally to the final average.

2. **Weighted Arithmetic Mean** : This method is used when some observations are more important than others. It takes into account the weight of each observation. In general, the weighted arithmetic mean is calculated as follows:

$$\bar{x} = \frac{\sum w_i x_i}{\sum w_i}$$

where:

- $\bar{x}$ = Arithmetic Mean
- $\sum w_i x_i$ = Sum of weighted observations
- $\sum w_i$ = Total weight

This formula is used to calculate the weighted arithmetic mean of a dataset.



Property

1. **Sum of D** : The sum of all observations is equal to the sum of the weighted observations.

$$\Sigma x_i = \Sigma w_i \bar{x}$$

Example:

Their sum is 100.

2. **Sensitive** : The arithmetic mean is sensitive to extreme values. A single outlier can significantly change the mean.

Example:

The arithmetic mean will change.

Example:

change.

3. **Affected** : The mean is affected by extreme values. A single outlier can significantly change the mean.

The mean calculation is sensitive to extreme values.

Example:

extreme values.

4. **Algebraic** : If there are n observations, the arithmetic mean can be calculated without using the summation symbol.

If there are n observations, the arithmetic mean can be calculated without using the summation symbol.

without using the summation symbol.

Example:

combined.

Example:

combined.

2. **Weighted Arithmetic Mean:** The Weighted Arithmetic Mean is used when different observations have different levels of importance. Instead of giving equal importance to all values (as in the simple arithmetic mean), each value is assigned a weight based on its significance. Values with higher weights have a greater impact on the final mean.

In general, if $x_1, x_2, \dots, x_n$ are given 'n' observations, then their arithmetic mean is denoted by:

$$\bar{x} = \frac{\text{Sum of the observations}}{\text{Total No. of observations}}$$

where:

- $\bar{x}$ = Arithmetic Mean
- Sum of all observations ($x_1 + x_2 + \dots + x_n$) = $\sum x_i$
- n = Total number of observations

This formula provides a simple way to calculate the average value of a dataset by dividing the total sum of observations by the number of values.



Properties of Arithmetic Mean

1. Sum of Deviations from Mean is Zero

The sum of the deviations of all observations from the arithmetic mean is always zero.

$$\sum (x_i - \bar{x}) = 0$$

Example: If the marks are 10, 20, and 30, the mean is 20. The deviations are -10, 0, and +10. Their sum is 0.

2. Sensitive to Every Value in the Dataset

The arithmetic mean takes into account every observation in the dataset. Any change in a single value will change the mean.

Example: If the average height of 4 students is 150 cm, and one height changes, the mean will also change.

3. Affected by Extreme Values

The mean is influenced by very high or very low values (outliers) because all values are included in the calculation.

Example: Incomes ₹20,000, ₹22,000, ₹25,000, and ₹1,00,000 have a much higher mean due to the extreme ₹1,00,000 value.

4. Algebraic Property of Combined Mean

If there are two or more groups with their means and sizes, the combined mean can be calculated without using the raw data:

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2 + \dots}{n_1 + n_2 + \dots}$$

Example: If the mean marks of two classes (30 students, mean = 60) and (20 students, mean = 70) are combined, the overall mean can be found directly.

5. Not Affected by Change of Origin and Scale

- If the same number is added to or subtracted from all values, the mean changes by the same number.
- If all values are multiplied or divided by the same number, the mean is multiplied or divided by that number.

Example: Adding 5 to each score increases the mean by 5.

6. Used for Further Statistical Analysis

The arithmetic mean is often used in other statistical measures like variance, standard deviation, and coefficient of variation.

7. Applicable to Quantitative Data Only

The mean can only be calculated for numerical data, not for qualitative data such as colours, opinions, or categories.

8. Unique Value for a Given Dataset

For a given set of numerical data, there is only one arithmetic mean. It is a definite and fixed value.



Merits of Arithmetic Mean

- Arithmetic mean is easy to calculate and understand.
- It considers all values in the dataset.
- It is defined by a fixed mathematical formula.
- It can be used in further statistical calculations.
- It remains stable across different samples.
- It provides a fair and balanced measure when data is evenly distributed.
- It is widely used in various fields such as economics, business, and science.



Demerits of Arithmetic Mean

- It is affected by very high or very low values.
- Arithmetic mean cannot be computed when class intervals have open ends. If the dataset has open-ended groups (like "above 50"), assumptions are needed to compute the mean.
- It does not work for qualitative data such as opinions or preferences.
- It cannot be determined by observation or graphs.
- It is not useful in destructive testing, where samples are destroyed (Example, testing product durability).
- It may not always represent the most typical value in a highly skewed dataset.
- It is not suitable for data with extreme fluctuations, as it can give misleading results.

	Ungrouped Data Raw data/Individual Data	Grouped Data Discrete Data/ Continuous Data
Direct Method	$\bar{x} = \frac{\sum x}{n}$	$\bar{x} = \frac{\sum fx}{N}$
Shortcut Method	$\bar{x} = A + \frac{\sum d}{n}$	$\bar{x} = A + \frac{\sum fd}{N}$
Step-Deviation Method	$\bar{x} = A + \left[\frac{\sum d'}{n} \right] \times i$	$\bar{x} = A + \left[\frac{\sum fd'}{N} \right] \times i$

$\bar{x}$ = Arithmetic Mean,

n = Total no. of observations

A = Assumed Mean

$d' = \frac{x - A}{i}$ [Simplified deviation from assumed mean]

i = The difference between the upper and lower boundaries of a class.

$\sum x$ = Total observations,

$N = \sum f$ = Total frequency,

$d = x - A$ [deviation from assumed mean]

Individual Series :

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = \frac{\sum x}{n}$$

Discrete Series or Continuous Series:

$$\bar{x} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{f_1 + f_2 + \dots + f_n}$$

$$= \frac{\sum fx}{\sum f} \text{ or } \frac{\sum fx}{N}$$

Problems on Mean for Raw Data /Individual Series (Direct Method)

Illustration 1

The wages of 9 workers in rupees are as follows:

155, 85, 130, 70, 90, 140, 100, 90, 130.

Find the arithmetic mean of the wages.

Solution

Direct Method : $\sum x = 155 + 85 + 130 + 70 + 90 + 140 + 100 + 90 + 130 = 990$

$$\bar{x} = \frac{\sum x}{n} = \frac{990}{9} = 110$$

$$\therefore \bar{x} = 110$$

Illustration 2

Calculate Arithmetic Mean from the following data.

- a. Marks obtained by 10 students 30, 61, 47, 25, 50, 39, 56, 66, 12, 24
 b. Income of 7 families ₹ 500, 490, 720, 890, 435, 595, 570

Solution

a.

$$\bar{x} = \frac{\sum x}{n}$$

$$\bar{x} = \frac{30 + 61 + 47 + 25 + 50 + 39 + 56 + 66 + 12 + 24}{10} = \frac{410}{10} = 41$$

∴ **Mean Marks = 41.**

b.

$$\bar{x} = \frac{\sum x}{n}$$

$$\bar{x} = \frac{500 + 490 + 720 + 890 + 435 + 595 + 570}{7} = \frac{4200}{7} = 600$$

∴ **Mean Income = 600.**

Illustration 3

Calculate arithmetic mean by direct method, for the following data. 40, 55, 55, 78, 53.

Solution

$$\bar{x} = \frac{\sum x}{n} = \frac{40 + 55 + 55 + 78 + 53}{5} = \frac{281}{5} = 56.2$$

∴ **$\bar{x} = 56.2$**

Illustration 4

The monthly income of 10 persons are as follows 1400, 1500, 1560, 1440, 1475, 1490, 1520, 1470, 1485 Find the average monthly income.

Solution

$$\bar{x} = \frac{\sum x}{n}$$

$$= \frac{1400 + 1500 + 1560 + 1440 + 1475 + 1490 + 1575 + 1520 + 1470 + 1485}{10}$$

$$= \frac{14915}{10} = 1491.5$$

∴ **Average Monthly Income = 1491.5**

Illustra

If sum of

Solution

Given:

∴

Illustra

If mean o

Solution

Given :

∴ Sum

Illustra

Marks ob
step devi

Solution

Calculati

Marks

30

41

47

54

23

34

37

51

53

47

 $\sum x = 41$

Note: A =

Illustration 5

If sum of 15 observations is 600. Find its Mean.

Solution

Given: $n = 15, \Sigma x = 600$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{600}{15} = 40$$

Illustration 6

If mean of 50 observations is 72. Find the sum of observations.

Solution

Given: $n = 50$
Mean = 72

$$\bar{x} = \frac{\Sigma x}{n} \Rightarrow 72 = \frac{\Sigma x}{50} \Rightarrow \Sigma x = 72 \times 50 = 3600$$

Sum of observations = 3,600

Illustration 7

Marks obtained by 10 students are given, calculate arithmetic mean using direct, shortcut & step deviation method. 30, 41, 47, 54, 23, 34, 37, 51, 53, 47.

Solution

Calculation of arithmetic mean using Direct, Shortcut and Step Deviation Method.

Marks (x)	Let A = 34 $d = x - A$	$d' = d/i$ [Let $i = 2$]	Direct Method
30	-4	-2	$\bar{x} = \frac{\Sigma x}{n} = \frac{417}{10} = 41.7$
41	7	3.5	$\bar{x} = 41.7$
47	13	6.5	
54	20	10	
23	-11	-5.5	
34	0	0	
37	3	1.5	
51	17	8.5	
53	19	9.5	
47	13	6.5	
$\Sigma x = 417$	$\Sigma d = 77$	$\Sigma d' = 38.5$	
			Shortcut Method
			$\bar{x} = A + \frac{\Sigma d}{n} = 34 + \frac{77}{10} = 34 + 7.7 = 41.7$
			$\bar{x} = 41.7$
			Step Deviation Method
			$\bar{x} = A + \left[\frac{\Sigma d'}{n} \right] \times i = 34 + \left[\frac{38.5}{10} \right] \times 2$
			$= 34 + [3.85] 2 = 41.7$
			$\bar{x} = 41.7$

Note: A = assumed mean, we can assume any number from the given data.

Problems on Arithmetic Mean for Discrete Series

Illustration 1

Calculate the arithmetic mean from the following data.

x	5	10	15	20	25	30	35	40
f	5	7	9	10	8	6	5	2

Solution

Calculation of Arithmetic Mean

x	f	fx
5	5	25
10	7	70
15	9	135
20	10	200
25	8	200
30	6	180
35	5	175
40	2	80
	N = 52	Σfx = 1065

$$\bar{x} = \frac{\Sigma fx}{N}$$

$$\bar{x} = \frac{1,065}{52} = 20.48$$

$$\therefore \bar{x} = 20.48$$

Illustration 2

Compute arithmetic mean for the following data

x	2	3	4	5	6	7	8	9	10	11
f	3	8	10	12	16	14	10	8	7	5

Solution

Calculation of Arithmetic Mean

x	f	fx
2	3	6
3	8	24
4	10	40
5	12	60
6	16	96
7	14	98
8	10	80
9	8	72
10	7	70
11	5	55
	N = 93	Σfx = 601

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{601}{93} = 6.46$$

$$\therefore \bar{x} = 6.46$$

Illustration

Calculate the

Marks obtained

No. of Students

Solution

Marks obtained

4

8

12

16

20

Direction

Find the

Find the

Illustration

A total of 1 Fund as fol

• 15 st

• 12 st

• 28 st

• 20 st

• 15 st

• 10 st

Find the ave

To find the

contributed

the total num

Illustration 3

Calculate mean marks for the following:

Marks Obtained	4	8	12	16	20
No. of Students	6	12	18	15	9

Solution**Calculation of Arithmetic Mean**

Marks obtained (x)	No of students (f)	fx	Let A = 12 d = x - A	fd	(Let i = 4) d' = d/i	fd'
4	6	24	-8	-48	-2	-12
8	12	96	-4	-48	-1	-12
12	18	216	0	0	0	0
16	15	240	4	60	1	15
20	9	180	8	72	2	18
	N = 60	$\Sigma fx = 756$		$\Sigma fd = 36$		$\Sigma fd' = 9$

Direct Method	Shortcut Method	Step Deviation Method
$\bar{x} = \frac{\Sigma fx}{N} = \frac{756}{60} = 12.6$ $\therefore \bar{x} = 12.6$	$\bar{x} = A + \frac{\Sigma fd}{N} = 12 + \frac{36}{60}$ $= 12 + 0.6 = 12.6$ $\therefore \bar{x} = 12.6$	$\bar{x} = A + \left[\frac{\Sigma fd'}{N} \right] \times i$ $= 12 + \left[\frac{9}{60} \right] \times 4$ $= 12 + [0.15]4 = 12 + 0.6 = 12.6$ $\therefore \bar{x} = 12.6$

Illustration 4

A total of 100 students from Bangalore University contributed to the Prime Minister's Relief Fund as follows:

- 15 students contributed ₹ 3,000 each
- 12 students contributed ₹ 4,000 each
- 28 students contributed ₹ 5,000 each
- 20 students contributed ₹ 6,000 each
- 15 students contributed ₹ 7,000 each
- 10 students contributed ₹ 8,000 each

Find the average contribution made by each student.

To find the average contribution per student, multiply the number of students by the amount they contributed in each group. Then add all the total contributions together. Divide this total amount by the total number of students. This will give the average contribution per student.

Solution

Calculation of Arithmetic Mean

Contribution (x)	No. of students (f)	fx	d = x - A Let A = 5000	fd	d' = d/i Let i = 1000	fd'
3000	15	45000	-2,000	-30000	-2	-30
4000	12	48000	-1,000	-12000	-1	-12
5000	28	140000	0	0	0	0
6000	20	120000	1000	20000	1	20
7000	15	105000	2000	30000	2	30
8000	10	80000	3000	30000	3	30
	N = 100	$\Sigma fx = 538000$		$\Sigma fd = 38000$		$\Sigma fd' = 38$

Direct Method	Shortcut Method	Step Deviation Method
$\bar{x} = \frac{\Sigma fx}{N} = \frac{538000}{100}$ $= 5380$ $\therefore \bar{x} = 5380$	$\bar{x} = A + \frac{\Sigma fd}{N} = 5000 + \frac{38000}{100}$ $= 5000 + 380$ $= 5380$ $\therefore \bar{x} = 5380$	$\bar{x} = A + \left[\frac{\Sigma fd'}{N} \right] \times i$ $= 5000 + \left[\frac{38}{100} \right] \times 1000$ $= 5000 + (0.38) 1000$ $= 5000 + 380$ $= 5380$ $\therefore \bar{x} = 5380$

Problems on Arithmetic Mean for Continuous Series

Illustration 1

From the following information calculate Arithmetic Mean.

Income (₹)	5000-6000	6000-7000	7000-8000	8000-9000	9000-10000
No. of people	7	11	18	26	38

Solution

This is a continuous series. In a continuous series, data is grouped into class intervals without gaps. Each class interval follows the other in a sequence. To calculate the arithmetic mean for this data, first find the mid-point of each class. Then multiply each mid-point by the frequency of the class. Add all these values and divide the total by the sum of frequencies.

Class Interval

5000-6000
6000-7000
7000-8000
8000-9000
9000-10000

Illustration

Calculate average

Marks
No of Students

Solution

Marks (C.I.)

0-10
10-20
20-30
30-40
40-50
50-60
60-70

Illustration

Calculate Arithmetic Mean

Income
No of Persons

Solution

Income (C.I.)	No of Persons
10-20	
20-30	

Calculation of Arithmetic Mean

Class Interval	Frequency (f)	Mid value (x)	fx
5000-6000	7	$(5000+6000)/2 = 5500$	38500
6000-7000	11	$(6000+7000)/2 = 6500$	71500
7000-8000	18	$(7000+8000)/2 = 7500$	135000
8000-9000	26	$(8000+9000)/2 = 8500$	221000
9000-10000	38	$(9000+10000)/2 = 9500$	361000
	N = 100		$\Sigma fx = 827000$

$$\begin{aligned}\bar{x} &= \frac{\Sigma fx}{N} \\ &= \frac{827000}{100} = 8270 \\ \therefore \bar{x} &= 8270\end{aligned}$$

Illustration 2

Calculate average marks of the following students.

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70
No of Students (f)	5	12	15	25	11	6	2

Solution

Marks (C.I.)	No of Students (f)	Mid value (x)	fx
0-10	5	5	25
10-20	12	15	180
20-30	15	25	375
30-40	25	35	875
40-50	11	45	495
50-60	6	55	330
60-70	2	65	130
	N = 76		$\Sigma fx = 2410$

$$\begin{aligned}\bar{x} &= \frac{\Sigma fx}{N} \\ &= \frac{2410}{76} = 31.71 \\ \therefore \bar{x} &= 31.71\end{aligned}$$

Illustration 3

Calculate Arithmetic Mean of Income from the following

Income	10-20	20-30	30-40	40-50	50-60	60-70
No of Persons	4	7	16	20	15	8

Solution

Calculation of Arithmetic Mean of Income

Income (C.I.)	No of Persons (f)	Mid point (x)	fx	(let A = 35) d = x - A	fd	(Let i = 10) d' = d/i	fd'
10-20	4	15	60	-20	-80	-2	-8
20-30	7	25	175	-10	-70	-1	-7

30-40	16	35	560	0	0	0
40-50	20	45	900	10	200	1
50-60	15	55	825	20	300	2
60-70	8	65	520	30	240	3
	N = 70		Σfx = 3040		Σfd = 590	

Direct Method	Shortcut Method	Step Deviation Method
$\bar{x} = \frac{\Sigma fx}{N} = \frac{3040}{70} = 43.42$ $\therefore \bar{x} = 43.42$	$\bar{x} = A + \frac{\Sigma fd}{N}$ $= 35 + \frac{590}{70}$ $= 35 + 8.42 = 43.42$ $\therefore \bar{x} = 43.42$	$\bar{x} = A + \left[\frac{\Sigma fd'}{N} \right] \times i$ $= 35 + \left[\frac{59}{70} \right] \times 10$ $= 35 + (0.842) \times 10$ $= 35 + 8.42 = 43.42$ $\therefore \bar{x} = 43.42$

Note: Any values of x can be considered for Assumed mean

Illustration 4

Calculate Arithmetic Mean for the following

Class Interval	4 - 10	10 - 16	16 - 22	22 - 28	28 - 34	34 - 40	40 - 46
Frequency	2	8	19	35	18	14	4

Solution

Calculation of Arithmetic Mean

CI	f	Midpoint (x)	fx	Let A = 25 d = x - A	fd	Let i = 6 d' = d/i	fd'
4 - 10	2	7	14	-18	-36	-3	-6
10 - 16	8	13	104	-12	-96	-2	-16
16 - 22	19	19	361	-6	-114	-1	-19
22 - 28	35	25	875	0	0	0	0
28 - 34	18	31	558	6	108	1	18
34 - 40	14	37	518	12	168	2	28
40 - 46	4	43	172	18	72	3	12
	N = 100		Σfx = 2602		Σfd = 102		Σfd' = 10

Direct Method	Shortcut Method	Step Deviation Method
$\bar{x} = \frac{\sum fx}{N} = \frac{2602}{100}$ $\bar{x} = 26.02$	$\bar{x} = A + \frac{\sum fd}{N}$ $= 25 + \frac{102}{100} = 25 + 1.02$ $\bar{x} = 26.02$	$\bar{x} = A + \left[\frac{\sum fd'}{N} \right] \times i$ $= 25 + \left[\frac{17}{100} \right] \times 6$ $= 25 + (0.17)6 = 25 + 1.02$ $\bar{x} = 26.02$

Properties of Mean

1. The algebraic sum of deviations of a given set of observation taken from their mean is zero.

$$\text{ie., } \sum (x - \bar{x}) = 0$$

2. The sum of squares of deviations from their arithmetic mean is least $\Rightarrow \sum (x - \bar{x})^2$ is least.

3. Combined arithmetic mean;

If N_1 and N_2 are sizes of two series and their means are $\bar{x}_1$ and $\bar{x}_2$ respectively, then the combined mean can be obtained by,

$$\bar{x}_{12} = \frac{\bar{x}_1 N_1 + \bar{x}_2 N_2}{N_1 + N_2}$$

Problems on Less than Cumulative Frequency Distribution**Illustration 1**

For the following data calculate mean marks

Marks	Below 10	Below 20	Below 30	Below 40	Below 50	Below 60	Below 70	Below 80	Below 90	Below 100
No. of Students	15	35	60	84	96	127	198	250	275	300

Solution

Calculation of mean marks by converting the data into simple frequency distribution

Marks	No. of Students	Exclusive Class Interval (ECI)	Frequency (f)	Mid Point (x)	fx
Below 10	15	0-10	15	5	75
Below 20	35	10-20	(35-15)=20	15	300
Below 30	60	20-30	(60-35)=25	25	625
Below 40	84	30-40	(84-60)=24	35	840
Below 50	96	40-50	(96-84)=12	45	540

Below 60	127	50-60	(127-96)=31	55	1705
Below 70	198	60-70	(198-127)=71	65	4615
Below 80	250	70-80	(250-198)=52	75	3900
Below 90	275	80-90	(275-250)=25	85	2125
Below 100	300	90-100	(300-275)=25	95	2375
			N = 300		$\Sigma fx = 17100$

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{17100}{300} = 57$$

$$\bar{x} = 57$$

Illustration 2

Calculate mean marks for the following data.

Marks	< 10	< 20	< 30	< 40	< 50	< 60	< 70	< 80
No. of Students	25	40	60	75	95	125	190	240

Solution

Calculation of Arithmetic Mean

Marks	No. of Students	Exclusive Class Interval (ECI)	Frequency (f)	Mid Point (x)	fx
Less than 10	25	0-10	25	5	125
Less than 20	40	10-20	(40 - 25) = 15	15	225
Less than 30	60	20-30	(60 - 40) = 20	25	500
Less than 40	75	30-40	(75 - 60) = 15	35	525
Less than 50	95	40-50	(95 - 75) = 20	45	900
Less than 60	125	50-60	(125 - 95) = 30	55	1650
Less than 70	190	60-70	(190 - 125) = 65	65	4225
Less than 80	240	70-80	(240 - 190) = 50	75	3750
			N = 240		$\Sigma fx = 11900$

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{11900}{240} = 49.58$$

$$\bar{x} = 49.58$$

Problems on

Illustration 1

Find Arithmetic Mean

Marks	No. of Students
-------	-----------------

Solution

Marks

More than 10
More than 20
More than 30
More than 40
More than 50
More than 60

Illustration 2

Calculate mean

Wages (₹)

No. of Workers

Solution

Wages (₹)

Above 0
Above 10
Above 20
Above 30
Above 40

Problems on More than Cumulative Frequency Distribution

Illustration 1

Find Arithmetic Mean marks from the following

Marks	> 10	> 20	> 30	> 40	> 50	> 60
No. of Students	50	46	40	20	10	3

Solution

Calculation of Arithmetic Mean Marks under Direct Method

Marks	No. of Students	Exclusive Class Interval (ECI)	Frequency (f)	Mid Point (x)	fx
More than 10	50	10-20	(50-46)=4	15	60
More than 20	46	20-30	(46-40)=6	25	150
More than 30	40	30-40	(40-20)=20	35	700
More than 40	20	40-50	(20-10)=10	45	450
More than 50	10	50-60	(10-3)=7	55	385
More than 60	3	60-70	3	65	195
			N = 50		$\Sigma fx = 1940$

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{1940}{50} = 38.8$$

$$\bar{x} = 38.8$$

Illustration 2

Calculate mean wages for the following

Wages (₹)	Above 0	Above 10	Above 20	Above 30	Above 40	Above 50	Above 60	Above 70
No. of Workers	650	500	425	375	300	275	250	100

Solution

Calculation of Mean Wages

Wages (₹)	No. of Workers	Exclusive Class Interval (ECI)	Frequency (f)	Mid value (x)	fx
Above 0	650	0-10	(650 - 500) = 150	5	750
Above 10	500	10-20	(500 - 425) = 75	15	1125
Above 20	425	20-30	(425 - 375) = 50	25	1250
Above 30	375	30-40	(375 - 300) = 75	35	2625
Above 40	300	40-50	(300 - 275) = 25	45	1125

Above 50	275	50-60	$(275 - 250) = 25$	55	1375
Above 60	250	60-70	$(250 - 100) = 150$	65	9750
Above 70	100	70-80	100	75	7500
			$N = 650$		$\Sigma fx = 25500$

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{25500}{650} = 39.23$$

$$\bar{x} = 39.23$$

Problems on Arithmetic Mean for Open End Class Interval Frequency Distribution

Illustration 1

Calculate Mean under direct, shortcut & step deviation method for open-end class interval.

Marks	Below 10	10 - 30	30 - 60	60 - 100	100 - 150	Above 150
No. of Students	4	6	8	10	4	8

Solution

Calculation of Arithmetic Mean under direct, shortcut & step deviation method

Marks (C.I)	No. of Students (f)	ECI	Mid Point (x)	fx	d = x - A (A = 80)	fd	d' = $\frac{d}{i}$ (i = 2)	fd'
Below 10	4	0 - 10	5	20	-75	-300	-37.5	-150
10 - 30	6	10 - 30	20	120	-60	-360	-30	-180
30 - 60	8	30 - 60	45	360	-35	-280	-17.5	-140
60 - 100	10	60 - 100	80	800	0	0	0	0
100 - 150	4	100 - 150	125	500	45	180	22.5	90
Above 150	8	150 - 210	180	1440	100	800	50	400
	$N = 40$			$\Sigma fx = 3240$		$\Sigma fd = 40$		$\Sigma fd' = 20$

Direct Method	Shortcut Method	Step Deviation Method
$\bar{x} = \frac{\Sigma fx}{N} = \frac{3240}{40} = 81$ $\bar{x} = 81$	$\bar{x} = A + \frac{\Sigma fd}{N}$ $= 80 + \frac{40}{40}$ $= 80 + 1 \Rightarrow \bar{x} = 81$	$\bar{x} = A + \left[\frac{\Sigma fd'}{N} \right] \times i$ $= 80 + \left[\frac{20}{40} \right] \times 2$ $= 80 + 1 \Rightarrow \bar{x} = 81$

Note : When the common differences are not equal, 'i' will be assumed as 2.

Illustration

Calculate mean

Monthly Income

No. of Families

Solution

ECI	
0 - 100	
100 - 200	
200 - 300	
300 - 400	
400 - 500	
500 - 600	
	N

Direct Method

$$\bar{x} = \frac{\Sigma fx}{N}$$

$$= 209$$

$$\bar{x} = 209$$

Problem

Illustration

Find out missing

Height (in)

No. of Students

Solution

Let the 2 missing

Convert the given

Illustration 2

Calculate mean monthly income for following

Monthly Income	Below 100	100 - 200	200 - 300	300 - 400	400 - 500	500 & Above
No. of Families	50	500	555	100	3	2

Solution**Calculation of Mean Monthly Income**

ECI	f	Mid point(x)	fx	A = 150 d = x - A	fd	d' = d/i (i = 100)	fd'
0 - 100	50	50	2500	-100	-5000	-1	-50
100 - 200	500	150	75000	0	0	0	0
200 - 300	555	250	138750	100	55500	1	555
300 - 400	100	350	35000	200	20000	2	200
400 - 500	3	450	1350	300	900	3	9
500 - 600	2	550	1100	400	800	4	8
	N = 1210		$\Sigma fx = 253700$		$\Sigma fd = 72200$		$\Sigma fd' = 722$

Direct Method	Shortcut Method	Step Deviation Method
$\bar{x} = \frac{\Sigma fx}{N} = \frac{253700}{1210}$ $= 209.66$ $\therefore \bar{x} = 209.66$	$\bar{x} = A + \frac{\Sigma fd}{N}$ $= 150 + \frac{72200}{1210}$ $= 150 + 59.66 = 209.66$ $\therefore \bar{x} = 209.66$	$\bar{x} = A + \left[\frac{\Sigma fd'}{N} \right] \times i$ $= 150 + \left[\frac{722}{1210} \right] \times 100$ $= 150 + 59.66 = 209.66$ $\therefore \bar{x} = 209.66$

Problems on Missing Frequency**Illustration 1**

Find out missing frequency for the following data if the Arithmetic Mean is 67.45 and N = 100

Height (inch)	60 - 62	63 - 65	66 - 68	69 - 71	72 - 74
No. of Students	5	18	?	?	8

Solution

Let the 2 missing frequencies be 'a' & 'b'.

Convert the given data into exclusive class interval.

Calculation of missing frequencies

Height in inches (C.I)	No. of Students (f)	ECI	Mid Point (x)	fx
60 - 62	5	59.5 - 62.5	61	
63 - 65	18	62.5 - 65.5	64	305
66 - 68	a	65.5 - 68.5	67	1152
69 - 71	b	68.5 - 71.5	70	67a
72 - 74	8	71.5 - 74.5	73	70b
	$N = 31 + a + b$			584
				$\Sigma fx = 2041 + 67a + 70b$

$$N = 31 + a + b$$

$$100 = 31 + a + b$$

$$100 - 31 = a + b$$

$$69 = a + b$$

$$b = 69 - a$$

$$\bar{x} = \frac{\Sigma fx}{N}$$

$$67.45 = \frac{2041 + 67a + 70b}{100}$$

$$67.45 \times 100 = 2041 + 67a + 70b$$

$$6745 - 2041 = 67a + 70b$$

$$4704 = 67a + 70b$$

$$4704 = 67a + 70[69 - a] \text{ (substituting } b=69-a)$$

$$4704 = 67a + 4830 - 70a$$

$$4704 - 4830 = 67a - 70a$$

$$-126 = -3a$$

$$a = \frac{-126}{-3} = 42$$

Now, let's find the value of b.

We know that $b = 69 - a$

$$b = 69 - 42 \text{ (substituting } a = 42)$$

$$b = 27$$

Illustration 2

Calculate the missing frequency if the mean for the below distribution is 28

Marks	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60
No. of Students	12	18	27	?	17	6

Solution

Let missing frequency be 'a'

Marks	No. of Students (f)	Mid (x)	fx
0 - 10	12	5	60
10 - 20	18	15	270
20 - 30	27	25	675
30 - 40	a	35	35a
40 - 50	17	45	765
50 - 60	6	55	330
	$N = 80 + a$		$\Sigma fx = 2100 + 35a$

$$\text{Mean } (\bar{x}) = 28$$

$$\frac{\Sigma fx}{N} = 28$$

$$\frac{2100 + 35a}{80 + a} = 28$$

$$2100 + 35a = 28(80 + a)$$

$$2100 + 35a = 2240 + 28a$$

$$35a - 28a = 2240 - 2100$$

$$7a = 140$$

$$a = \frac{140}{7} = 20$$

Illustration

Find out the

Marks

No. of Students

Solution

Let the missing

Marks	freq
5 - 10	
10 - 15	
15 - 20	
20 - 25	
25 - 30	
	N =

Illustration

From the follow

Mid Value (x)
Frequency

Solution

Let the missing

Calcu

Mid Value (x)
5
15
25
35
45
55

Illustration 3

Find out the missing frequency if the mean = 19

Marks	5 - 10	10 - 15	15 - 20	20 - 25	25 - 30
No. of Students	2	2	?	4	4

Solution

Let the missing frequency be 'a'

Calculation of missing frequencies

Marks	frequency (f)	Mid Value (x)	fx
5 - 10	2	7.5	15
10 - 15	2	12.5	25
15 - 20	a	17.5	17.5a
20 - 25	4	22.5	90
25 - 30	4	27.5	110
	$N = 12 + a$		$\Sigma fx = 240 + 17.5a$

Given :

Mean ($\bar{x}$) = 19

$$\bar{x} = \frac{\Sigma fx}{N}$$

$$= \frac{240 + 17.5a}{12 + a} = 19$$

$$= 240 + 17.5a = 19(12 + a)$$

$$= 240 + 17.5a = 228 + 19a$$

$$= 240 - 228 = 19a - 17.5a$$

$$12 = 1.5a \Rightarrow a = \frac{12}{1.5} = 8$$

Illustration 4

From the following information find missing frequency if mean = 33

Mid Value (x)	5	15	25	35	45	55
Frequency	5	10	25	30	?	10

Solution

Let the missing frequency be 'a'

Calculation of missing frequencies

Mid Value (x)	Frequency (f)	fx
5	5	25
15	10	150
25	25	625
35	30	1050
45	a	45a
55	10	550
	$N = 80 + a$	$\Sigma fx = 2400 + 45a$

Given :

Mean ($\bar{x}$) = 33

$$\bar{x} = \frac{\Sigma fx}{N} \Rightarrow \frac{2400 + 45a}{80 + a} = 33$$

$$2400 + 45a = 33(80 + a)$$

$$2400 + 45a = 2640 + 33a$$

$$45a - 33a = 2640 - 2400$$

$$12a = 240$$

$$a = \frac{240}{12} \Rightarrow a = 20$$

Illustration 5

From the following information find out missing frequencies if mean = 25 and $N = 45$.

Class Interval	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
Frequency	5	?	15	?	5

Solution

Let the 2 missing frequencies be 'a' & 'b'

Class Interval	Frequency (f)	Mid Value (x)	fx
0 - 10	5	5	25
10 - 20	a	15	15a
20 - 30	15	25	375
30 - 40	b	35	35b
40 - 50	5	45	225
	$N = 25 + a + b$		$\Sigma fx = 625 + 15a + 35b$

$$N = 25 + a + b$$

$$45 = 25 + a + b$$

$$45 - 25 = a + b$$

$$20 = a + b$$

$$20 - a = b$$

$$b = 20 - a$$

$$\bar{x} = \frac{\Sigma fx}{N}$$

$$25 = \frac{625 + 15a + 35b}{45}$$

$$1125 = 625 + 15a + 35b$$

$$1125 - 625 = 15a + 35b$$

$$500 = 15a + 35b$$

$$500 = 15a + 35[20 - a] \text{ (substituting } b=20-a)$$

$$500 = 15a + 700 - 35a$$

$$500 - 700 = 15a - 35a$$

$$-200 = -20a$$

$$a = \frac{-200}{-20} = 10$$

$$a = 10$$

Now, let's find the value of b.

We know that $b = 20 - a$

$$a = 10$$

$$b = 20 - 10$$

$$b = 10$$

Combined Mean

The arithmetic means of several sets of data may be combined into a single arithmetic mean for combined sets of data. For two sets of data, the combined arithmetic mean may be defined as

$$\bar{x}_{12} = \frac{\bar{x}_1 N_1 + \bar{x}_2 N_2}{N_1 + N_2}$$

Where

 $\bar{x}_{12}$ = combined mean of two sets of data N_1 and N_2 $\bar{x}_1$ = arithmetic mean of the first set of data $\bar{x}_2$ = arithmetic mean of the second set of data N_1 = number of observations in the first set of data. N_2 = number of observations in the second set of data.

If we have to combine three or more sets of data, then the same formula can be generalized as

$$\bar{x}_{123} = \frac{\bar{x}_1 N_1 + \bar{x}_2 N_2 + \bar{x}_3 N_3 + \dots}{N_1 + N_2 + N_3 + \dots}$$

Problems on Combined Mean

Illustration 1

The mean of marks scored by 30 girls of a class is 44%. The mean for 20 boys of the same class is 42%. Find the mean for the whole class.

Solution

$$N_1 = 30, \quad N_2 = 20, \quad \bar{x}_1 = 44\%, \quad \bar{x}_2 = 42\%.$$

$$\bar{x}_a = \frac{\bar{x}_1 N_1 + \bar{x}_2 N_2}{N_1 + N_2}$$

$$= \frac{(44)(30) + (42)(20)}{30 + 20} = \frac{1320 + 840}{50} = \frac{2160}{50} = 43.2\%$$

$\therefore$ The combined mean for the whole class is 43.2%

Illustration 2

Find combined Mean if

$$\bar{x}_1 = 25, \bar{x}_2 = 40, N_1 = 20, N_2 = 10$$

Solution

$$\bar{x}_{12} = \frac{\bar{x}_1 N_1 + \bar{x}_2 N_2}{N_1 + N_2}$$

$$= \frac{25(20) + 40(10)}{20 + 10} = \frac{500 + 400}{30} = \frac{900}{30} = 30\%$$

$\therefore$ The combined mean for the whole class is 30%

Illustration 3

In a class of 50 students, 10 have failed & their average marks is 2.5. The total marks secured by entire class was 281. Find the average marks of those who have passed.

Solution

Total No. of Students = 50

Total Marks secured by 50 Students = 281

$$\therefore \text{Average marks of 50 students} = \bar{x}_{12} = \frac{281}{50} = 5.62$$

Given:

$$(\text{Failed}) \bar{x}_1 = 2.5$$

$$(\text{Pass}) \bar{x}_2 = ?$$

$$N_1 = 10$$

$$N_2 = 50 - 10 = 40$$

$$\bar{x}_{12} = 5.62$$

$$\bar{x}_{12} = \frac{\bar{x}_1 N_1 + \bar{x}_2 N_2}{N_1 + N_2} \Rightarrow 5.62 = \frac{(2.5)(10) + \bar{x}_2(40)}{10 + 40} \Rightarrow 5.62 = \frac{25 + 40\bar{x}_2}{50}$$

$$(5.62)(50) = 25 + 40\bar{x}_2 \Rightarrow 281 = 25 + 40\bar{x}_2$$

$$281 - 25 = 40\bar{x}_2 \Rightarrow 256 = 40\bar{x}_2$$

$$\frac{256}{40} = \bar{x}_2 \Rightarrow \bar{x}_2 = 6.4$$

The average marks of those who have passed is 6.4

Illustration 4

The mean weight of 100 students in a class is 50 kgs. Mean weight of boys in class is 55 kgs that of girls is 45 kgs. Find the no. of boys & girls in the class.

Solution

Given:

$$(\text{Boys}) \bar{x}_1 = 55$$

$$N_1 = ?$$

$$\bar{x}_{12} = 50$$

$$(\text{Girls}) \bar{x}_2 = 45$$

$$N_2 = ?$$

$$N_1 + N_2 = 100$$

$$\bar{x}_{12} = \frac{\bar{x}_1 N_1 + \bar{x}_2 N_2}{N_1 + N_2}$$

$$50 = \frac{(55)N_1 + (45)N_2}{100}$$

$$(50)(100) = 55N_1 + 45N_2$$

$$5000 = 55N_1 + 45(100 - N_1)$$

$$5000 = 55N_1 + 4500 - 45N_1$$

$$5000 - 4500 = 55N_1 - 45N_1$$

$$500 = 10N_1$$

$$\frac{500}{10} = N_1$$

$$N_1 = 50$$

$$\therefore N_1 + N_2 = 100$$

$$N_2 = 100 - N_1$$

$$\text{Substitute } N_1 = 50$$

$$N_2 = 100 - 50$$

$$\therefore N_2 = 50$$

No. of boys = 50 and No. of girls = 50

Illustration 5

The mean weight of group of 6 students is 119 pounds. The individual weight of 5 of them are 113, 107, 131, 115 & 112 pounds. What is the weight of 6th student?

Solution

Given:

The individual weight of first 5 students are 113, 107, 131, 115, 112

∴ Average weight of first 5 students

$$= \frac{113+107+131+115+112}{5} = \frac{578}{5} = 115.6$$

Given:

$$\bar{x}_1 = 115.6 \quad N_1 = 5$$

$$\bar{x}_2 = ? \quad N_2 = 1$$

$$\bar{x}_{12} = 119$$

$$\bar{x}_{12} = \frac{\bar{x}_1 N_1 + \bar{x}_2 N_2}{N_1 + N_2}$$

$$119 = \frac{(115.6)(5) + \bar{x}_2(1)}{5+1}$$

$$(119)(6) = 578 + \bar{x}_2$$

$$714 = 578 + \bar{x}_2$$

$$\bar{x}_2 = 714 - 578$$

$$\bar{x}_2 = 136$$

∴ Weight of 6th student is 136 pounds.

Problems on Correct Mean from Incorrect Mean

$$\text{Correct Mean} = \frac{[\text{Wrong Mean} \times \text{No. of observations}] - [\text{Wrong Values}] + [\text{Correct Values}]}{\text{Total no. of observations}}$$

Illustration 1

The mean of 200 items was 50, later on it was discovered that 2 items were wrongly read as 90 & 10 instead of 192 & 88. Find out the correct mean.

Solution

Given:

Wrong mean = 50

No. of observations = 200

Wrong values = 90 & 10

Correct values = 192 & 88

$$\begin{aligned}
 \text{Correct Mean} &= \frac{[\text{Wrong Mean} \times \text{No. of observations}] - [\text{Wrong Values}] + [\text{Correct Values}]}{\text{Total no. of observations}} \\
 &= \frac{(50 \times 200) - (90 + 10) + (192 + 88)}{200} \\
 &= \frac{10000 - 100 + 280}{200} \\
 &= \frac{10180}{200}
 \end{aligned}$$

Correct Mean = 50.9

Illustration 2

The average income of a group of 50 persons working in a factory was calculated to be ₹ 169. It was later discovered that one figure was misread as 124, instead of correct value 143. Calculate the correct average income.

Solution

Given : Wrong mean = 169
 No. of observations = 50
 Wrong values = 124
 Correct value = 143

$$\begin{aligned}
 \text{Correct Mean} &= \frac{[\text{Wrong Mean} \times \text{No. of observations}] - [(\text{Wrong Values}) + (\text{Correct Values})]}{\text{Total no. of observations}} \\
 &= \frac{(169 \times 50) - 124 + 143}{50} \\
 &= \frac{8450 - 124 + 143}{50} \\
 &= \frac{8326 + 143}{50} = \frac{8469}{50}
 \end{aligned}$$

Correct Mean = 169.38

Illustration 3

An Arithmetic Mean of marks of 100 students is given as 45. But later on it was found that marks of students getting 64 was wrongly included as 46. Find out correct mean.

Solution

Given : Wrong mean = 45
 No. of observations = 100
 Wrong value = 46
 Correct value = 64

Corre

Correc

Weighted

In the calcu
 situations,
 result. In su
 importance
 less weight.

Example:

To calculate
 managers) h
 would show
 the wages o
 arithmetic m

Weighted ar
 and must be

Weighted A
 to each value
 by the total c

Formulae fo

Let $x_1, x_2, \dots$
 mean denote

Proble

Illustratio

The candidat
 English - 60, I
 Admission to
 different subj
 Calculate the

$$\text{Correct Mean} = \frac{[\text{Wrong Mean} \times \text{No. of observations}] - [(\text{Wrong Values}) + (\text{Correct Values})]}{\text{Total no. of observations}}$$

$$= \frac{(45 \times 100) - 46 + 64}{100} = \frac{4500 - 46 + 64}{100} = \frac{4518}{100}$$

$$\text{Correct Mean} = 45.18$$

Weighted Arithmetic Mean

In the calculation of simple arithmetic mean, equal importance is given to all the values. But in real situations, not all items carry the same importance. So, using a simple average may not give a correct result. In such cases, it becomes necessary to give different weights to different items based on their importance. Items that are more important are given more weight, and less important items are given less weight.

Example:

To calculate the average wage of workers in a factory, using a simple mean may not be right if a few people (like managers) have very high wages, while most workers earn lower wages. In such a case, the simple average would show a higher wage, which does not reflect the true average for the group. To get a realistic average, the wages of managers should be given less weight. The average calculated this way is called the weighted arithmetic mean.

Weighted arithmetic mean is helpful in many cases where the items have different levels of importance and must be treated differently in calculations.

Weighted Arithmetic Mean is the average of a set of values calculated by giving appropriate weight to each value according to its relative importance. To find this, the total of weighted values is divided by the total of weights.

Formulae for Weighted Arithmetic Mean (Direct Method)

Let $x_1, x_2, \dots, x_n$ be n values with their respective weights $w_1, w_2, \dots, w_n$. Their weighted arithmetic mean denoted as $\bar{x}_w$ is given by,

$$\bar{x}_w = \frac{\sum x_i w_i}{\sum w_i} \quad \text{or} \quad \bar{x}_w = \frac{\sum xw}{\sum w}$$

Problems on Weighted Arithmetic Mean

Illustration 1

The candidate obtained the following marks in PU exam is given below:

English - 60, Kannada - 50, Physics - 80, Chemistry - 70, Maths - 90, Biology - 70

Admission to the science institute is based on weighted mean of marks where weights for the different subjects are 1, 1, 3, 3, 3 & 2 respectively.

Calculate the weighted mean marks:

Solution

Calculation of Weighted Arithmetic Mean Marks

Subjects	Marks (x)	Weights (w)	xw
English	60	1	60
Kannada	50	1	50
Physics	80	3	240
Chemistry	70	3	210
Maths	90	3	270
Biology	70	2	140
		$\Sigma w = 13$	$\Sigma xw = 970$

$$\bar{x}_w = \frac{\Sigma xw}{\Sigma w} = \frac{970}{13} = 74.61$$

$$\therefore \bar{x}_w = 74.61$$

Illustration 2

Calculate the Weighted Arithmetic Mean of marks of candidates A & B

Subject	Kannada	English	Maths	Statistics
Weights	1	2	3	4
Candidate A	70	65	58	63
Candidate B	80	64	56	60

Solution

Calculation of Weighted Arithmetic Mean of marks two candidates.

Subjects	Candidate A (x)	Weights (w)	xw
Kannada	70	1	70
English	65	2	130
Maths	58	3	174
Statistics	63	4	252
		$\Sigma w = 10$	$\Sigma xw = 626$

Subjects	Candidate B (x)	Weights (w)	xw
Kannada	80	1	80
English	64	2	128
Maths	56	3	168
Statistics	60	4	240
		$\Sigma w = 10$	$\Sigma xw = 616$

Candidate A

$$\bar{x}_w = \frac{\Sigma xw}{\Sigma w} = \frac{626}{10} = 62.6$$

$$\therefore \bar{x}_w = 62.6$$

Candidate B

$$\bar{x}_w = \frac{\Sigma xw}{\Sigma w} = \frac{616}{10} = 61.6$$

$$\therefore \bar{x}_w = 61.6$$

Illustration 3

Compute Weighted Arithmetic Mean

Group	Index [x]	Weights [W]
Food		
Clothing		
Fuel		
Power		
Rent		
Other Exp.		

Solution

Calculation

Group	Index [x]	Weights [W]
Food		
Clothing		
Fuel		
Power		
Rent		
Other Exp.		

The weighted arithmetic mean is calculated as follows:

2.3 Median

Median is another measure of central tendency. It divides the data into two equal parts, i.e., half are less than or equal to the median and half are greater than or equal to the median. If the number of observations is odd, then the median is the middle value. If the number of observations is even, then the median is the average of the two middle values.

1. Median is a measure of central tendency.
2. It is also known as the middle value.
3. It divides the data into two equal parts.
4. It is suitable for qualitative data.
5. It is very useful in the study of income distribution.
6. It gives a better picture of the data.



Property

1. **Middle Value**
The median is the middle value of the data arranged in ascending or descending order.
2. **Position**
The median is the position of the middle value of the data arranged in ascending or descending order.

Illustration 3

Compute Weighted Arithmetic Mean of index number from the following data

Group	Food	Clothing	Fuel	Power	Rent	Other Exp.
Index [x]	130	101	155	140	170	180
Weights [W]	4	2	3	2	3	2

Solution**Calculation of Weighted Arithmetic Mean**

Group	Index (x)	Weight (w)	xw
Food	130	4	520
Clothing	101	2	202
Fuel	155	3	465
Power	140	2	280
Rent	170	3	510
Other Exp.	180	2	360
		$\Sigma w = 16$	$\Sigma xw = 2337$

$$\bar{x}_w = \frac{\Sigma xw}{\Sigma w} = \frac{2337}{16} = 146.06$$

$$\therefore \bar{x}_w = 146.06$$

$\therefore$ The weighted Arithmetic mean of the above index number is 146.06

2.3 Median

Median is another method to find the central value in a set of data. Median is that value which divides the data into two equal halves. Half of the values in the data are more than the median and the other half are less than the median. Median is the value of the middle item when all values are arranged in order. If the number of values is odd, then the median is the middle value. If the number of values is even, then the median is the average of the two middle values. The symbol for median is M.

1. Median is the middle value of the arranged data set.
2. It is also called "Positional Average" as it depends on the position of the data.
3. It divides the data into two equal parts with the same number of values on both sides.
4. It is suitable in cases where open-end class intervals are given.
5. It is very useful in understanding the central point in income and wage distributions.
6. It gives a better idea when the data is skewed or not evenly spread.

**Properties of Median****1. Middle Value of Ordered Data**

The median is the central value that divides an arranged dataset into two equal parts. Half the values lie below the median, and half lie above it.

Example: In the data 12, 15, 20, 25, 30, the median is 20.

2. Positional Measure

The median is called a positional average because its value depends on the position of the data when arranged in order, not on the numerical values themselves.

Example: Changing extreme values like the highest or lowest observation will not change the position of the median unless they fall in the middle.

3. Not Affected by Extreme Values

Since the median depends only on the middle position, very high or very low values (outliers) do not influence it significantly.

Example: Incomes ₹20,000, ₹22,000, ₹25,000, and ₹10,00,000 have a median of ₹23,000, which is not pulled up by the extreme ₹10,00,000 value.

4. Suitable for Skewed Distributions

The median gives a better central value when the data is not evenly distributed or is skewed.

Example: In income distribution, where a few people earn very high amounts, the median gives a better idea of a typical income.

5. Applicable to Open-End Class Intervals

The median can be determined even when the data has open-end intervals such as "Below 10" or "Above 50", unlike the mean.

Example: In a wage table where the highest category is "Above ₹50,000", the median can still be calculated.

6. Requires Data to be Arranged

To find the median, the dataset must be arranged in ascending or descending order before calculation.

7. Does Not Use All Observations in Calculation

The median is found using only the middle value(s). The other values do not directly influence the calculation.

8. Unique Value for a Dataset

For any given dataset, there is only one median value, making it a definite measure of central tendency.

9. Graphical Location Possible

The median can be located using graphs such as less than and more than ogives, where their intersection point gives the median.

10. Suitable for Qualitative and Quantitative Data

The median can be used for both numerical data (e.g., marks, heights) and ordinal data (e.g., ratings).



Merits of Median

- It is simple to understand and easy to calculate, especially in ungrouped data.
- Median can be found even when the data contains extreme values or outliers.
- It is the most suitable average for data that has open-end class intervals.
- The median can be easily located using graphical methods such as ogive curves.
- Median is not affected by very large or very small values in the data.
- It is useful in measuring typical values in skewed distributions like income or wealth.



Demerits of Median

- It does not consider all observations.
- Different sets of data can have the same median.
- It cannot be used for qualitative data.
- There is no provision for importance.
- Median is not suitable for small data sets.
- The value of median is not affected by extreme values.

Calculation of Median

1. Ungrouped Data

Step 1:

Step 2:

Step 3:

2. Grouped Data

a. Discrete Data

Step 1:

Step 2:

Step 3:

Step 4:

b. Continuous Data

Step 1:

Step 2:

Step 3:

Step 4:

Demerits of Median

- It does not consider all the values in the data while calculating the result.
- Different sets of values may give the same median even if their total values are very different.
- It cannot be used for mathematical or algebraic calculations like mean.
- There is no proper method to calculate a weighted median, so it is not used when values carry different importance.
- Median is not suitable for comparing two or more data sets when exact accuracy is needed.
- The value of median changes easily if the middle values are changed, so it is sensitive to small changes in small data sets.

Calculation of Median**1. Ungrouped Data or Raw Data**

Step 1: Arrange the given data in an order

Step 2: Find $\left[\frac{n+1}{2}\right]$ where 'n' denotes the number of observations

Step 3: $M = \text{size of } \left[\frac{n+1}{2}\right]^{\text{th}} \text{ Item / Value / Term / Observation.}$

2. Grouped Data**a. Discrete Frequency Distribution**

Step 1: Find cumulative frequencies

Step 2: Find $\left[\frac{N+1}{2}\right]^{\text{th}}$ Item / Value / Term / Observation

Step 3: See in the cumulative frequencies the value just greater than $\left[\frac{N+1}{2}\right]^{\text{th}}$ item

Step 4: Then the corresponding value of x is median

b. Continuous frequency distribution

Step 1: Find cumulative frequencies

Step 2: Find $\frac{N}{2}$

Step 3: See in the cumulative frequency the value first greater than $\frac{N}{2}$. Then the

Corresponding class interval is called as Median class. Then apply the formula

Step 4: $M = L + \left[\frac{\frac{N}{2} - pcf}{f} \right] \times i$

Where,

L = Lower limit of the median class

pcf = Cumulative frequency preceding the median class

f = frequency of median class

i = width of the median class



Note:

For the computation of median, inclusive class intervals must be converted into exclusive class interval.

Problems on Median for Raw Data

Illustration 1

The following are the marks scored by 8 students in an examination. 50 85 15 07 19 25 32 07
Find the median.

Solution

Sorted Array : 07 15 19 25 32 36 50 85

Here,

$n = 8$. Therefore median is

M = Size of $\left[\frac{n+1}{2}\right]^{\text{th}}$ term in array

$$\left[\frac{8+1}{2}\right]^{\text{th}} \text{ term in array} \Rightarrow \left[\frac{9}{2}\right]^{\text{th}} \text{ term in array}$$

$$= (4.5)^{\text{th}} \text{ term in array}$$

$$= 4^{\text{th}} \text{ value} + 0.5 (5^{\text{th}} \text{ value} - 4^{\text{th}} \text{ value}) = 25 + 0.5 (32 - 25) = 25 + 0.5 (7) = 28.5$$

$$M = 28.5$$

(Since the number of values is even, take the two middle values and find their average. The 4th value is 25 and the 5th value is 32. So, $(25+32)/2 = 28.5$)

Illustration 2

Calculate the value of median from the following figures.

$X : 5 \ 7 \ 9 \ 12 \ 11 \ 8 \ 7 \ 15 \ 25$

Solution

Sorted Array : 5, 7, 7, 8, 9, 11, 12, 15, 25

$$M = \text{Size of } \left[\frac{n+1}{2}\right]^{\text{th}} \text{ term in array} = \left[\frac{9+1}{2}\right]^{\text{th}} \text{ term in array} = \left[\frac{10}{2}\right]^{\text{th}} \text{ term in array}$$

$$M = (5)^{\text{th}} \text{ term in array}$$

$$M = 9 \text{ (since the number of values is odd, the middle value is the median)}$$

Illustration 1

Find the median

Solution

Sorted Array: 4

Problems

Illustration 1

The following data

No. of Apples

No. of Boxes

Solution

Calc

No. of apples (x)	b
115	
125	
135	
145	
155	
165	
175	
185	
195	
	N =

Illustration 3

Find the median for the data given below 4, 5, 9, 12, 18

Solution

Sorted Array: 4, 5, 9, 12, 18

$$M = \text{Size of } \left[\frac{n+1}{2} \right]^{\text{th}} \text{ term in array}$$

$$= \left[\frac{5+1}{2} \right]^{\text{th}} \text{ term in array} = \left[\frac{6}{2} \right]^{\text{th}} \text{ term in array}$$

$$M = (3)^{\text{rd}} \text{ term in array}$$

$$M = 9$$

Problems on Median for Discrete Data**Illustration 1**

The following data relates to the number of apples per box in 390 boxes find the median.

No. of Apples	115	125	135	145	155	165	175	185	195
No. of Boxes	6	25	48	72	116	60	38	22	3

Solution**Calculation of Median**

No. of apples (x)	No. of boxes (f)	Cumulative frequency
115	6	6
125	25	(25 + 6) = 31
135	48	(31 + 48) = 79
145	72	(79 + 72) = 151
155	116	(151 + 116) = 267
165	60	(267 + 60) = 327
175	38	(327 + 38) = 365
185	22	(365 + 22) = 387
195	3	(387 + 3) = 390
	N = 390	

Here $N = 390$

Therefore, median is

$$M = \text{size of } \left[\frac{N+1}{2} \right]^{\text{th}} \text{ Value}$$

$$= \left[\frac{390+1}{2} \right]^{\text{th}} \text{ value} = 195.5^{\text{th}} \text{ Value}$$

$$= (195.5)^{\text{th}} \text{ value}$$

(The next highest cumulative frequency of 195.5 is 267 and corresponding 'X' value is 155)

$$M = 155 \text{ apples / box}$$

Illustration 2

Calculate Median from the following data.

Marks	50	40	30	20	10
Frequency	10	40	20	12	16

Solution**Calculation of Median**

Marks (x)	Frequency (f)	Cumulative frequency
50	10	10
40	40	(10 + 40) = 50
30	20	(50 + 20) = 70
20	12	(70 + 12) = 82
10	16	(82 + 16) = 98
	N = 98	

Here $N = 98$

$\therefore$ Median is

$$M = \text{Size of } \left[\frac{N+1}{2} \right]^{\text{th}} \text{ Value} = \left[\frac{98+1}{2} \right]^{\text{th}} \text{ Value}$$

$$= (49.5)^{\text{th}} \text{ value}$$

(The next highest cumulative frequency of 49.5 is and corresponding 'X' value is 40)

$$M = 40$$

Illustration 3

Calculate median from the following data.

x	5	10	15	20	25	30
f	3	7	6	2	8	4

Solution**Calculation of Median**

x	f	cf
5	3	3
10	7	(3 + 7) = 10
15	6	(10 + 6) = 16
20	2	(16 + 2) = 18
25	8	(18 + 8) = 26
30	4	(26 + 4) = 30
	N=30	

$$M = \text{Size of } \left[\frac{N+1}{2} \right]^{\text{th}} \text{ value}$$

$$M = \text{Size of } \left[\frac{30+1}{2} \right]^{\text{th}} \text{ value} = \left[\frac{31}{2} \right]^{\text{th}} \text{ value}$$

$M = 15.5^{\text{th}}$ value lies in the cf of 16

(The next highest cumulative frequency of 15.5 is 16 corresponding 'X' value is 15)

$$\therefore M = 15$$

Problems on Median for Continuous Frequency Distribution

Illustration 1

The frequency distribution of weight in grams of mangoes of a given variety is given below. Calculate median:

Weight in grams	409.5 - 419.5	419.5 - 429.5	429.5 - 439.5	439.5 - 449.5	449.5 - 459.5	459.5 - 469.5	469.5 - 479.5
Number of mangoes	10	24	42	54	45	15	10

Solution

Calculation of Median

Weight in Grams (C.I)	No. of Mangoes(f)	Cumulative Frequency
409.5 - 419.5	10	10
419.5 - 429.5	24	(10 + 24) = 34
429.5 - 439.5	42	(34 + 42) = 76
439.5 - 449.5	54	(76 + 54) = 130
449.5 - 459.5	45	(130 + 45) = 175
459.5 - 469.5	15	(175 + 15) = 190
469.5 - 479.5	10	(190 + 10) = 200
	N = 200	

$$N = 200$$

$$\text{size of } \left[\frac{N}{2} \right]^{\text{th}} \text{ Item} = 100^{\text{th}} \text{ Item}$$

The cumulative frequency > 100 = 130

$\therefore$ Corresponding median class: 439.5-449.5

L = lower limit of the median class = 439.5

pcf = previous cumulative frequency of
Median class = 76

f = frequency of the median class = 54

i = width of the Median class = 10

$$\begin{aligned} \text{Median} &= L + \left[\frac{\frac{N}{2} - \text{pcf}}{f} \right] \times i = 439.5 + \left[\frac{100 - 76}{54} \right] \times 10 \\ &= 439.5 + \left[\frac{24}{54} \right] \times 10 = 439.5 + (0.44) \times 10 \\ &= 439.5 + 4.44 = 443.94 \end{aligned}$$

Median weight = 443.94 grams

Illustration 2

Following is the distribution of marks in statistics obtained by students. Calculate the median marks.

Marks	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60
No. of Students	4	6	20	10	8	2

Solution

Calculation of Median

C.I	f	cf
0 - 10	4	
10 - 20	6	4
20 - 30	20	(4 + 6) = 10
30 - 40	10	(10 + 20) = 30
40 - 50	8	(30 + 10) = 40
50 - 60	2	(40 + 8) = 48
	N = 50	(48 + 2) = 50

$$N = 50$$

$$\text{Size of } \left[\frac{N}{2} \right]^{\text{th}} \text{ Item} = \frac{50}{2} = 25^{\text{th}} \text{ item}$$

The cumulative frequency > 25 = 30

$\therefore$ Corresponding median class: 20-30

L = lower limit of the median class = 20

pcf = previous cumulative frequency of

Median class = 10

f = frequency of the median class = 20

i = width of the Median class = 10

$$M = L + \left[\frac{\frac{N}{2} - \text{pcf}}{f} \right] \times i = 20 + \left[\frac{25 - 10}{20} \right] \times 10 = 20 + 7.5 = 27.5$$

$\therefore$ Median Marks = 27.5

Illustration 3

Following data relates to daily wages of person working in a factory. Compute the median wages.

Daily wages in ₹ (x)	55-60	55-60	55-50	45-50	40-45	35-40	30-35	25-30
No of workers (f)	7	7	13	12	23	3	33	28

Solution

Calculation of Median

Daily wages in ₹ (x)	No of workers (f)	cf
55-60	7	7
55-50	13	(07 + 13) = 20
45-50	12	(20 + 12) = 32
40-45	23	(32 + 23) = 55
35-40	3	(55 + 03) = 58
30-35	33	(58 + 33) = 91
25-30	28	(91 + 28) = 119
20-25	14	(119 + 14) = 133
	N = 133	

$$N = 133$$

$$\text{size of } \left[\frac{N}{2} \right]^{\text{th}} \text{ Item} = \left[\frac{133}{2} \right]^{\text{th}} = 66.5^{\text{th}} \text{ item}$$

The cumulative frequency > 66.5 = 91

$\therefore$ Corresponding median class: 30-35

L = lower limit of the median class

pcf = previous cumulative frequency

Median Class = 58

f = frequency of the median class

i = width of the Median class = 5

$$M = L + \left[\frac{\frac{N - pcf}{2}}{f} \right] \times i = 30 + \left[\frac{66.5 - 58}{33} \right] \times 5$$

$$= 30 + \left[\frac{8.5}{33} \right] \times 5 = 30 + 1.29 = 31.29$$

Median wages = 31.29

Illustration 4

Find median of the following data.

Wages	60-70	50-60	40-50	30-40	20-30
No of Workers	7	21	10	7	5

Solution

Calculation of Median

Wages (C.I)	No. of Workers (f)	cf
60-70	7	7
50-60	21	(7 + 21) = 28
40-50	10	(28 + 10) = 38
30-40	7	(38 + 7) = 45
20-30	5	(45 + 5) = 50
	N = 50	

$$N = 50$$

$$\text{size of } \frac{N^{\text{th}}}{2} \text{ Item} = \frac{50^{\text{th}}}{2} = 25^{\text{th}} \text{ Item}$$

The cumulative frequency > 25 = 28

∴ Corresponding median class: 50-60

L = lower limit of the median class = 21

pcf = previous cumulative frequency of

Median Class = 7

f = frequency of the median class = 21

i = width of the median class = 10

$$M = L + \left[\frac{\frac{N - pcf}{2}}{f} \right] \times i = 21 + \left[\frac{25 - 7}{21} \right] \times 10$$

$$= 21 + 0.857 \times 10 = 21 + 8.57 = 29.57$$

∴ Median wages = 29.57

Problems on Median for Less than Cumulative Frequency Distribution

Illustration 1

The following table shows the Marks distribution of students in a particular region. Find the median Marks.

Marks	< 10	< 20	< 30	< 40	< 50	< 60	< 70	< 80
No. of Students (in thousand)	2	10	16	20	25	38	64	70

Solution

Since the formula for median is based on continuous frequency distribution, we shall convert given inclusive class interval in to exclusive class interval.

Calculation of Median

Marks	No. of Students	Class Interval	Frequency (f)	cf
< 10	2	0 - 10	2	2
< 20	10	10 - 20	(10 - 2) = 8	10
< 30	16	20 - 30	(16 - 10) = 6	16
< 40	20	30 - 40	(20 - 16) = 4	20
< 50	25	40 - 50	(25 - 20) = 5	25
< 60	38	50 - 60	(38 - 25) = 13	38
< 70	64	60 - 70	(64 - 38) = 26	64
< 80	70	70 - 80	(70 - 64) = 6	70
			N = 70	

$$\text{size of } \left[\frac{N}{2} \right]^{\text{th}} \text{ item} = \frac{70}{2} = 35$$

The cf is just greater than 35 is 38. Hence the corresponding class (50-60) is treated as median class.

L = lower limit of the median class = 50

pcf = previous cumulative frequency of Median class = 25

f = frequency of the median class = 13

i = width of the Median class = 10

$$M = L + \left[\frac{\frac{N}{2} - pcf}{f} \right] \times i = 50 + \left[\frac{35 - 25}{13} \right] \times 10 = 50 + \left[\frac{10}{13} \right] \times 10 = 50 + 7.69 = 57.69$$

∴ Median Marks = 57.69.

Illustration 2

Calculate Median from the following series:

Class interval	0 - 4	5 - 9	10 - 14	15 - 19	20 - 24	25 - 29	30 - 34	35 - 39
Frequency	5	18	41	76	102	120	135	143

Solution

Calculation of Median

C.I	f	c.f
-0.5 - 4.5	5	5
4.5 - 9.5	13	(5 + 13) = 18
9.5 - 14.5	23	(18 + 23) = 41
14.5 - 19.5	35	(41 + 35) = 76
19.5 - 24.5	26	(76 + 26) = 102
24.5 - 29.5	18	(102 + 18) = 120
29.5 - 34.5	15	(120 + 15) = 135
34.5 - 39.5	8	(135 + 8) = 143
	N = 143	

$$N = 143$$

$$\text{size of } \left[\frac{N}{2} \right]^{\text{th}} \text{ Item} = \frac{143}{2} = 71.5^{\text{th}} \text{ value}$$

The cumulative frequency > 71.5 = 76

$\therefore$ **Corresponding median class: 14.5 - 19.5**

L = lower limit of the median class = 14.5

pcf = previous cumulative frequency of

Median Class = 41

f = frequency of the median class = 35

i = width of the Median class = 5

$$M = L + \left[\frac{\frac{N}{2} - \text{pcf}}{f} \right] \times i = 14.5 + \left[\frac{71.5 - 41}{35} \right] \times 5$$

$$= 14.5 + 4.35 = 18.86$$

$$M = 18.86$$

Problems on Median for more than Cumulative Frequency Distribution

Illustration 1

Calculate median for the following

Wages	Above 0	Above 10	Above 20	Above 30	Above 40	Above 50	Above 60	Above 70
No. of Laborers	650	500	425	375	300	275	250	100

Solution

Calculation of Median

Wages	More than Cumulative Frequency	Frequency (f)	ECI	Cumulative Frequency (cf)
Above 0	650	(650 - 500) = 150	0-10	150
Above 10	500	(500 - 425) = 75	10-20	(150 + 75) = 225
Above 20	425	(425 - 375) = 50	20-30	(225 + 50) = 275
Above 30	375	(375 - 300) = 75	30-40	(275 + 75) = 350
Above 40	300	(300 - 275) = 25	40-50	(350 + 25) = 375
Above 50	275	(275 - 250) = 25	50-60	(375 + 25) = 400
Above 60	250	(250 - 100) = 150	60-70	(400 + 150) = 550
Above 70	100	100	70-80	(550 + 100) = 650
Total		N = 650		

$$N = 650$$

$$\text{Size of } \left[\frac{N}{2} \right]^{\text{th}}$$

$$\text{term} = \left[\frac{650}{2} \right]^{\text{th}} \text{ term}$$

$$= (325)^{\text{th}} \text{ term}$$

$$\text{The } cf > 325 = 350$$

$$\therefore \text{Corresponding median class} = 30 - 40$$

L = lower limit of the median class
 pcf = previous cumulative frequency
 Median class = 275

f = frequency of the median class
 i = width of the Median class = 10

$$\text{Median} = L + \left[\frac{\frac{N}{2} - pcf}{f} \right] \times i$$

$$= 30 + \left[\frac{325 - 275}{75} \right] 10$$

$$= 30 + \left[\frac{50}{75} \right] 10 = 30 + \left[\frac{500}{75} \right] = 30 + 6.67$$

$$\therefore \text{Median wages} = 36.67$$

Illustration 2

Calculate Median for the following

Income (1000's)	> 0	> 10	> 20	> 30	> 40	> 50	> 60	> 70
No. of families	100	91	78	54	38	26	18	10

Solution

Calculation of Median

Income	ECI	MCF	f	cf
> 0	0-10	100	$(100 - 91) = 9$	9
> 10	10-20	91	$(91 - 78) = 13$	$(9 + 13) = 22$
> 20	20-30	78	$(78 - 54) = 24$	$(22 + 24) = 46$
> 30	30-40	54	$(54 - 38) = 16$	$(46 + 16) = 62$
> 40	40-50	38	$(38 - 26) = 12$	$(62 + 12) = 74$
> 50	50-60	26	$(26 - 18) = 8$	$(74 + 8) = 82$
> 60	60-70	18	$(18 - 10) = 8$	$(82 + 8) = 90$
> 70	70-80	10	10	$(90 + 10) = 100$
Total			$N = 100$	

$$N = 100$$

$$\text{Size of } \left[\frac{N}{2} \right]^{\text{th}} \text{ term} = \left[\frac{100}{2} \right]^{\text{th}} \text{ term}$$

$$= 50^{\text{th}} \text{ term}$$

$$\text{The } cf > 50 = 62$$

$$\therefore \text{Corresponding Median class} = 30 - 40$$

$$L = \text{lower limit of the median class} = 30$$

$$pcf = \text{previous cumulative frequency of Median class} = 46$$

$$f = \text{frequency of the median class} = 16$$

$$i = \text{width of the Median class} = 10$$

Problem

Illustration

Calculate the

Mar

No. of St

Solution

Note: The s

The cf

Corr

$$\text{Median} = L + \left[\frac{\frac{N}{2} - \text{pcf}}{f} \right] \times i$$

$$= 30 + \left[\frac{50 - 46}{16} \right] \times 10 = 30 + \left[\frac{4}{16} \right] 10 = 30 + \left[\frac{40}{16} \right] = 30 + 2.5$$

$$\text{Median} = 32.5$$

Problems on Median for Open end Class Interval

Illustration 1

Calculate the suitable average for the following data.

Marks	Less than 10	10-20	20-30	30-40	40 & above
No. of Students	3	10	15	5	2

Solution

Note: The suitable average for open end class interval is median

Calculation of Median

Marks (C.I)	No. of Students (f)	cf
Less than 10	3	3
10-20	10	(3 + 10) = 13
20-30	15	(13 + 15) = 28
30-40	5	(28 + 5) = 33
40 & above	2	(33 + 2) = 35
	N = 35	

$$N = 35$$

$$\text{Size of } \left[\frac{N}{2} \right]^{\text{th}} \text{ term}$$

$$= \left[\frac{35}{2} \right]^{\text{th}} \text{ term}$$

$$= 17.5^{\text{th}} \text{ term}$$

$$\text{The } cf > 17.5 = 28$$

Corresponding Median class

$$= 20 - 30$$

L = lower limit of the median class = 20

pcf = previous cumulative frequency of

Median class = 13

f = frequency of the median class = 15

i = width of the Median class = 10

$$\text{Median} = L + \left[\frac{\frac{N}{2} - \text{pcf}}{f} \right] \times i = 20 + \left[\frac{17.5 - 13}{15} \right] \times 10$$

$$= 20 + \left[\frac{4.5}{15} \right] 10 = 20 + \left[\frac{45}{15} \right] = 20 + 3 = 23$$

∴ Median Marks = 23

Illustration 2

Calculate median for the following data.

Marks	<10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80 & above
No. of Students	5	10	20	30	25	18	14	6	3

Solution**Calculation of Median**

Marks	f	cf
Less than 10	5	5
10-20	10	(5 + 10) = 15
20-30	20	(15 + 20) = 35
30-40	30	(35 + 30) = 65
40-50	25	(65 + 25) = 90
50-60	18	(90 + 18) = 108
60-70	14	(108 + 14) = 122
70-80	6	(122 + 6) = 128
80 & above	3	(128 + 3) = 131
	N = 131	

$$N = 131$$

$$\text{Size of } \left[\frac{N}{2} \right]^{\text{th}} \text{ term}$$

$$= \left(\frac{131}{2} \right)^{\text{th}} \text{ term}$$

$$= (65.5)^{\text{th}} \text{ term}$$

The $cf > 65.5 = 90$

$\therefore$ Corresponding Median class = 40 - 50

L = lower limit of the median class = 40

pcf = previous cumulative frequency of

Median class = 65

f = frequency of the median class = 25

i = width of the Median class = 10

$$\text{Median} = L + \left[\frac{\frac{N}{2} - pcf}{f} \right] \times i$$

$$= 40 + \left(\frac{65.5 - 65}{25} \right) 10$$

$$= 40 + \left(\frac{0.5}{25} \right) 10 = 40 + \left(\frac{5}{25} \right) = 40 + 0.2$$

$\therefore$ **Median Marks = 40.2**

Problem**Illustration**

Obtain media

Mid value

Frequency

Solution

Calculation of

Size of

The $cf > 6$ $\therefore$ Corresponding

Problems on Median on Mid Point & Frequency

Illustration 1

Obtain median value for the following distribution.

Mid value	10	20	30	40	50
Frequency	6	25	72	22	4

Solution

Calculation of Median

Mid value	f	ECI	cf
10	6	5-15	6
20	25	15-25	(6 + 25) = 31
30	72	25-35	(31 + 72) = 103
40	22	35-45	(103 + 22) = 125
50	4	45-55	(125 + 4) = 129
	N = 129		

$$N = 129$$

$$\text{Size of } \left[\frac{N}{2} \right]^{\text{th}} \text{ term}$$

$$= \left[\frac{129}{2} \right]^{\text{th}} \text{ term}$$

$$= (64.5)^{\text{th}} \text{ term}$$

$$\text{The } cf > 64.5 = 103$$

$$\therefore \text{Corresponding Median class} \\ = 25 - 35$$

$$L = \text{lower limit of the median class} = 25$$

$$pcf = \text{previous cumulative frequency of} \\ \text{median class} = 31$$

$$f = \text{frequency of the median class} = 72$$

$$i = \text{width of the median class} = 10$$

$$\text{Median} = L + \left[\frac{\frac{N}{2} - pcf}{f} \right] \times i$$

$$= 25 + \left[\frac{64.5 - 31}{72} \right] 10$$

$$= 25 + \left[\frac{33.5}{72} \right] 10$$

$$= 25 + \left[\frac{335}{72} \right] = 25 + 4.65$$

$$\therefore \text{Median} = 29.65$$

Illustration 2

Calculate median for the following

Mid point	115	125	135	145	155	165	175	185
Frequency	6	25	48	72	116	60	38	22
								3

Solution**Calculation of Median**

Mid point	f	cf	Eci
115	6	6	110-120
125	25	(6 + 25) = 31	120-130
135	48	(31 + 48) = 79	130-140
145	72	(79 + 72) = 151	140-150
155	116	(151 + 116) = 267	150-160
165	60	(267 + 60) = 327	160-170
175	38	(327 + 38) = 365	170-180
185	22	(365 + 22) = 387	180-190
195	3	(387 + 3) = 390	190-200
	N = 390		

$$N = 390$$

$$\text{Size of } \left[\frac{N}{2} \right]^{\text{th}} \text{ term}$$

$$= \left[\frac{390}{2} \right]^{\text{th}} \text{ term}$$

$$= (195)^{\text{th}} \text{ term}$$

$$\text{The } cf > 195 = 267$$

 $\therefore$ Corresponding Median class

$$= 150 - 160$$

$$L = \text{lower limit of the median class} = 150$$

$$pcf = \text{previous cumulative frequency of}$$

$$\text{median class} = 151$$

$$f = \text{frequency of the median class} = 116$$

$$i = \text{width of the median class} = 10$$

$$\text{Median} = L + \left[\frac{\frac{N}{2} - pcf}{f} \right] \times i$$

$$= 150 + \left[\frac{195 - 151}{116} \right] 10$$

$$= 150 + \left[\frac{44}{116} \right] 10$$

$$= 150 + \left[\frac{440}{116} \right] = 150 + 3.79$$

$$\therefore \text{Median} = 153.79$$

Problems on Median on Missing Frequency

Illustration 1

For the following distribution Median is 33.5, find the missing frequency

CI	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70
f	4	6	60	?	40	6	4

Solution

Let the missing frequency be x

Calculation of missing frequency

CI	f	cf
0 - 10	4	4
10 - 20	6	10
20 - 30	60	70
30 - 40	x	70 + x
40 - 50	40	110 + x
50 - 60	6	116 + x
60 - 70	4	120 + x
	N = 120 + x	

$$M = 33.5$$

$$\text{Median Class} = 30 - 40$$

$$\text{Here, } L = 30, M = 33.5, \text{pcf} = 70, i = 10,$$

$$\frac{N}{2} = \frac{120 + x}{2}$$

$$M = L + \left[\frac{\frac{N}{2} - \text{pcf}}{f} \right] \times i$$

$$33.5 = 30 + \left[\frac{\frac{120 + x}{2} - 70}{x} \right] \times 10$$

$$33.5 - 30 = \left[\frac{120 + x - 140}{2x} \right] \times 10$$

$$(3.5)2x = 10(x - 20)$$

$$7x = 10x - 200$$

$$3x = 200$$

$$x = 200/3 = 66.67$$

$$x = 67$$

Illustration 2

From the following details find out the missing frequencies of median value is 30 and N = 100

Marks	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60	
No. of Students	10	?	25	30	?	10	N = 100

Solution

Let the 2 missing frequency be 'x' & 'y'

Calculation of Missing Frequency		
Marks (C.I)	f	cf
0-10	10	10
10-20	x	10+x
20-30	25	35+x
30-40	30	65+x
40-50	y	65+x+y
50-60	10	75+x+y
	N = 100	

$$N = 75 + x + y$$

Given: Median = 30

Median class = 30 - 40

$$N = 75 + x + y$$

$$100 = 75 + x + y$$

$$100 - 75 = x + y$$

$$25 = x + y$$

$$y = 25 - x$$

$$M = L + \left[\frac{\frac{N}{2} - pcf}{f} \right] \times i$$

$$30 = 30 + \left[\frac{\frac{100}{2} - (35 + x)}{30} \right] \times 10$$

$$30 - 30 = \left[\frac{50 - (35 + x)}{30} \right] \times 10$$

$$0 = \left[\frac{50 - 35 - x}{30} \right] \times 10$$

$$0 \times 30 = (15 - x) 10$$

$$0 = 10(15 - x)$$

$$150 = 10x$$

$$x = \frac{150}{10} \Rightarrow x = 15$$

Now, let's find the value of y

$$y = 25 - x$$

$$y = 25 - 15 \Rightarrow y = 10$$

Illustration 3

From the following data calculate missing frequencies if median is 50.

Expenditure	0-20	20-40	40-60	60-80	80-100	
No. of families	14	?	27	?	15	N = 100

Solution

Let the 2 missing frequency be 'x' & 'y'

Calculation of Missing Frequency

CI	f	cf
0-20	14	14
20-40	x	14+x

Given: Me
Median c

100

2.4 Mod

Mode is the
frequently. T
5, 6, 7, 8, 8, 8
by the symbo



Defin

1. Accor
the ite
2. Accor
neigh

The concept
is desired to
size of a rea
expenditure
duration of c

40-60	27	41+x
60-80	y	41+x+y
80-100	15	56+x+y
	N = 100	

$$N = 56 + x + y$$

Given: Median = 50,

Median class = 40 - 60

$$N = 56 + x + y$$

$$100 = 56 + x + y$$

$$100 - 56 = x + y$$

$$44 = x + y$$

$$44 - x = y$$

$$y = 44 - x$$

$$M = L + \left[\frac{\frac{N}{2} - pcf}{f} \right] \times i$$

$$= 50 = 40 + \left[\frac{\frac{100}{2} - (14 + x)}{27} \right] \times 20$$

$$50 - 40 = \left[\frac{50 - 14 - x}{27} \right] \times 20$$

$$10 = \left[\frac{36 - x}{27} \right] \times 20$$

$$10(27) = (36 - x) 20$$

$$270 = 20(36 - x)$$

$$270 = 720 - 20x$$

$$270 - 720 = -20x$$

$$-450 = -20x; x = \frac{450}{20}$$

$$x = 22.5$$

Now, let's find the value of y

$$y = 44 - x$$

$$y = 44 - 22.5$$

$$y = 22.5$$

2.4 Mode

Mode is the most common or repeated value in a set of data. It is the value that appears most frequently. The word mode means 'most usual' or 'most repeated'. For example, in the series 2, 3, 5, 5, 6, 7, 8, 8, 8, 9 - the mode is 8, because it comes more times than any other number. Mode is shown by the symbol Z.



Definitions: Mode

1. According to Croxton and Cowden, "The mode of a distribution is the value at the point around which the items tend to be most heavily concentrated. It may be regarded the most typical of a series of values".
2. According to A.M. Tuttle, "Mode is the value which has the greatest frequency density in its immediate neighborhood."

The concept of mode, as a measure of central tendency, is preferable to mean and median when it is desired to know the most typical value, e.g., the most common size of shoes, the most common size of a ready-made garment, the most common size of income, the most common size of pocket expenditure of a college student, the most common size of a family in a locality, the most common duration of cure of viral-fever, the most popular candidate in an election, etc.

The frequency distribution having one mode is called as 'unimodal'. The frequency distribution having two mode is called as 'bimodal'. The frequency distribution having more than two mode is called as 'multimodal' or 'no mode' at all.

Computation of Mode

1. Ungrouped Data (or) Raw Data / Individual Series

It is found by mere inspection. The highest repeated value in the series is mode.

2. Discrete Series:

The corresponding value of X for the maximum frequency is the value of mode.

3. Continuous Frequency Distribution (Grouped Data)

When the data under consideration is grouped into class intervals, mode can be determined using a formula, But only if the classes are of equal width

Steps : 1. Identify the highest frequency ' f_1 '

2. Determine the modal class = class interval corresponding to f_1

$$3. \quad Z = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times i$$

L = Lower limit of the modal class

i = width of the modal class

f_1 = Frequency of the modal class

f_0 = Preceding Frequency of the modal class.

f_2 = succeeding frequency of the modal class.



Note

1. For computation of mode, inclusive class interval of the modal class must be converted into exclusive class interval.
2. Mode cannot be computed for class intervals of unequal width. When unequal width class intervals are given, it ought to be regrouped into class intervals of equal width either by grouping of class intervals or by division of class intervals.



Properties of Mode

1. Most Frequent Value

The mode is the value or item that occurs most frequently in a dataset. It represents the most common observation.

Example: In the dataset 2, 4, 4, 5, 7, 4, 8, the mode is 4 because it appears most often.

2. Positional and Frequency-Based Measure

The mode is a positional measure because it depends on the position of the highest frequency in the dataset, not on the magnitude of all values.

3. Not Affected by Extreme Values

Extreme high or low values do not affect the mode because it is determined by frequency, not magnitude.

Example: In incomes ₹10,000, ₹12,000, ₹15,000, ₹50,000, ₹12,000, the mode is ₹12,000, unaffected by the extreme ₹50,000.

4. Represents Most Typical Item

The mode indicates the most popular or typical value in a dataset, making it useful in business, marketing, and consumer studies.

Example: In a shoe size survey, the mode shows the size most people wear.

5. Can Be Determined for Qualitative Data

The mode can be used for non-numerical data as well, such as the most preferred colour, brand, or product type.

Example: If "Red" is chosen most often in a colour preference survey, it is the mode.

6. Suitable for Open-End Distributions

The mode can be calculated even when the data contains open-ended intervals like "Below 10" or "Above 50".

7. Requires No Ordering of Data

Unlike the median, the data need not be arranged in ascending or descending order to find the mode; only frequency counts are needed.

8. Can Be Located Graphically

The mode in continuous data can be found using a histogram. The highest bar in the histogram indicates the modal class.

9. Relation with Mean and Median in Symmetrical Distributions

In a perfectly symmetrical distribution,

$$\text{Mean} = \text{Median} = \text{Mode}$$

In moderately skewed distributions, Karl Pearson's relation is often used:

$$\text{Mode} = 3 \times \text{Median} - 2 \times \text{Mean}$$



Merits of Mode

- It is easy to understand and simple to calculate.
- It is not affected by extreme values hence is a good representative of the series
- It cannot be affected by abnormal extreme values like median.
- It can be easily calculated for open end class internals.
- It can be found graphically also by using histogram.
- It can also be used in case of quantitative phenomenon.



Demerits of Mode

- It is not based on all the values.
- It is not as stable as arithmetic mean.
- Mode can not be determined if the series is bimodal or multimodal
- Mode is not rigidly defined. Solving the problem by different methods won't get the same results.

Problems on Mode for Raw Data / Individual Series

Illustration 1

Find mode for the following data. 20, 35, 40, 17, 27, 50, 20, 30, 20

Solution

Here, 20 has repeated thrice. That is the highest repetition value. So mode = 20. It is unimodal.

Illustration 3

Find mode 32, 49, 36, 24, 17, 46

Solution

No mode at all as no values are repeated.

Problems on Mode for Discrete Series

Illustration 1

Find mode for the following distribution.

X	10	20	30	40	50
f	11	21	28	45	36

Solution

Mode = The corresponding 'X' value of the highest frequency

As, the highest frequency is 45, the corresponding value of 'X' is 40. So, Mode = 40.

Illustration 2

Find mode from the values. 25, 51, 37, 25, 40, 37, 41

Solution

It is bimodal distribution, since the values 25 and 37 repeated equal number of times (twice).

Illustration 4

Calculate mode from the following data. 10, 9, 7, 5, 6, 10, 9, 14, 8, 5, 6, 8, 11, 12

Solution

Mode = 8 because value 8 is repeated 3 number of times.

Illustration 2

Find mode for the following distribution

X	40	50	60	70
f	5	11	34	29

Solution

Mode = The corresponding 'X' value of the highest frequency

As, the highest frequency is 34, the corresponding value of 'X' is 60. So, Mode = 60.

Illustration 3

Find mode for the following data

X	10	20	30	40
f	11	21	28	45

Solution

Mode = The value of 'X' corresponding to the highest frequency

As, the highest frequency is 45, the corresponding value of 'X' is 40. So, Mode = 40.

Problems

Illustration 1

Calculate mode from the following data

Marks	10-20	20-30	30-40	40-50	50-60	60-70
No of Students	11	21	28	45	36	29

Solution

Calculate mode from the following data

Marks (C.I)	10-20	20-30	30-40	40-50	50-60	60-70
No of Students	11	21	28	45	36	29

f_1 = frequency

f_2 = Successive frequency

i = width of class

Illustration 3

Find mode for the following distribution.

X	5	10	15	20	25
f	7	13	25	19	8

Solution

Mode = The corresponding 'X' value of the highest frequency
As, the highest frequency is 25, the corresponding value of 'X' is 15. So, Mode = 15.

Problems on Mode for Continuous Series

Illustration 1

Calculate mode from the following distribution.

Marks	10-20	20-30	30-40	40-50	50-60	60-70
No of Students	5	11	18	12	8	4

Solution

Calculation of Mode

Marks (C.I)	No of Students (f)
10-20	5
20-30	11 (f_0)
30-40	18 (f_1)
40-50	12 (f_2)
50-60	8
60-70	4
	N = 58

Since highest frequency = 18, its correspondence x is 30 - 40

$\therefore$ Modal class = 30 - 40

$$Z = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times i$$

L = Lower limit of modal class = 30,

f_0 = Preceding frequency of modal class 11,

f_1 = frequency of modal class = 18,

f_2 = Succeeding frequency of modal class 12,

i = width/ class Intervals of modal class 10

$$Z = 30 + \left[\frac{18 - 11}{2(18) - 11 - 12} \right] \times 10$$

$$= 30 + \left[\frac{7}{36 - 11 - 12} \right] \times 10 = 30 + \left[\frac{7}{13} \right] \times 10$$

$$= 30 + (0.538 \times 10) = 30 + 5.38 = 35.38$$

$\therefore$ Mode = 35.38

Illustration 2

Calculate mode from the following data :

Wages	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60
No. of workers	2	18	30	45	35	20

Solution**Calculation of Mode**

Wages (C.I)	No. of Workers (f)
0 - 10	2
10 - 20	18
20 - 30	30 (f_0)
30 - 40	45 (f_1)
40 - 50	35 (f_2)
50 - 60	20
	N = 150

Since highest frequency = 45, its correspondence x is 30.
 $\therefore$ Modal class = 30 - 40

$$Z = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times i$$

L = Lower limit of modal class = 30,

f_0 = Preceding frequency of modal class = 30,

f_1 = frequency of modal class = 45,

f_2 = Succeeding frequency of modal class = 35

i = width/ class intervals of modal class = 10

$$= 30 + \left[\frac{45 - 30}{2(45) - 30 - 35} \right] \times 10$$

$$= 30 + \left[\frac{15}{90 - 30 - 35} \right] \times 10$$

$$= 30 + \left[\frac{150}{25} \right] = 30 + 6 = 36$$

$\therefore$ Mode = 36

Illustration 3

Calculate mode from the following data.

Marks	< 100	< 200	< 300	< 400	< 500
No. of Students	10	35	65	85	100

Solution

Convert the given data into continuous form.

Calculation of Mode

Marks (C.I)
0 - 100
100 - 200
200 - 300
300 - 400
400 - 500

Illustration 4

Calculate mode from the following data.

Income (in Rs.)
No. of persons

Solution

Convert the given data into continuous form.

Calculation of Mode

Income
1000 - 1500
1500 - 2000
2000 - 2500
2500 - 3000
3000 - 3500
3500 - 4000
4000 - 4500

Solution

Convert the given less than cumulative frequency distribution into simple frequency distribution.

Calculation of Mode

Marks (C.I)	No. of Students (f)
0 - 100	10
100 - 200	35 - 10 = 25 (f_0)
200 - 300	65 - 35 = 30 (f_1)
300 - 400	85 - 65 = 20 (f_2)
400 - 500	100 - 85 = 15
	N = 100

Since Highest frequency = 30, Modal class = (200 - 300)

$$\begin{aligned}
 Z &= L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times i \\
 &= 200 + \left[\frac{30 - 25}{2(30) - 25 - 20} \right] \times 100 \\
 &= 200 + \left[\frac{5}{60 - 25 - 20} \right] \times 100 = 200 + \left[\frac{500}{15} \right]
 \end{aligned}$$

$$Z = 200 + 33.33 = 233.33$$

$$\therefore Z = 233.33$$

Illustration 4

Calculate mode for following data.

Income (in ₹) (above values)	1000	1500	2000	2500	3000	3500	4000
No. of Persons	80	62	33	16	8	4	2

Solution

Convert the given more than cumulative frequency distribution into simple frequency distribution.

Calculation of Mode

Income (C.I)	No. of Persons (f)
1000 - 1500	80 - 62 = 18 (f_0)
1500 - 2000	62 - 33 = 29 (f_1)
2000 - 2500	33 - 16 = 17 (f_2)
2500 - 3000	16 - 8 = 8
3000 - 3500	8 - 4 = 4
3500 - 4000	4 - 2 = 2
4000 - 4500	2 - 0 = 2
	N = 80

Since highest frequency is 29.

Modal Class = 1500 - 2000

$$\begin{aligned}
 Z &= L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times i \\
 &= 1500 + \left[\frac{29 - 18}{2(29) - 18 - 17} \right] \times 500 \\
 &= 1500 + \left[\frac{11}{23} \right] \times 500 \\
 &= 1500 + 239.1304 = 1739.13
 \end{aligned}$$

$$\therefore Z = 1739.13$$

Calculate mode for the following distribution.

C.I	20 - 29	30 - 39	40 - 49	50 - 59	60 - 69
Frequency (f)	9	17	21	18	10

Solution

Calculation of Mode

Convert the given inclusive class intervals into exclusive class intervals using correction factor,

$$\text{Correction factor} = \frac{30 - 29}{2} = \frac{1}{2} = 0.5$$

Subtract (0.5) from lower limit and add (0.5) to upper limit.

C.I (Exclusive)	Frequency (f)
19.5 - 29.5	9
29.5 - 39.5	17 (f_0)
39.5 - 49.5	21 (f_1)
49.5 - 59.5	18 (f_2)
59.5 - 69.5	10
	N = 75

Since highest frequency is 21, Modal class = (39.5- 49.5)

$$Z = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times i$$

$$= 39.5 + \left[\frac{21 - 17}{2(21) - 17 - 18} \right] \times 10$$

$$= 39.5 + \left[\frac{4}{7} \right] \times 10 = 39.5 + 5.714 = 45.214$$

$$\therefore Z = 45.214$$

Illustration 6

From the following distribution of hourly wages (in ₹) of firm employees find the modal wage

Hourly Wage	Below 10	Below 20	Below 30	Below 40	Below 50	Below 60	Below 70	Below 80
No of Employees	2	15	60	82	95	122	140	158

Solution

Convert the given data into exclusive class interval.

Calculation of Mode

Here highest frequency is 45

Modal class = (20-30)

Hourly Wage (C.I)	No of Employees (f)
0-10	2
10-20	13 (f_0)
20-30	45 (f_1)
30-40	22 (f_2)

$$Z = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times i$$

$$= 20 + \left[\frac{45 - 13}{2(45) - 13 - 22} \right] \times 10$$

This means
this formula

Mod

But when
cases, Karl

There is a
moderate

In a perfect
curve is ba

But when
cases, Karl

Mod

2.5 E

14-1
16-1
18 & ab

Calculat

Soluti

Calcula

Illust

40-50	13
50-60	27
60-70	18
70-80	10
	N = 150

$$= 20 + \left[\frac{32}{90 - 35} \right] \times 10 \Rightarrow 20 + 0.5818 (10)$$

$$= 20 + 5.818 = 25.818$$

$$\therefore \text{Mode} = 25.818$$

Illustration 7

Calculate mode for the following distribution

CI	Below 10	10-12	12-14	14-16	16-18	18 & above
f	3	15	27	20	3	2

Solution

Calculation of Mode

Here highest frequency is 27

Modal class = (12 - 14)

$$Z = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times i$$

Given : $L = 12$, $f_1 = 27$, $f_2 = 20$, $f_0 = 15$, $i = 2$

$$= 12 + \left[\frac{27 - 15}{2(27) - 15 - 20} \right] \times 2$$

$$= 12 + \left[\frac{12}{54 - 35} \right] \times 2 = 12 + \left[\frac{12}{19} \right] \times 2$$

$$= 12 + 0.6315 (2) = 12 + 1.2632 = 13.2632$$

$$\therefore \text{Mode} = 13.2632$$

2.5 Empirical Relation among Mean, Median and Mode

There is a well-known formula that shows the relationship between Mean, Median, and Mode in a moderately skewed distribution. This is called **Karl Pearson's empirical formula**.

In a perfectly symmetrical distribution, the Mean = Median = Mode. This happens when the frequency curve is balanced on both sides.

But when the distribution is skewed (not symmetrical), the three values are not the same. In such cases, Karl Pearson gave the following formula to estimate the mode:

$$\text{Mode} = 3 \times \text{Mean} - 2 \times \text{Median}$$

$$Z = 3M - 2\bar{x}$$

This means, if the values of mean and median are known, the value of mode can be estimated using this formula.

Problems on Empirical Relation between Mean, Median and Mode

Illustration 1

Compute Mode, if Median = 25 and Mean = 20

Solution

Given : $M = 25, \bar{x} = 20$

$$Z = 3M - 2\bar{x}$$

$$= 3(25) - 2(20) = 75 - 40$$

$$\therefore Z = 35$$

Illustration 2

Find mean, if mode is 38.82 and median

Solution

$$Z = 3M - 2\bar{x}$$

$$38.82 = 3(40) - 2\bar{x}$$

$$2\bar{x} = 120 - 38.82$$

$$2\bar{x} = 81.18$$

$$\bar{x} = 81.18/2$$

$$\therefore \bar{x} = 40.59$$

Illustration 3

If $Z = 90$ and Median = 40 find the $\bar{x}$.

Solution

$$Z = 3 \text{ Median} - 2 \text{ Mean}$$

$$90 = 3(40) - 2\bar{x}$$

$$90 = 120 - 2\bar{x}$$

$$30 = 2\bar{x}$$

$$\therefore \bar{x} = \frac{30}{2} = 15$$

Illustration 4

If Median = 42 and $Z = 40$, find $\bar{x}$.

Solution

$$Z = 3 \text{ Median} - 2 \text{ Mean}$$

$$40 = 3(42) - 2\bar{x}$$

$$40 = 126 - 2\bar{x} \Rightarrow 40 - 126 = -$$

$$-86 = -2\bar{x}$$

$$\therefore \bar{x} = \frac{86}{2} = 43$$

Illustration 5

If $\bar{x} = 12, Z = 13$ find Median

Solution

$$Z = 3M - 2\bar{x}$$

$$13 = 3M - 2(12) = 3M - 24$$

$$3M = 13 + 24 = 37$$

$$\therefore \text{Median} = \frac{37}{3} = 12.33$$

Illustration 6

In a frequency distribution, if Mean = 24.6 and Mode = 26.1 find the value of median.

Solution

Given : Mode = 3 Median - 2 Mean
Mode = 26.1, Mean = 24.6,
Median = ?

$$26.1 = 3(M) - 2(24.6)$$

$$26.1 = 3M - 49.2$$

$$26.1 + 49.2 = 3M$$

$$75.3 = 3M$$

$$M = \frac{75.3}{3} = 25.1$$

$$\therefore \text{Median} = 25.1$$

Illustration 7

For a distribution if Z is 20.5 and second quartile is 22. Find mean.

Solution

Given : Mode = 20.5, Median = 22

Note : Second quartile is same as median

$$\text{Mode} = 3\text{Median} - 2\text{Mean}$$

$$20.5 = 3(22) - 2\text{Mean}$$

$$20.5 = 66 - 2\text{Mean}$$

$$2\text{Mean} = 66 - 20.5$$

$$2\text{Mean} = 45.5$$

$$\text{Mean} = \frac{45.5}{2}$$

$$\therefore \text{Mean} = 22.75$$

Problems on Mean, Median and Mode

Illustration 1

The following is the number of books issued by a college library on ten different days 92, 146, 88, 200, 99, 100, 75, 200, 95, 105. Calculate Mean, Median and mode.

Solution

$$\text{Mean} = \frac{\Sigma x}{n} = \frac{92 + 146 + 88 + 200 + 99 + 100 + 75 + 200 + 95 + 105}{10} = \frac{1200}{10} = 120$$

$$\text{Median} = \text{Size of } \left[\frac{n+1}{2} \right]^{\text{th}} \text{ item}$$

Sorted Array : 75, 88, 92, 95, 99, 100, 105, 146, 200, 200

$$\text{Median} = \text{size of } \left[\frac{10+1}{2} \right]^{\text{th}} \text{ item}$$

$$= \text{Size of } (5.5)^{\text{th}} \text{ item}$$

$$= 5^{\text{th}} \text{ term} + 0.5 (6^{\text{th}} \text{ term} - 5^{\text{th}} \text{ term})$$

$$= 99 + 0.5 (100 - 99) \Rightarrow 99 + 0.5$$

$$M = 99.5$$

Mode : Hence, the value 200 is repeated twice

$$\text{Mode} = 200$$

Illustration 2

Find Mean, Median and Mode

C.I	200-400	400-600	600-800	800-1000	1000-1200	1200-1400	1400-1600	1600-1800	1800-2000
f	6	9	11	14	20	15	10	8	7

Solution

Calculation of Mean, Median and Mode

C.I	f	M.V (x)	fx	cf
200 - 400	6	300	1800	6
400 - 600	9	500	4500	15
600 - 800	11	700	7700	26
800 - 1000	14 (f_0)	900	12,600	40
1000 - 1200	20 (f_1)	1,100	22,000	60
1200 - 1400	15 (f_2)	1,300	19,500	75
1400 - 1600	10	1,500	15,000	85
1600 - 1800	8	1,700	13,600	93
1800 - 2000	7	1,900	13,300	100
	N=100		$\Sigma fx = 1,10,000$	

$$\text{Mean} = \frac{\Sigma fx}{N} = \frac{110000}{100} = 1100$$

$$\text{Median} = \text{Size of } \left[\frac{N}{2} \right]^{\text{th}} \text{ item}$$

$$\text{Size of } \left[\frac{100}{2} \right]^{\text{th}} \text{ item}$$

Size of (50)th item50th item lies in the cf of 60**Median class = 1000 - 1200**

$$L = 1000, \frac{N}{2} = 50, pcf = 40, f = 20, i = 200$$

$$M = L + \left[\frac{\frac{N}{2} - pcf}{f} \right] \times i$$

$$= 1000 + \left[\frac{50 - 40}{20} \right] \times 200 \Rightarrow 1000 + \left[\frac{10}{20} \right] \times 200 \Rightarrow 1000 + 0.5(200)$$

$$1000 + 100 = 1100$$

Median = 1100**Mode** - Highest frequency is 20**Modal class = 1000 - 1200**

$$L = 1000, f_1 = 20, f_2 = 15, f_0 = 14, i = 200$$

$$Z = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times i$$

$$= 1000 + \left[\frac{20 - 14}{2(20) - 14 - 15} \right] \times 200 \Rightarrow 1000 + \left[\frac{6}{40 - 29} \right] \times 200 \Rightarrow 1000 + (0.5455)200$$

$$= 1000 + 109.09 = 1109.09$$

Mode = 1109.09

Illustration 3

Find mean, median & mode from the following data.

Profit (in lakhs)	4 - 7	8 - 11	12 - 15	16 - 19	20 - 23	24 - 27	28 - 31	32 - 35	36 - 39
No. of Companies	6	10	18	30	15	12	10	6	2

Solution

C.I	f	Mid Point (x)	fx	cf
3.5 - 7.5	6	5.5	33	6
7.5 - 11.5	10	9.5	95	16
11.5 - 15.5	18(f_0)	13.5	243	34
15.5 - 19.5	30(f_1)	17.5	525	64
19.5 - 23.5	15(f_2)	21.5	322.5	79
23.5 - 27.5	12	25.5	306	91
27.5 - 31.5	10	29.5	295	101
31.5 - 35.5	6	33.5	201	107
35.5 - 39.5	2	37.5	75	109
	N = 109		$\Sigma fx = 2095.5$	

$$\text{Mean} = \bar{X} = \frac{\Sigma fx}{N} = \frac{2095.5}{109} = 19.22$$

$$\begin{aligned} \text{Median} &= \frac{N}{2}^{\text{th}} \text{ value} \\ &= \left[\frac{109}{2} \right]^{\text{th}} \text{ value} = [54.5]^{\text{th}} \text{ value} \end{aligned}$$

$$\text{Here, } L = 15.5, cf = 34, f = 30, i = 4$$

$$\text{Median class} = 15.5 - 19.5$$

$$\begin{aligned} M &= L + \left[\frac{\frac{N}{2} - pcf}{f} \right] \times i = 15.5 + \left[\frac{54.5 - 34}{30} \right] \times 4 \\ &= 15.5 + \left[\frac{20.5}{30} \right] \times 4 = 15.5 + 2.733 = 18.233 \end{aligned}$$

Mode: Highest frequency = 30, Modal class = 15.5 - 19.5

$$\text{Here, } L = 15.5, f_1 = 30, f_0 = 18, f_2 = 15, i = 4$$

$$\begin{aligned} Z &= L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times i = 15.5 + \left[\frac{30 - 18}{2(30) - 18 - 15} \right] \times 4 \\ &= 15.5 + \left[\frac{12}{27} \right] \times 4 = 15.5 + 1.777 \end{aligned}$$

$$Z = 17.278$$

Illustration 4

Calculate mode using empirical relation for the given data

x	2	4	6	8	10	12
f	9	14	14	8	3	2

Solution

Given distribution is bimodal distribution Using empirical relation, we can find mode.

x	f	fx	c.f
2	9	18	9
4	14	56	9 + 14 = 23
6	14	84	23 + 14 = 37
8	8	64	37 + 8 = 45
10	3	30	45 + 3 = 48
12	2	24	48 + 2 = 50
N = 50		Σfx = 276	

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{276}{50} = 5.52$$

$$\text{Median} = \left[\frac{N+1}{2} \right]^{\text{th}} \text{ item} = \left[\frac{50+1}{2} \right]^{\text{th}} \text{ item} = 25.5^{\text{th}} \text{ item}$$

CF just greater than 25.5 is 37. Corresponding to 37, the value of x is 6. So, **Median = 6**

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

$$= 3(6) - 2(5.52) = 18 - 11.04 = 6.96$$

$$\therefore \text{Mode} = 6.96$$

Illustration 5

Calculate mode using empirical relation

CI	0 - 4	4 - 8	8 - 12	12 - 16	16 - 20
f	5	11	15	15	9

Solution

Given distribution is a bimodal distribution Using empirical relation we can find mode.

CI	f	x	fx	cf
0 - 4	5	2	10	5
4 - 8	11	6	66	5 + 11 = 16
8 - 12	15	10	150	16 + 15 = 31
12 - 16	15	14	210	31 + 15 = 46
16 - 20	9	18	162	46 + 9 = 55
	N = 55		Σfx = 598	

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{598}{55} = 10.872$$

$$\left[\frac{N}{2} \right]^{\text{th}} \text{ item} = \left[\frac{55}{2} \right]^{\text{th}} \text{ item} = 27.5^{\text{th}} \text{ item}$$

It lies in cf of 31 the median class is 8 - 12

$$\text{Median} = L + \left[\frac{\frac{N}{2} - \text{pcf}}{f} \right] \times i = 8 + \left[\frac{27.5 - 16}{15} \right] \times 4 = 8 + \left[\frac{11.5}{15} \right] \times 4$$

$$\text{Median} = 8 + 3.066 = 11.066$$

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

$$= 3(11.066) - 2(10.872) = 33.198 - 21.744 = 11.454$$

$$\text{Mode} = 11.454$$

2.6 Review**Short Answer Type**

1. Define central tendency.
2. What is a measure of central tendency?
3. State any two measures of central tendency.
4. Write any two measures of central tendency.
5. Define arithmetic mean.
6. Name the measures of central tendency.
7. Write the formula for arithmetic mean.
8. State any two measures of central tendency.
9. Mention two measures of central tendency.
10. Mention two measures of central tendency.
11. Define weighted arithmetic mean.
12. What is a measure of central tendency?
13. State two measures of central tendency.
14. Write two measures of central tendency.
15. Write two measures of central tendency.
16. Define mode.
17. Write two measures of central tendency.
18. Mention two measures of central tendency.
19. Mention two measures of central tendency.

Long Answer Type

1. Explain the measures of central tendency.
2. Explain the measures of central tendency.
3. Write the formula for arithmetic mean.
4. Explain the measures of central tendency.
5. Define arithmetic mean.
6. State the measures of central tendency.
7. Define mode.
8. Write the formula for arithmetic mean.
9. Define median.
10. Write the formula for arithmetic mean.
11. Explain the measures of central tendency.
12. Compute the measures of central tendency.

Wa
Number

2.6 Review Questions

Short Answer Type Questions

1. Define central tendency.
2. What is an average? Give an example.
3. State any two objectives of measures of central tendency.
4. Write any two requisites of an ideal average.
5. Define arithmetic mean.
6. Name the different types of averages.
7. Write the formula for arithmetic mean.
8. State any two properties of arithmetic mean.
9. Mention two merits of arithmetic mean.
10. Mention two demerits of arithmetic mean.
11. Define weighted arithmetic mean.
12. What is a median?
13. State two properties of median.
14. Write two merits of median.
15. Write two demerits of median.
16. Define mode.
17. Write two properties of mode.
18. Mention two merits of mode.
19. Mention two demerits of mode.

Long Answer Type Questions

1. Explain the meaning and definitions of central tendency.
2. Explain the objectives of measures of central tendency.
3. Write the requisites of an ideal average.
4. Explain the different types of averages.
5. Define arithmetic mean and explain its properties.
6. State the merits and demerits of arithmetic mean.
7. Define median and explain its properties.
8. Write the merits and demerits of median.
9. Define mode and explain its properties.
10. Write the merits and demerits of mode.
11. Explain the relationship between mean, median, and mode.
12. Compute mean wages of worker from the following data:

Wages	48-56	56-64	64-72	72-80	80-88	88-96	96-104
Number of Workers	8	3	11	14	5	7	2

13. Calculate mode from the following data:

X	0-9	10-19	20-29	30-39	40-49	50-59	60-69
f	678	720	664	598	524	378	244

14. From a continuous frequency table. The marks scored by 50 student sin an examination given below, taking class interval of 10-20, 20-30 etc.

Prepare frequency table and calculate Median:

48	30	31	39	18	54	33	10	29
62	38	41	43	51	37	71	62	34
55	29	43	64	43	52	64	44	
55	45	22	32	21	59	61	22	
74	19	46	73	33	85	85	51	
63	58	27	44	32	31	47	18	

15. Calculate Arithmetic Mean.

Marks	0 - 10	10-30	30-60	60-100
Students	7	13	22	8

16. Calculate Median from the following data:

Marks:	50	40	30	20	10
Frequency	10	40	20	12	16

17. Calculate Mean, Median and mode from the following data.

Central size	10	15	20	25	30	35	40	45
Frequency	10	25	40	50	25	17	8	5

18. Calculate mode and Median from the following data:

X	0-10	10-20	20-30	30-50	50-70	70-100
Y	5	10	17	40	62	60

19. Find Mean, Median and mode from the following data:

Profits (in lakhs)	4 - 7	8 - 11	12 - 15	16 - 9	20 - 23	24 - 27	28 - 31	32 - 35	36 - 39
No. of Companies	6	10	18	30	15	12	10	6	2

20. Calculate Arithmetic Mean.

Marks	0-10	10-30	30-60	60-100
Students	7	13	22	8

21. The weekly pocket money (₹) of 12 students is: 48, 60, 55, 52, 66, 70, 45, 58, 62, 49, 53, 57. Find the arithmetic mean.

22. Marks of 10 students are 31, 45, 37, 49, 33, 41, 28, 52, 47, 39. (i) Find the mean using A = 40 shortcut method. (ii) Find the mean using step-deviation method with $i = 2$.

23. The following table gives the marks obtained by students. Compute the mean.

X	5	10	15	20	25	30	35	40
f	4	6	9	12	10	5	3	1

24. Calculate the mean by shortcut method and step-deviation method taking $A = 25$ and $i = 5$.

X	5	10	15	20	25
f	6	10	18	14	12

25. The daily wages of 60 workers in a factory are given below. Find the arithmetic mean.

Wages (₹)	100-120	120-140	140-160	160-180	180-200	200-220	220-240
No. of Workers	5	9	12	15	10	6	3

26. Compute the mean using shortcut and step-deviation methods taking $A = 165$ and $i = 10$.

Class Interval	140-150	150-160	160-170	170-180	180-190
Frequency	8	14	20	12	6

27. Convert the cumulative frequencies into a simple frequency distribution and find the mean.

Marks	<20	<40	<60	<80	<100	<120
No. of Students	8	22	47	70	86	100

28. Convert the following cumulative frequency into a simple frequency distribution and calculate the mean.

Income (₹ '000)	>0	>10	>20	>30	>40	>50
No. of Families	120	100	76	52	28	10

29. Calculate the average from the following data (open-end classes).

Class Interval	Below 20	20-40	40-70	70-110	Above 110
Frequency	6	12	20	10	7

30. The mean of the following data is 42. Find the missing frequency.

Marks	10-20	20-30	30-40	40-50	50-60	60-70
Frequency	6	12	?	16	10	6

31. The mean of the following distribution is 35 and the total number of observations is 120. Find the missing frequencies a and b .

Class Interval	0-10	10-20	20-30	30-40	40-50
Frequency	10	a	45	b	15

32. The mean of 150 observations was found to be 62. Later it was found that two values were misread as 48 and 75 instead of 84 and 57. Find the correct mean.

32. The mean marks of 36 students of Section A is 58 and that of 24 students of Section B is 66. Find the combined mean.

33. The mean marks of 80 students are 54. The mean marks of 30 girls are 58. Find the mean marks of boys.

34. The mean weight of 7 students is 56 kg. The weights of six students are 62, 51, 55, 59, 49, and 58. Find the weight of the seventh student.

35. The marks of a student in five subjects are given below. Find the weighted mean when weights are given as shown.

Subject	English	Maths	Science	Social	Computer
Marks (x)	72	84	78	69	90
Weights(w)	2	3	3	2	4

36. The marks obtained by two candidates P and Q in four subjects are given below. The weights for the subjects are 1, 2, 3, and 4 respectively. Compute the weighted mean for each candidate and state who ranks higher.

Subject	A	B	C	D
Candidate P	68	70	61	64
Candidate Q	72	66	58	62

37. The group indices and weights of six groups are given below. Find the weighted average index.

Group	Food	Clothing	Fuel	Rent	Misc	Education
Index (x)	128	104	150	136	172	160
Weight (w)	4	2	3	3	2	2

38. Find the median of the following data: 14, 25, 19, 33, 41, 27, 22.

39. Find the median of the following data: 12, 15, 20, 25, 28, 31, 34, 40.

40. Find the median of the following distribution.

Marks	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	5	8	20	22	11	4

41. Convert the cumulative frequency into simple frequency and find the median.

Weight (kg)	<40	<50	<60	<70	<80	<90
Persons	6	20	44	70	88	100

42. Convert the following more than cumulative distribution into simple frequency and find the median.

Income (₹)	>0	>10,000	>20,000	>30,000	>40,000	>50,000
Families	180	150	110	70	35	10

43. Find the median of the following open-end class distribution.

Class Interval	Below 2000	2000-4000	4000-7000	7000-11000	Above 11000
Frequency	12	26	40	18	9

44. Find the median of the following data where mid-values are given.

Mid-value	15	25	35	45	55
Frequency	8	18	30	22	12

45. The median of the following distribution is 35 and the total number of observations is 100. Find the missing frequencies x and y.

Class Interval	10-20	20-30	30-40	40-50	50-60
Frequency	10	x	40	y	14

46. Find the mode of the following data: 21, 25, 19, 25, 18, 21, 25, 23, 19, 21.

47. Find the mode of the following data:

x	10	15	20	25	30	35	40	45	50
Frequency	10	15	20	25	30	35	40	45	50

48. Find the mode of the following data:

Marks	10	15	20	25	30	35	40	45	50
Frequency	10	15	20	25	30	35	40	45	50

49. Convert the following data into simple frequency distribution:

Sales (₹'000)	10	15	20	25	30	35	40	45	50
Shops	10	15	20	25	30	35	40	45	50

50. Convert the following data into simple frequency distribution:

Wage (₹)	10	15	20	25	30	35	40	45	50
Workers	10	15	20	25	30	35	40	45	50

51. The following is a frequency distribution. Find the median.

Class Interval	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	10	15	20	25	30	35

52. The following is a frequency distribution. Find the median to estimate the mode.

x	10	15	20	25	30	35	40	45	50
Frequency	10	15	20	25	30	35	40	45	50

53. In a distribution, the mean is 35 and the standard deviation is 10. Find the coefficient of variation.

54. In a distribution, the mean is 35 and the standard deviation is 10. Find the coefficient of variation.

55. In a distribution, the mean is 35 and the standard deviation is 10. Find the coefficient of variation.

47. Find the mode of the following distribution.

x	5	10	15	20	25
Frequency	7	12	18	14	9

48. Find the mode of the following distribution.

Marks	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	6	14	28	24	12	6

49. Convert the following cumulative frequencies into a simple distribution and find the mode.

Sales (₹'000)	<20	<40	<60	<80	<100
Shops	8	22	45	70	90

50. Convert the following more than cumulative distribution into simple frequencies and find the mode.

Wage (₹)	>0	>100	>200	>300	>400
Workers	200	170	120	70	20

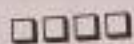
51. The following data is given in inclusive form. Convert into exclusive and calculate the mode.

Class Interval	20-29	30-39	40-49	50-59	60-69
Frequency	9	17	24	16	11

52. The following frequency distribution is bimodal. Use mean, median, and the empirical relation to estimate the mode.

x	2	4	6	8	10	12
Frequency	10	14	14	8	3	1

53. In a distribution, the mean is 42.8 and the median is 46. Find the mode using the empirical relation.
54. In a distribution, the mode is 38.5 and the median is 40.2. Find the mean using the empirical relation.
55. In a distribution, the mean is 24.6 and the mode is 26.1. Find the median using the empirical relation.



MEASURES OF DISPERSION

3.1 Introduction to Measures of Dispersion

A measure of central tendency is a single value that shows the central or typical value of a dataset. It gives a general idea of the whole data. But this single value is not enough to fully describe the data.

Sometimes, two or more sets of data can have the same average, but the way their values are spread out may be very different. On the other hand, some data sets may look similar in pattern, but their averages may be different. So, just knowing the average does not always give a complete picture of the data.

To understand how the values are spread out or scattered around the average, another measure is used. This is called the **measure of dispersion** or **measure of variation**.

Dispersion is the state of getting dispersed or spread. Statistical dispersion means the extent to which a numerical data is likely to vary about an average value. In other words, dispersion helps to understand the distribution of the data.

The following example will illustrate this view point.

Student A	6	6	6	6	6
Student B	4	6	9	7	5
Student C	3	8	10	4	11

All three students have the same average marks. But the marks of Student A are all the same. This means the marks are very consistent. In Student B and C, the marks go up and down. So, they are less consistent and more spread out. This shows that averages are not enough to describe the full nature of the data. That is why we also need to study dispersion to get a complete idea of the data.



Definitions : Dispersion

The concept of dispersion is related to the extent of scatter or variability in observations. The variability, in an observation, is often measured as its deviation from a central value. A suitable average of all such deviations is called the measure of dispersion.

1. According to A.L. Bowley, "Dispersion is the measure of the variation of the items".
2. According to B.C Brooks and W. F. L Dick, "Dispersion or Spread is the degree of the scatter or variation of the variables about a central value".
3. According to L.R. Connor, "Dispersion is the measure of extent to which individual items vary".

4. According to Simpson and Kafka, "The measure of the scatteredness of the mass of figures in a series about an average is called the measure of variation or dispersion".
5. According to Spiegel, "The degree to which numerical data tend to spread about an average value is called variation or dispersion of the data."



Did You Know?

1. **Measures of central tendency are known as the averages of first order.**
 - First order averages are those which give a single representative value of the entire data.
 - These values tell where the center or average of the data lies.
 - Examples include Mean, Median, and Mode.
 - They are called "first order" because they are the first step in understanding a dataset.
 - These measures help to get a quick idea of the general trend or middle point of the data.
2. **Measures of dispersion are known as the averages of second order.**
 - Second order averages describe how the data is spread out around the average.
 - They give information about the consistency, variation, or stability of the data.
 - Examples include Range, Quartile Deviation, Mean Deviation, and Standard Deviation.
 - They are called "second order" because they come after the first step of finding the average.
 - These measures help to know how much the data values differ from the central tendency.

3.1.1 Objectives or Significance of the Measures of Dispersion

The main objectives of studying dispersion are

1. **To Find out the Reliability of an Average:** A measure of dispersion helps to know how dependable or trustworthy the average is. If the dispersion is low, it means the values are close to each other. This shows a higher level of similarity in the data. So, the average can be taken as a good and reliable representative of the data.
2. **To Control the Variation of the Data from the Central Value :** One key purpose of dispersion is to find out whether the data values are close to the average or scattered far apart. It helps to know if the observations are uniform or not.
3. **To Compare two or More Sets of Data Regarding their Variability:** Dispersion helps to compare how much two or more datasets vary. If one set has lower dispersion, it means the data is more consistent or stable than the other.
4. **To Find other Statistical Measures for further Analysis of Data :** Measures of dispersion are also useful in doing further statistical analysis like correlation, regression, confidence intervals, control charts, and test statistics. These methods need dispersion to give accurate results.

3.1.2 Characteristics for an Ideal Measures of Dispersion

1. **It should be Clearly and Properly Defined**

The measure must have a fixed meaning and should not change from person to person.

2. It should be Simple to Understand and Easy to Calculate

The method should be easy enough for students, teachers, and analysts to use without confusion.

3. It should be based on all Values in the Dataset

A good measure must take into account every observation to give a complete idea of variation.

4. It should allow further Mathematical Use

The measure should be suitable for use in other statistical formulas like standard deviation, regression, and correlation.

5. It should not change much with Small Changes in the Sample

A small change in the data should not affect the value of dispersion too much.

6. It should not be Greatly Influenced by Extreme Values

Outliers or very large/small values should not affect the result too much.

7. It should help in Comparison

A good measure of dispersion should be useful to compare the variability between two or more datasets.

8. It should be Reliable in both Large and Small Datasets

The method should give meaningful results whether the number of observations is few or many.

3.1.3 Types of Measures of Dispersion

There are two main types of dispersion methods in statistics which are:

1. **Absolute Measure of Dispersion** : These are the measures of dispersion that are shown in the original units of the data. For example, if the data is in rupees, the dispersion will also be in rupees. These are useful to understand how values vary in a single dataset. But they cannot be used to compare two datasets that have different units (like marks and weight).
2. **Relative Measure of Dispersion** : These measures are shown in the form of ratios or percentages, so they do not depend on units. They are useful when comparing two or more datasets with different units or scales. Since they are pure numbers, they give a better idea of which dataset has more or less variation.

3.1.4 Types of Absolute Measures of Dispersion

The various types of absolute measures of dispersion are:

1. Range
2. Quartile Deviation
3. Mean Deviation
4. Standard Deviation

3.1.5 Types of Relative Measures of Variation

The various types of relative measures of dispersion are

1. Coefficient of Range
2. Coefficient of Quartile Deviation
3. Coefficient of Mean Deviation
4. Coefficient of Standard Deviation

3.2 Range and Coefficient of Range

1. **Range :** Range is the simplest measure of dispersion. It shows how spread out the values are by looking at just two numbers — the highest and the lowest values in the data. Range is denoted as 'R'.

Range = Highest value (H.V) - Lowest value (L.V)

$$R = HV - LV \quad \text{or} \quad R = L - S$$

Where L (or) H.V = Largest value or Highest Value, S (or) L.V = Smallest value or Lowest Value.

2. **Coefficient of Range :** It is a relative measure and is calculated to compare the range between different datasets. Formula is usually given as:

$$= \frac{H.V - L.V}{H.V + L.V} \quad \text{or} \quad \frac{L - S}{L + S}$$

Points to Remember :

- Range is easy to calculate and simple to understand.
- But it is not very reliable because it only uses the highest and lowest values.
- It does not consider all other values in the data.
- It is easily affected by extreme values (very high or very low).
- Range can change a lot from one sample to another, even if most values are similar.
- In open-ended classes, range cannot be calculated, as there is no clear highest or lowest value.

3.2.1 Inter Quartile Range

Interquartile Range (IQR) is a way to measure how spread out the middle values of a data set are. It shows the range where the middle 50% of the values lie. It gives an idea of how tightly or loosely the central values are grouped.

Understanding Quartiles

When the data is arranged in ascending order, it is divided into four equal parts using three quartile points:

1. Q_1 (First Quartile): Value below which 25% of the data falls
2. Q_2 (Second Quartile or Median): Value below which 50% of the data falls
3. Q_3 (Third Quartile): Value below which 75% of the data falls
4. Q_4 is not a separate value but simply refers to the highest part of the data, from Q_3 to the maximum value. So, Q_4 covers the last 25% of the data.

The three quartile values Q_1 , Q_2 , and Q_3 divide the data into four parts, each part having 25% of the total data. The quartiles are then the highest values in each of these four parts, and the interquartile range is the difference between the values of the first and third quartiles:

$$\text{Inter Quartile Range} = Q_3 - Q_1$$

The below figure shows

The space between the Range (IQR). The

3.2.2 Quartile Deviation

The quartile deviation is known as semi-interquartile range. It is known as 'Quartile Deviation'.

Quartile

Coefficient

3.3 Mean

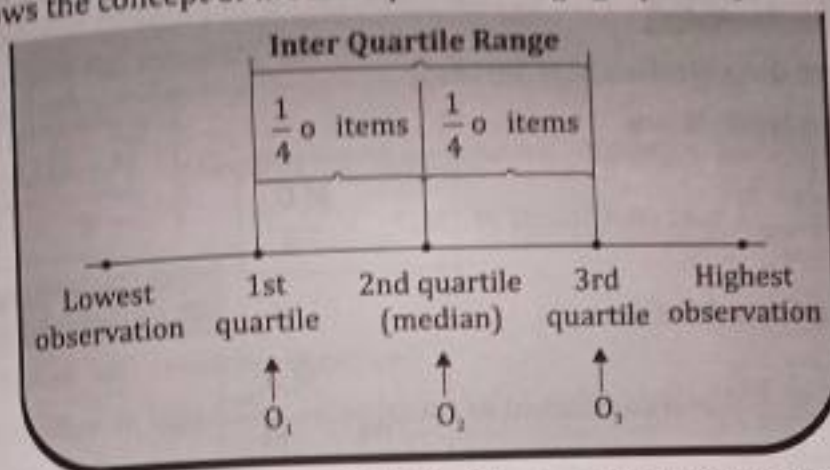
As already discussed, the mean is calculated on all values in the data. To solve this, we take into account all the values, mean, median,



Mean

- **Mean** is the average of all the values. It shows the central tendency. Only the values are taken into account.
- **Mean** is calculated by adding all the values and dividing by the number of values (mean).
- In the case of grouped data, the mean is calculated by using the formula: $\text{Mean} = \frac{\sum f x}{\sum f}$

The below figure shows the concept of the interquartile range graphically.



The space between Q_1 and Q_3 shows the middle 50% of the data. This distance is called the Interquartile Range (IQR). The last section, from Q_3 to the highest value, is the fourth part (Q_4) of the data.

3.3.2

Quartile Deviation or Semi Inter Quartile Range

The quartile deviation is half of the distance between the third and the first quartile and hence is also known as semi inter quartile range. The distance between first and third quartile divided by two is known as 'Quartile Deviation'. Thus

$$\text{Quartile Deviation} = \frac{Q_3 - Q_1}{2}$$

$$\text{Coefficient of Quartile deviation} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

3.3 Mean Deviation / Average Deviation

As already discussed, the two methods of dispersion - Range and Quartile Deviation - are not based on all values in the data. They also do not clearly show how much the values differ from an average. To solve this, we use Mean Deviation, also known as Average Deviation. This method takes into account all the values in the dataset and measures how far they are from a central value like the mean, median, or mode.



Meaning and Definition : Mean Deviation

- **Mean Deviation** is the average of the differences between each observation and a central value. It shows how spread out the values are, by calculating how much each value differs from the average. Only the size of the difference is used — the sign (positive or negative) is ignored.
- Mean deviation is the arithmetic mean of the absolute deviations of the data values from a central value (mean, median, or mode). It is a true measure of dispersion because:
 - It uses all the observations in the dataset
 - It shows how much each value varies from the average
 - It gives a better idea of the overall scatteredness in the distribution
- In short, mean deviation is a more complete and accurate measure of variation compared to range and quartile deviation.

Calculation of Mean Deviation

Ungrouped Data [Raw data (Individual Series)]

1. Mean Deviation from Mean

$$M.D(\bar{x}) = \frac{\sum |x - \bar{x}|}{n} \text{ and coefficient of } M.D(\bar{x}) = \frac{M.D(\bar{x})}{\bar{x}}$$

2. Mean Deviation from Median

$$M.D(M) = \frac{\sum |x - M|}{n} \text{ and coefficient of } M.D(M) = \frac{M.D(M)}{M}$$

3. Mean Deviation from Mode

$$M.D(Z) = \frac{\sum |x - Z|}{n} \text{ and coefficient of } M.D(Z) = \frac{M.D(Z)}{Z}$$

Grouped Data

1. Mean Deviation from Mean

$$M.D(\bar{x}) = \frac{\sum f|x - \bar{x}|}{N} \text{ and Coefficient of } M.D(\bar{x}) = \frac{MD \text{ from Mean}}{\text{Mean}}$$

2. Mean Deviation from Median

$$M.D(M) = \frac{\sum f|x - M|}{N} \text{ and Coefficient of } M.D(M) = \frac{MD \text{ from Median}}{\text{Median}}$$

3. Mean Deviation from Mode

$$M.D(Z) = \frac{\sum f|x - Z|}{N} \text{ and coefficient of } M.D(Z) = \frac{MD \text{ from Mode}}{\text{Mode}}$$

3.4 Standard Deviation

Standard Deviation (SD) is one of the most widely used and accurate measures of dispersion. It is especially useful in research, experiments, and statistical analysis, where consistency and precision are important.



Meaning of Standard Deviation

Standard deviation shows how much the values in a dataset differ from the mean (average). It is found by taking the square root of the average of the squared deviations from the mean.

Unlike mean deviation, which ignores the signs (positive or negative), standard deviation squares all deviations to avoid negative values. Also, standard deviation is always calculated from the mean, not from the median or mode.

Standard deviation, usually denoted by σ (small sigma) of the Greek alphabet. If $x_1, x_2, x_3, \dots, x_n$ is a set of n observations, then the standard deviation is given by :

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

$$\text{Coefficient of Standard Deviation} = \frac{\sigma}{\bar{x}} = \frac{\text{Standard Deviation}}{\text{Mean}}$$

Properties of Standard Deviation

1. Standard deviation will never be negative
2. If the same number is added to or subtracted from every value in the data, the standard deviation does not change.
3. If each value in the dataset is multiplied or divided by the same number, the standard deviation also changes in the same way. For example, if every value is multiplied by 2, SD is also multiplied by 2.
4. Among all possible values used as a central point, the arithmetic mean gives the least value of SD. That is why SD is always taken from the mean.
5. If all values of the series are the same then the standard deviation is zero (coz each value is equal to the Mean).

Calculation of Standard Deviation

Un-grouped Data

Direct Method

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

$$\sigma = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

Grouped Data

(a) Direct method

$$\sigma = \sqrt{\frac{\sum f(x - \bar{x})^2}{N}} \quad \sigma = \sqrt{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2}$$

(b) Indirect Method

$$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2}$$



Note

Variance = σ^2 = Square of Standard deviation.

Standard Deviation = $\sqrt{\text{variance}}$



Merits of Standard Deviation

- It is based on all the observations.
- It is well defined by mathematical formula.
- It is capable of further mathematical treatment.
- It possesses sampling stability.



Demerits of Standard Deviation

- It is difficult to calculate.
- It is affected by abnormal extreme values.

3.4.1 Coefficient of Variation

Standard deviation is considered the most accurate and reliable measure of dispersion. To compare the homogeneity (similarity) or heterogeneity (variation) between two or more data sets, we use the coefficient of standard deviation. But for better comparison, especially when the data are in different units or scales, we use the Coefficient of Variation (CV).

$$CV = \frac{\sigma}{\bar{x}} \times 100$$

Where, σ = Standard Deviation and $\bar{x}$ = Mean



Definition : Coefficient of Variation

According to Karl Pearson, "Coefficient of Variation is the percentage variation in the mean. Standard deviation is considered as the total variation in the mean."

This means that the coefficient of variation (CV) tells how much the data is varying as a percentage of mean. It shows the relative spread of the data, making it useful for comparing different datasets.



Note

- CV is expressed as a percentage.
- It is a relative measure, so it does not depend on units.
- Lower CV means more consistency or homogeneity in the data.
- Higher CV means more variation or heterogeneity.
- So, when comparing two or more distributions, the one with the lower CV is considered more uniform or stable.

Problems on Individual Series

Illustration 1

Values: 20, 28, 40, 12, 30, 15, 50

Calculate the following for Raw Data

- Range
- Co-efficient of range
- Inter Quartile range
- Quartile deviation/Semi Inter Quartile Range
- Co-efficient of Quartile deviation
- Mean deviation from mean & its co-efficient
- Mean Deviation from median.
- Standard deviation under direct method.
- Standard deviation under assumed mean method.

- j. Standard deviation from deviation from actual mean.
- k. Co-efficient of variation.

Solution**Calculation of range & its coefficient**

Sorted Array : 12, 15, 20, 28, 30, 40, 50

a. Range [R] = HV - LV (Highest value - Lowest value)
 $= 50 - 12 = 38$

b. Coefficient of Range

$$= \frac{H.V - L.V}{H.V + L.V} = \frac{50 - 12}{50 + 12} = \frac{38}{62} = 0.6129$$

c. Calculation of Inter Quartile Range, QD & Coefficient of QD**Calculation of Q_1 [Lower Quartile]**

$$\begin{aligned} Q_1 &= \left[\frac{n+1}{4} \right]^{\text{th}} \text{ term in array} \\ &= \left[\frac{7+1}{4} \right]^{\text{th}} \text{ term in array} \\ &= \left[\frac{8}{4} \right]^{\text{th}} \text{ term in array} = 2^{\text{nd}} \text{ term in array} \end{aligned}$$

$$Q_1 = 15$$

Calculation of Q_3

$$\begin{aligned} Q_3 &= 3 \left[\frac{n+1}{4} \right]^{\text{th}} \text{ term in array} \\ &= 3[2]^{\text{th}} \text{ term in array} = 6^{\text{th}} \text{ term in array} \end{aligned}$$

$$Q_3 = 40$$

Inter Quartile range

$$= Q_3 - Q_1 = 40 - 15 = 25$$

d. Calculation of Quartile Deviation

$$\begin{aligned} \text{Quartile Deviation} &= \frac{Q_3 - Q_1}{2} \\ &= \frac{40 - 15}{2} = \frac{25}{2} = 12.5 \end{aligned}$$

e. Calculation of Coefficient of QD

$$\begin{aligned}\text{Coefficient of Quartile deviation} &= \frac{Q_3 - Q_1}{Q_3 + Q_1} \\ &= \frac{40 - 15}{40 + 15} = \frac{25}{55} = 0.4545\end{aligned}$$

f. Calculation of Mean deviation from mean & its co-efficient

Values (x)	$ x - \bar{x} $
20	7.85
28	0.15
40	12.15
12	15.85
30	2.15
15	12.85
50	22.15
$\Sigma x = 195$	$\Sigma x - \bar{x} = 73.15$

$$\begin{aligned}\bar{x} &= \frac{\Sigma x}{n} \\ &= \frac{195}{7} \\ \bar{x} &= 27.85\end{aligned}$$

$$\begin{aligned}\text{M.D}(\bar{x}) &= \frac{\Sigma |x - \bar{x}|}{n} \\ &= \frac{73.15}{7} = 10.45\end{aligned}$$

$$\begin{aligned}\text{Co-efficient of MD from Mean} &= \frac{\text{MD from mean}}{\text{mean}} \\ &= \frac{10.45}{27.85} = 0.3752\end{aligned}$$

g. Calculation of mean deviation from median & its co-efficient

Array : 12, 15, 20, 28, 30, 40, 50

$$\begin{aligned}M &= \left[\frac{N+1}{2} \right]^{\text{th}} \text{ term in array} \\ &= \left[\frac{7+1}{2} \right]^{\text{th}} \text{ term in array} = 4^{\text{th}} \text{ term in array}\end{aligned}$$

$$M = 28$$

Calculation of Mean Deviation from Median

x	x - M
20	8
28	0
40	12
12	16
30	2
15	13
50	22
$\Sigma x = 195$	$\Sigma x - M = 73$

$$\begin{aligned}\text{MD from Median} &= \frac{\Sigma |x - M|}{n} \\ &= \frac{73}{7} = 10.42\end{aligned}$$

$$\begin{aligned}\text{Co-efficient of MD from median} &= \frac{\text{MD from median}}{\text{median}} \\ &= \frac{10.42}{28} = 0.3721\end{aligned}$$

b. Calculation of Standard Deviation under direct method

x	x^2
20	400
28	784
40	1600
12	144
30	900
15	225
50	2500
$\Sigma x = 195$	$\Sigma x^2 = 6553$

$$\begin{aligned}\sigma &= \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2} = \sqrt{\frac{6553}{7} - \left(\frac{195}{7}\right)^2} \\ &= \sqrt{936.14 - (27.85)^2} = \sqrt{936.14 - 775.62} \\ \sigma &= \sqrt{160.52} = 12.65\end{aligned}$$

i. Calculation of Standard Deviation under assumed mean method

x	A = 40 d = x - A	d^2
20	-20	400
28	-12	144
40	0	0
12	-28	784
30	-10	100
15	-25	625
50	10	100
	$\Sigma d = -85$	$\Sigma d^2 = 2153$

$$\begin{aligned}\sigma &= \sqrt{\frac{\Sigma d^2}{n} - \left(\frac{\Sigma d}{n}\right)^2} = \sqrt{\frac{2153}{7} - \left(\frac{-85}{7}\right)^2} \\ &= \sqrt{307.57 - [-12.14]^2} = \sqrt{307.57 - 147.37} \\ \sigma &= \sqrt{160.2} = 12.66\end{aligned}$$

j. Calculation of Standard Deviation under deviation from actual mean method

x	$x - \bar{x}$	$(x - \bar{x})^2$
20	-7.85	61.6225
28	0.15	0.0225
40	12.15	147.6225
12	-15.85	251.2225
30	2.15	4.6225
15	-12.85	165.1225
50	22.15	490.6225
$\Sigma x = 195$		$\Sigma(x - \bar{x})^2 = 1120.8575$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{195}{7} = \bar{x} = 27.85$$

$$\sigma = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n}} = \sqrt{\frac{1120.8575}{7}}$$

$$\sigma = \sqrt{160.1225} = 12.66$$

k. Calculation of co-efficient of variation

Co-efficient of variation

$$\begin{aligned} &= \frac{\sigma}{\bar{x}} \times 100 = \frac{12.66}{27.85} \times 100 \\ &= 0.4545 \times 100 = 45.45\% \end{aligned}$$

Illustration 2

Marks - 35, 38, 40, 43, 45, 46, 50, 55

Calculate the following for Raw Data

- Range
- Co-efficient of range
- Inter Quartile range
- Quartile deviation/Semi Inter Quartile Range
- Co-efficient of Quartile deviation
- Mean deviation from mean & its co-efficient
- Mean Deviation from median.
- Standard deviation under direct method.
- Standard deviation under assumed mean method.
- Standard deviation from actual mean.
- Co-efficient of variation.

Solution

Calculation of range & its coefficient

Sorted Array: 35, 38, 40, 43, 45, 46, 50, 55

$$\begin{aligned} \text{a. Range} &= L - S \text{ (Large value - Small value)} \\ &= 55 - 35 = 20 \end{aligned}$$

b. Co-efficient of range

$$= \frac{L-S}{L+S} = \frac{55-35}{55+35} = \frac{20}{90} = 0.23$$

c. Calculation of inter-quartile range, QD & its co-efficient

$$Q_1 = \left[\frac{n+1}{4} \right]^{\text{th}} \text{ term in array}$$

$$= \left[\frac{8+1}{4} \right]^{\text{th}} \text{ term in array}$$

$$= \left[\frac{9}{4} \right]^{\text{th}} \text{ term in array} = (2.25)^{\text{th}} \text{ term in array}$$

$$= 2^{\text{nd}} \text{ term} + 0.25 (3^{\text{rd}} \text{ term} - 2^{\text{nd}} \text{ term})$$

$$= 38 + 0.25 (40 - 38) = 38 + 0.25 (2) = 38 + 0.5 = 38.5$$

$$Q_3 = 3 \left[\frac{n+1}{4} \right]^{\text{th}} \text{ term in array} = 3 (2.25)^{\text{th}} \text{ term in array} = (6.75)^{\text{th}} \text{ term in array}$$

$$= 6^{\text{th}} \text{ term} + 0.75 (7^{\text{th}} \text{ term} - 6^{\text{th}} \text{ term})$$

$$= 46 + 0.75 (50 - 46) = 46 + 0.75 (4) = 46 + 3 = 49$$

$$\text{Inter Quartile range} = Q_3 - Q_1 = 49 - 38.5 = 10.5$$

$$\text{d. Quartile Deviation} = \frac{Q_3 - Q_1}{2} = \frac{49 - 38.5}{2} = \frac{10.5}{2} = 5.25$$

$$\text{e. Co-efficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{49 - 38.5}{49 + 38.5} = \frac{10.5}{87.5} = 0.12$$

f. Calculation of Mean Deviation from mean & its co-efficient

x	$ x - \bar{x} $
35	9
38	6
40	4
43	1
45	1
46	2
50	6
55	11
$\Sigma x = 352$	$\Sigma x - \bar{x} = 40$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{352}{8} \Rightarrow \bar{x} = 44$$

$$\text{MD from Mean} = \frac{\Sigma |x - \bar{x}|}{n} = \frac{40}{8} = 5$$

Co-efficient of MD from mean

$$= \frac{\text{MD from mean}}{\text{mean}} = \frac{5}{44} = 0.1136$$

g. Calculation of MD from median & its co-efficient

$$M = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ term in array} = \left(\frac{8+1}{2} \right)^{\text{th}} \text{ term in array}$$

$$= \left(\frac{9}{2} \right)^{\text{th}} \text{ term in array}$$

$$= 4.5^{\text{th}} \text{ term in array}$$

$$= 4^{\text{th}} \text{ term} + 0.5 (5^{\text{th}} \text{ term} - 4^{\text{th}} \text{ term})$$

$$= 43 + 0.5 (45 - 43) = 43 + 0.5(2) = 43 + 1$$

$$M = 44$$

x	x - M
35	9
38	6
40	4
43	1
45	1
46	2
50	6
55	11
$\Sigma x = 352$	$\Sigma x - M = 40$

$$\text{MD from median} = \frac{\Sigma |x - M|}{n} = \frac{40}{8} = 5$$

Co-efficient of MD from median

$$\frac{\text{MD from Median}}{\text{Median}} = \frac{5}{44} = 0.1136$$

h. Calculation of SD under direct method

x	x^2
35	1225
38	1444
40	1600
43	1849
45	2025
46	2116
50	2500
55	3025
$\Sigma x = 352$	$\Sigma x^2 = 15784$

$$\sigma = \sqrt{\frac{\Sigma x^2}{n} - \left[\frac{\Sigma x}{n} \right]^2} = \sqrt{\frac{15784}{8} - \left[\frac{352}{8} \right]^2}$$

$$= \sqrt{1973 - (44)^2} = \sqrt{1973 - 1936}$$

$$\sigma = \sqrt{37} = 6.0827$$

i. Calculation of SD

x	A = 45 d = x - A
35	-10
38	-7
40	-5
43	-2
45	0
46	1
50	5
55	10
	$\Sigma d = -8$

j. Calculation of SD

x	(x - $\bar{x}$)
35	-9
38	-6
40	-4
43	-1
45	1
46	2
50	6
55	11

k. Calculation of SD

$$= \frac{\sigma}{x} \times 100 =$$

Problems on Discussion

Illustration 1

Calculation of measures of central tendency

- Range
- Co-efficient of range
- Inter Quartile range
- Quartile Deviation
- Mean Deviation

i. Calculation of SD under deviation from assumed mean:

x	A = 45 d = x - A	Σd^2
35	-10	100
38	-7	49
40	-5	25
43	-2	4
45	0	0
46	1	1
50	5	25
55	10	100
$\Sigma d = -8$		$\Sigma d^2 = 304$

$$\sigma = \sqrt{\frac{\Sigma d^2}{n} - \left[\frac{\Sigma d}{n}\right]^2} = \sqrt{\frac{304}{8} - \left[\frac{-8}{8}\right]^2}$$

$$= \sqrt{38 - (-1)^2} = \sqrt{38 - 1}$$

$$\sigma = \sqrt{37} = 6.0827$$

j. Calculation of SD under actual mean

x	$(x - \bar{x})$	$(x - \bar{x})^2$
35	-9	81
38	-6	36
40	-4	16
43	-1	1
45	1	1
46	2	4
50	6	36
55	11	121
		$\Sigma (x - \bar{x})^2 = 296$

when

$\bar{x} = 44$ (previously calculated)

$$\sigma = \sqrt{\frac{\Sigma (x - \bar{x})^2}{n}} = \sqrt{\frac{296}{8}} = \sqrt{37}$$

$$\sigma = 6.0827$$

k. Calculation of co-efficient of variation

$$= \frac{\sigma}{\bar{x}} \times 100 = \frac{6.0827}{44} \times 100 = 0.1382 \times 100 = 13.82\%$$

Problems on Discrete Series

Illustration 1

Calculation of measures of dispersion for discrete data:

- Range
- Co-efficient of range
- Inter Quartile range
- Quartile Deviation & its coefficient
- Mean Deviation from mean & its coefficient

- f. Mean Deviation from median & its coefficient
- g. Standard Deviation from direct method
- h. Standard Deviation from deviation from assumed mean
- i. Standard Deviation from deviation from actual mean
- j. Coefficient of variation

Marks	10	20	30	40	50	60	70
No. of Students	5	11	19	22	15	6	2

Solution**a. Calculation of range**

Marks	f
10	5
20	11
30	19
40	22
50	15
60	6
70	2

$$\text{Range} = L - S = 70 - 10$$

$$\text{Range} = 60$$

b. Calculation of Co-efficient of range

$$= \frac{L-S}{L+S} = \frac{70-10}{70+10} = \frac{60}{80} = 0.75$$

c. Calculation inter Quartile range QD & its co-efficient

Calculation of Q_1 (lower Quartile) & Q_3 upper quartile

x	f	cf
10	5	5
20	11	5+11=16
30	19	16+19=35
40	22	35+22=57
50	15	57+15=72
60	6	72+6=78
70	2	78+2=80

$$Q_1 = \left[\frac{N+1}{4} \right]^{\text{th}} \text{ term}$$

$$= \left[\frac{80+1}{4} \right]^{\text{th}} \text{ term}$$

$$= 20.25^{\text{th}} \text{ term}$$

$$\text{The cf} > 20.25 = 35$$

$\therefore$ Corresponding class is

$$Q_1 = 30$$

$$Q_3 = 3 \left[\frac{N+1}{4} \right]^{\text{th}} \text{ term}$$

$$= 3 \left[\frac{80+1}{4} \right]^{\text{th}} \text{ term}$$

$$= 3[20.25]^{\text{th}} \text{ term}$$

$$= 60.75^{\text{th}} \text{ term}$$

$$\text{The cf} > 60.75 = 72$$

$\therefore$ Corresponding class is $Q_3 = 60$

d. Inter quartile range

$$= Q_3 - Q_1 = 50 - 30 = 20$$

$$QD = \frac{Q_3 - Q_1}{2} = \frac{50 - 30}{2} = \frac{20}{2} = 10$$

Co-efficient of QD

$$= \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{50 - 30}{50 + 30}$$

$$= \frac{20}{80} = 0.25$$

e. Calculation of Mean Deviation from mean & its co-efficient:

x	f	fx	$ x - \bar{x} $	$f x - \bar{x} $
10	5	50	27.125	135.625
20	11	220	17.125	188.375
30	19	570	7.125	135.375
40	22	880	2.875	63.25
50	15	750	12.875	193.125
60	6	360	22.875	137.25
70	2	140	32.875	65.75
N = 80		$\Sigma fx = 2970$		$\Sigma f x - \bar{x} = 918.75$

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{2970}{80}$$

$$= \bar{x} = 37.125$$

MD from mean

$$= \frac{\Sigma f|x - \bar{x}|}{N} = \frac{918.75}{80} = 11.4843$$

Co-efficient of MD from mean

$$= \frac{\text{MD from mean}}{\text{mean}}$$

$$= \frac{11.48}{37.125} = 0.3092$$

f. Calculation of Mean Deviation from Median & its co-efficient.

x	f	cf	$ x - M $	$f x - M $
10	5	5	30	150
20	11	16	20	220
30	19	35	10	190
40	22	57	0	0
50	15	72	10	150
60	6	78	20	120
70	2	80	30	60
N = 80				$\Sigma f x - M = 890$

$$M = \left[\frac{N+1}{2} \right]^{\text{th}} \text{ term}$$

$$= \left[\frac{80+1}{2} \right]^{\text{th}} \text{ term}$$

$$= \left[\frac{81}{2} \right]^{\text{th}} \text{ term in cf} = 40.5^{\text{th}} \text{ term}$$

$$M = 40$$

Calculation of MD from Median

$$\begin{aligned} \text{MD from Median} &= \frac{\Sigma f|x - M|}{N} \\ &= \frac{890}{80} = 11.125 \end{aligned}$$

Co-efficient of MD from Median

$$= \frac{\text{MD from Median}}{\text{Median}}$$

$$= \frac{11.125}{40} = 0.2781$$

x	f	fx	x ²	fx ²
10	5	50	100	500
20	11	220	400	4400
30	19	570	900	17100
40	22	880	1600	35200
50	15	750	2500	37500
60	6	360	3600	21600
70	2	140	4900	9800
N = 80		Σfx = 2970		Σfx ² = 126100

$$\begin{aligned}
 \sigma &= \sqrt{\frac{\Sigma fx^2}{N} - \left(\frac{\Sigma fx}{N}\right)^2} \\
 &= \sqrt{\frac{126100}{80} - \left(\frac{2970}{80}\right)^2} \\
 &= \sqrt{1576.25 - (37.125)^2} \\
 &= \sqrt{1576.25 - 1378.2656} \\
 &= \sqrt{197.9844} = 14.07
 \end{aligned}$$

h. Calculation of Standard Deviation under Assumed Mean

x	f	A = 40 d = x - A	fd	d ²	fd ²
10	5	-30	-150	900	4500
20	11	-20	-220	400	4400
30	19	-10	-190	100	1900
40	22	0	0	0	0
50	15	10	150	100	1500
60	6	20	120	400	2400
70	2	30	60	900	1800
N = 80			Σfd = -230		Σfd ² = 16500

$$\begin{aligned}
 \sigma &= \sqrt{\frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2} \\
 &= \sqrt{\frac{16500}{80} - \left(\frac{-230}{80}\right)^2} \\
 &= \sqrt{206.25 - (2.875)^2} \\
 &= \sqrt{206.265 - 8.2656} \\
 &= \sqrt{197.9843} \\
 \sigma &= 14.07
 \end{aligned}$$

i. Calculation of Standard Deviation under deviation from Actual Mean

x	f	fx	(x - $\bar{x}$)	(x - $\bar{x}$) ²	f(x - $\bar{x}$) ²
10	5	50	-27.125	735.7656	3678.828
20	11	220	-17.125	293.2656	3225.9216
30	19	570	-7.125	50.7656	964.5465
40	22	880	2.875	8.2656	181.8432
50	15	750	12.875	165.7656	2486.484
60	6	360	22.875	523.2656	3139.5036
70	2	140	32.875	1080.7656	2161.5312
N = 80		Σfx = 2970			Σf(x - $\bar{x}$) ² = 15,838.6581

$$\begin{aligned}
 \bar{x} &= \frac{\Sigma fx}{N} = \frac{2970}{80} \\
 \bar{x} &= 37.125 \\
 \sigma &= \sqrt{\frac{\Sigma f(x - \bar{x})^2}{N}} \\
 &= \sqrt{\frac{15838.6581}{80}} \\
 &= \sqrt{197.9822} \\
 \sigma &= 14.07
 \end{aligned}$$

i. Calculation of co-efficient of variation

Co-efficient of variation

$$= \frac{\sigma}{x} \times 100 = \frac{14.07}{37.125} \times 100$$

$$= 0.3789 \times 100 = 37.89\%$$

Illustration 2

Calculation of measures of dispersion for Discrete Series

x	2	3	4	5	6	7	8	9
f	6	3	9	11	7	14	12	4

Solution

a. Calculation of Range

x	f
2	6
3	3
4	9
5	11
6	7
7	14
8	12
9	4

$$\text{Range} = L - S$$

$$= 9 - 2$$

$$\text{Range} = 7$$

b. Calculation of Co-efficient of Range

$$= \frac{L - S}{L + S} = \frac{9 - 2}{9 + 2} = \frac{7}{12} = 0.6363$$

c. Calculation of Inter Quartile Range, QD & its Co-efficient

x	f	cf
2	6	6
4	3	6+3=9
4	9	9+9=18
6	11	18+11=29
6	7	29+7=36
7	14	36+14=50
8	12	50+12=62
9	4	62+4=66
	N = 66	

$$Q_1 = \left[\frac{N+1}{4} \right]^{\text{th}} \text{ term}$$

$$= \left[\frac{66+1}{4} \right]^{\text{th}} \text{ term}$$

$$= \left[\frac{67}{4} \right]^{\text{th}} \text{ term}$$

$$Q_1 = (16.75)^{\text{th}} \text{ term}$$

The cf > 16.75 = 18

∴ Corresponding class is $Q_1 = 4$

$$Q_3 = 3 \left[\frac{N+1}{4} \right]^{\text{th}} \text{ term} \\ = 3 \left[\frac{66+1}{4} \right]^{\text{th}} \text{ term} = 3(16.75)^{\text{th}} \text{ term}$$

(50.25)th term

The c.f. 50.25 = 62

 $\therefore$ Corresponding class is $Q_3 = 8$ **d. Inter Quartile Range**

$$Q_3 - Q_1 = 8 - 4 = 4$$

$$\text{Quartile Deviation} = \frac{Q_3 - Q_1}{2} = \frac{8 - 4}{2} = \frac{4}{2} = 2$$

$$\text{Co-efficient of Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

$$= \frac{8 - 4}{8 + 4} = \frac{4}{12} = 0.3333$$

e. Calculation of Mean Deviation from Mean & its Co-efficient

x	f	fx	x - $\bar{x}$	f x - $\bar{x}$
2	6	12	3.81	22.86
3	3	9	2.81	8.43
4	9	36	1.81	16.29
5	11	55	0.81	8.91
6	7	42	0.19	1.33
7	14	98	1.19	16.66
8	12	96	2.19	26.28
9	04	36	3.19	12.76
	N = 66	$\Sigma fx = 384$		$\Sigma f x - \bar{x} = 113.52$

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{384}{66} = 5.81$$

$$\text{Mean Deviation} = \frac{\Sigma f|x - \bar{x}|}{N} = \frac{113.52}{66} = 1.72$$

from Mean

Co-efficient of Mean Deviation from Mean

$$= \frac{\text{M.D from Mean}}{\text{Mean}} = \frac{1.72}{5.81} = 0.2960$$

1. Calculation of Mean deviation from Median & its Co-efficient

x	f	cf	X - M	f X - M
2	6	6	4	24
3	3	9	3	9
4	9	18	2	18
5	11	29	1	11
6	7	36	0	0
7	14	50	1	14
8	12	62	2	24
9	4	66	3	12
N = 66				$\Sigma f X - M = 112$

Calculation of M.D from Median

$$\begin{aligned} \text{M.D from Median} &= \frac{\Sigma f|X - M|}{N} \\ &= \frac{112}{66} = 1.69 \end{aligned}$$

$$M = \left[\frac{N+1}{2} \right]^{\text{th}} \text{ term}$$

$$= \left[\frac{66+1}{2} \right]^{\text{th}} \text{ term}$$

$$= \left[\frac{67}{2} \right]^{\text{th}} \text{ term} = (33.5)^{\text{th}} \text{ term}$$

The cf > 33.5 = 36

∴ Corresponding Median class = 6

$$\text{Co-efficient of M.D from Median} = \frac{\text{M.D from M}}{\text{Median}} = \frac{1.69}{6} = 0.2816$$

2. Calculation of S.D under Direct Method

x	f	fx	x ²	fx ²
2	6	12	4	24
3	3	9	9	27
4	9	36	16	144
5	11	55	25	275
6	7	42	36	252
7	14	98	49	686
8	12	96	64	768
9	4	36	81	324
N = 66		$\Sigma fx = 384$		$\Sigma fx^2 = 2500$

$$\sigma = \sqrt{\frac{\Sigma fx^2}{N} - \left[\frac{\Sigma fx}{N} \right]^2}$$

$$= \sqrt{\frac{2500}{66} - \left[\frac{384}{66}\right]^2} = \sqrt{37.87 - (5.818)^2}$$

$$= \sqrt{37.87 - 33.85} = \sqrt{4.02}$$

$$\sigma = 2.00$$

h. Calculation of S.D under Assumed Mean

x	f	[A = 5] d = x - A	fd	d ²	fd ²
2	6	-3	-18	9	54
3	3	-2	-6	4	12
4	9	-1	-9	1	9
5	11	0	0	0	0
6	7	1	7	1	7
7	14	2	28	4	56
8	12	3	36	9	108
9	4	4	16	16	64
N = 66			$\Sigma fd = 54$		$\Sigma fd^2 = 310$

$$\sigma = \sqrt{\frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2} = \sqrt{\frac{310}{66} - \left(\frac{54}{66}\right)^2}$$

$$= \sqrt{4.696 - (0.818)^2} = \sqrt{4.696 - 0.667}$$

$$= \sqrt{4.027} = 2.00$$

i. Calculation of S.D under deviation from Actual Mean

x	f	fx	(x - $\bar{x}$)	(x - $\bar{x}$) ²	f(x - $\bar{x}$) ²
2	6	12	-3.81	14.5161	87.0966
3	3	9	-2.81	7.8961	23.6883
4	9	36	-1.81	3.276	29.4849
5	11	55	-0.81	0.6561	7.2171
6	7	42	0.19	0.0361	0.2527
7	14	98	1.19	1.4161	19.8254
8	12	96	2.19	4.7961	57.5532
9	4	36	3.19	10.1761	40.7044
N = 66		$\Sigma fx = 384$			$\Sigma f(x - \bar{x})^2 = 265.8224$

$$\bar{x} = \frac{\sum fx}{N} = \frac{384}{66} = 5.81$$

$$\sigma = \sqrt{\frac{\sum f(x - \bar{x})^2}{N}} = \sqrt{\frac{265.8226}{66}} = \sqrt{4.0276} = 2.00$$

j. Calculation of Co-efficient of Variation.

$$\text{Co-efficient of variation} = \frac{\sigma}{\bar{x}} \times 100 = \frac{2}{5.81} \times 100 = 34.42\%$$

Problems on Continuous Series

Illustration 1

Calculate the following

CI	10-20	20-30	30-40	40-50	50-60	60-70
f	1	4	14	8	2	1

- Range
- Inter quartile range
- MD from mean & its co-efficient
- SD under direct method
- SD under actual mean

- Co-efficient of range
- QD & its co-efficient
- MD from median & its co-efficient
- SD under assumed mean method
- Co-efficient of variation

Solution

a. Calculation of Range

CI	f
10 - 20	1
20 - 30	4
30 - 40	14
40 - 50	8
50 - 60	2
60 - 70	1
	N = 30

$$\begin{aligned} R &= L - S \\ &= 70 - 10 \\ R &= 60 \end{aligned}$$

b. Calculation of Co-efficient of Range

$$\begin{aligned} \text{Co-efficient of Range} &= \frac{L - S}{L + S} \\ &= \frac{70 - 10}{70 + 10} = \frac{60}{80} \\ &= \frac{60}{80} = 0.75 \end{aligned}$$

c. Calculation of inter quartile range, QD & its co-efficient:

CI	f	cf
10 - 20	1	1
20 - 30	4	1+4=5
30 - 40	14	5+14=19
40 - 50	8	19+8=27
50 - 60	2	27+2=29
60 - 70	1	29+1=30
	N = 30	

Calculation of Q_1 & Q_3 :

$$Q_1 \text{ class} = \left[\frac{N}{4} \right]^{\text{th}} \text{ term}$$

$$= \left[\frac{30}{4} \right]^{\text{th}} \text{ term} = 7.5^{\text{th}} \text{ term}$$

The $cf > 7.5 = 19$ $\therefore$ Corresponding Q_1 class = 30 - 40

$$Q_3 \text{ class} = 3 \left[\frac{N}{4} \right]^{\text{th}} \text{ term}$$

$$= 3[7.5]^{\text{th}} \text{ term} = 22.5^{\text{th}} \text{ term}$$

The $cf > 22.5 = 27$ $\therefore$ Corresponding Q_3 class = 40 - 50

$$Q_1 = L + \left[\frac{\frac{N}{4} - pcf}{f} \right] \times i$$

L = Lower limit of Q_1 Class = 30

pcf = Preceding cumulative

frequency of Q_1 Class = 5,f = Frequency of Q_1 Class = 14,i = Class interval of Q_1 Class = 10

$$= 30 + \left(\frac{7.5 - 5}{14} \right) 10$$

$$= 30 + \left(\frac{2.5}{14} \right) 10$$

$$= 30 + \left(\frac{25}{14} \right) = 31.785$$

$$Q_3 = L + \left[\frac{3 \left(\frac{N}{4} \right) - pcf}{f} \right] \times i$$

L = Lower limit of Q_3 Class = 40

pcf = Preceding cumulative frequency

 Q_3 Class = 19,f = Frequency of Q_3 Class = 8,i = Class interval of Q_3 Class = 10

$$= 40 + \left[\frac{22.5 - 19}{8} \right] 10$$

$$= 40 + \left[\frac{3.5}{8} \right] 10$$

$$= 40 + 4.375 = 44.375$$

(d) Inter quartile range

$$= Q_3 - Q_1 = 44.375 - 31.785 = 12.59$$

$$QD = \frac{Q_3 - Q_1}{2} = \frac{44.375 - 31.785}{2}$$

$$= \frac{12.59}{2} = 6.295$$

$$\begin{aligned}\text{Coefficient of QD} &= \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{44.375 - 31.785}{44.375 + 31.785} \\ &= \frac{12.59}{76.16} = 0.1653\end{aligned}$$

(e) Calculation of MD from mean & its co-efficient

CI	f	Mid point (x)	fx	$ x - \bar{x} $	$f x - \bar{x} $
10 - 20	1	15	15	23	23
20 - 30	4	25	100	13	52
30 - 40	14	35	490	3	42
40 - 50	8	45	360	7	56
50 - 60	2	55	110	17	34
60 - 70	1	65	65	27	27
	N = 30		$\Sigma fx = 1140$		$\Sigma f x - \bar{x} = 234$

$$\begin{aligned}\bar{x} &= \frac{\Sigma fx}{N} \\ &= \frac{1140}{30} \\ \bar{x} &= 38\end{aligned}$$

$$\begin{aligned}\text{MD from mean} &= \frac{\Sigma f|x - \bar{x}|}{N} \\ &= \frac{234}{30} = 7.8\end{aligned}$$

$$\begin{aligned}\text{co-efficient of MD from mean} &= \frac{\text{MD from mean}}{\text{mean}} \\ &= \frac{7.8}{38} = 0.2052\end{aligned}$$

(f) Calculation of MD from median & its co-efficient:

CI	f	cf	x	$ x - M $	$f x - M $
10 - 20	1	1	15	22.14	22.14
20 - 30	4	5	25	12.14	48.56
30 - 40	14	19	35	2.14	29.96
40 - 50	8	27	45	7.86	62.88
50 - 60	2	29	55	17.86	35.72
60 - 70	1	30	65	27.86	27.86
	N = 30				$\Sigma f x - M = 227.12$

$$\text{Median class} = \left[\frac{N}{2} \right]^{\text{th}} \text{ term}$$

$$= \left[\frac{30}{2} \right]^{\text{th}} \text{ term} = 15^{\text{th}} \text{ term}$$

The $cf > 15 = 19$

$\therefore$ Corresponding Median class
= 30 - 40

$$M = L + \left[\frac{\frac{N}{2} - pcf}{f} \right] \times i$$

L = Lower limit of Median Class = 30
 pcf = Preceding cumulative frequency of Median Class = 5,

f = Frequency of Median Class = 14,

i = Class interval of Median Class = 10

$$= 30 + \left[\frac{15 - 5}{14} \right] 10 = 30 + \left[\frac{10}{14} \right] 10$$

$$= 30 + \left[\frac{100}{14} \right] = 30 + 7.1428$$

$$M = 37.14$$

MD from Median

$$= \frac{\sum f|x - M|}{N}$$

$$= \frac{227.12}{30} = 7.57$$

$$= 7.57$$

Co-efficient of M.D. from median

$$= \frac{\text{MD from median}}{\text{median}}$$

$$= \frac{7.57}{37.14} = 0.2038$$

$$= 0.2038$$

Co-efficient of M.D. from median

(g) Calculation of Standard Deviation under direct method:

CI	f	x (MV)	fx	x^2	fx^2
10 - 20	1	15	15	225	225
20 - 30	4	25	100	625	2500
30 - 40	14	35	490	1225	17150
40 - 50	8	45	360	2025	16200
50 - 60	2	55	110	3025	6050
60-70	1	65	65	4225	4225
	N = 30		$\Sigma fx = 1140$		$\Sigma fx^2 = 46350$

$$\sigma = \sqrt{\frac{\Sigma fx^2}{N} - \left[\frac{\Sigma fx}{N} \right]^2}$$

$$= \sqrt{\frac{46350}{30} - \left[\frac{1140}{30} \right]^2}$$

$$= \sqrt{1545 - (38)^2}$$

$$= \sqrt{1545 - 1444} = \sqrt{100}$$

$$\therefore \sigma = 10.0498$$

(h) Calculation of Standard Deviation under deviation from Assumed Mean

CI	f	x (MV)	A = 35 d = x - A	fd	d ²	fd ²
10-20	1	15	-20	-20	400	400
20-30	4	25	-10	-40	100	400
30-40	14	35	0	0	0	0
40-50	8	45	10	80	100	800
50-60	2	55	20	40	400	800
60-70	1	65	30	30	900	900
	N = 30			$\Sigma fd = 90$		$\Sigma fd^2 = 3300$

$$\begin{aligned}\sigma &= \sqrt{\frac{\Sigma fd^2}{N} - \left[\frac{\Sigma fd}{N} \right]^2} \\ &= \sqrt{\frac{3300}{30} - \left[\frac{90}{30} \right]^2} \\ &= \sqrt{110 - (3)^2} \\ &= \sqrt{110 - 9} = \sqrt{101} \\ \therefore \sigma &= 10.0498\end{aligned}$$

(i) Calculation of Standard Deviation under deviation from Actual Mean

CI	f	x (MV)	fx	(x - $\bar{x}$)	(x - $\bar{x}$) ²	f(x - $\bar{x}$) ²
10-20	1	15	15	-23	529	529
20-30	4	25	100	-13	169	676
30-40	14	35	490	-3	9	126
40-50	8	45	360	7	49	392
50-60	2	55	110	17	289	578
60-70	1	65	65	27	729	729
	N = 30		$\Sigma fx = 1140$			$\Sigma f(x - \bar{x})^2 = 3030$

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{1140}{30} = 38$$

$$\begin{aligned}\sigma &= \sqrt{\frac{\Sigma f(x - \bar{x})^2}{N}} = \sqrt{\frac{3030}{30}} \\ &= \sqrt{101} = 10.0498\end{aligned}$$

(j) Calculation of co-efficient of Variation:

Co-efficient of Variation

$$= \frac{\sigma}{\bar{x}} \times 100$$

$$= \frac{10.0498}{38} \times 100 = 0.2644 \times 100 = 26.44\%$$

Illustration 2

Calculate the following:

- Range
- Inter quartile range
- MD from mean and its co-efficient
- SD under direct method
- SD under actual mean
- Co-efficient of Range
- QD & its co-efficient
- MD from median and its co-efficient
- SD under assumed mean method
- Co-efficient of variation

C.I	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70
f	5	8	7	12	28	20	10

Solution

a. Calculation of Range:

C.I	f
0 - 10	5
10 - 20	8
20 - 30	7
30 - 40	12
40 - 50	28
50 - 60	20
60 - 70	10
70 - 80	10
	N = 100

$$\begin{aligned}
 R &= L - S \\
 &= 80 - 0 \\
 R &= 80
 \end{aligned}$$

b. Calculation of Co-efficient of Range

$$\begin{aligned}
 \text{Co-efficient of Range} &= \frac{L - S}{L + S} \\
 &= \frac{80 - 0}{80 + 0} = \frac{80}{80} = 1
 \end{aligned}$$

c. Calculation of inter quartile Range Q.D and its co-efficient.

C.I	f	c.f
0 - 10	5	5
10 - 20	8	5 + 8 = 13
20 - 30	7	13 + 7 = 20
30 - 40	12	20 + 12 = 32
40 - 50	28	32 + 28 = 60
50 - 60	20	60 + 20 = 80
60 - 70	10	80 + 10 = 90
70 - 80	10	90 + 10 = 100
	N = 100	

d. Inter

Calculation of Q_1 and Q_3

$$Q_1 \text{ class} = \left[\frac{N}{4} \right]^{\text{th}} \text{ term} = \left[\frac{100}{4} \right]^{\text{th}} \text{ term} \\ = (25)^{\text{th}} \text{ term}$$

The $cf > 25 = 32$

$\therefore$ Corresponding Q_1 class = 30 - 40

$$Q_3 \text{ class} = 3 \left[\frac{N}{4} \right]^{\text{th}} \text{ term} = 3 \left[\frac{100}{4} \right]^{\text{th}} \text{ term} \\ = 3(25)^{\text{th}} \text{ term} \\ = (75)^{\text{th}} \text{ term}$$

The $cf > 75 = 80$

$\therefore$ Corresponding Q_3 class = 50 - 60

$$Q_1 = L + \left[\frac{\frac{N}{4} - \text{pcf}}{f} \right] \times i$$

L = Lower limit of Q_1 Class = 30

pcf = Preceding cumulative

frequency of Q_1 Class = 20,

f = Frequency of Q_1 Class = 12,

i = Class interval of Q_1 Class = 10

$$= 30 + \left[\frac{25 - 20}{12} \right] \times 10$$

$$= 30 + \left[\frac{5}{12} \right] \times 10$$

$$= 30 + \left[\frac{50}{12} \right] = 30 + 4.16 = 34.16$$

$$Q_3 = L + \left[\frac{3 \left[\frac{N}{4} \right] - \text{pcf}}{f} \right] \times i$$

L = Lower limit of Q_3 Class = 50

pcf = Preceding cumulative frequency of

Q_3 Class = 60,

f = Frequency of Q_3 Class = 20,

i = Class interval of Q_3 Class = 10

$$= 50 + \left[\frac{75 - 60}{20} \right] \times 10$$

$$= 50 + \left[\frac{15}{20} \right] \times 10$$

$$= 50 + \left[\frac{150}{20} \right] = 50 + 7.5 = 57.5$$

d. Inter Quartile Range:

$$= Q_3 - Q_1 = 57.5 - 34.16 = 23.34$$

$$Q.D = \frac{Q_3 - Q_1}{2} = \frac{57.5 - 34.16}{2} = \frac{23.34}{2} = 11.67$$

$$\text{Co-efficient of Q.D} = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{57.5 - 34.16}{57.5 + 34.16} = \frac{23.34}{91.66} = 0.2546$$

e. Calculate of M.D from Mean and its Co-efficient

C.I	f	x	fx	$ x - \bar{x} $	$f x - \bar{x} $
0-10	5	5	25	40	200
10-20	8	15	120	30	240
20-30	7	25	175	20	140
30-40	12	35	420	10	120
40-50	28	45	1260	0	0
50-60	20	55	1100	10	200
60-70	10	65	650	20	200
70-80	10	75	750	30	300
	N = 100		$\Sigma fx = 4500$		$\Sigma f x - \bar{x} = 1400$

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{4500}{100}$$

$$\bar{x} = 45$$

$$\text{M.D from Mean} = \frac{\Sigma f|x - \bar{x}|}{N} = \frac{1400}{100} = 14$$

$$\text{Co-efficient of M.D. from Mean} = \frac{\text{M.D from Mean}}{\bar{x}} = \frac{14}{45} = 0.31$$

f. Calculation of M.D from Median and its Co-efficient

C.I	f	cf	x	$ x - M $	$f x - M $
0-10	5	5	5	41.42	207.10
10-20	8	13	15	31.42	251.36
20-30	7	20	25	21.42	149.49
30-40	12	32	35	11.42	137.04
40-50	28	60	45	1.42	39.76
50-60	20	80	55	8.58	171.60
60-70	10	90	65	18.58	185.80
70-80	10	100	75	28.58	285.80
	N = 100				$\Sigma f x - M = 1427.95$

$$\text{Median class} = \left[\frac{N}{2} \right]^{\text{th}} \text{ term}$$

$$= \left(\frac{100}{2} \right)^{\text{th}} \text{ term}$$

$$= (50)^{\text{th}} \text{ term}$$

The $cf > 50 = 60$

$\therefore$ Corresponding Median class
= 40 - 50

$$\text{M.D from Median} = \frac{\Sigma f|x - M|}{N} = \frac{1427.95}{100} = 14.27$$

$$M = L + \left[\frac{N/2 - pcf}{f} \right] \times i$$

L = Lower limit of Median Class = 40

pcf = Preceding cumulative frequency of Median Class = 32,

f = Frequency of Median Class = 28,

i = Class interval of Median Class = 10

$$= 40 + \left[\frac{50 - 32}{28} \right] \times 10 = 40 + \left[\frac{18}{28} \right] \times 10$$

$$= 40 + \left[\frac{180}{28} \right] = 40 + 6.42$$

$$M = 46.42$$

Calculation of S.D. under Direct Method.

CI	f	x	fx	x^2	fx^2
0-10	5	5	25	25	125
10-20	8	15	120	225	1800
20-30	7	25	175	625	4375
30-40	12	35	420	1225	14700
40-50	28	45	1260	2025	56700
50-60	20	55	1100	3025	60500
60-70	10	65	650	4225	42250
70-80	10	75	750	5625	56250
	N = 100		$\Sigma fx = 4500$		$\Sigma fx^2 = 236700$

$$\sigma = \sqrt{\frac{\Sigma fx^2}{N} - \left[\frac{\Sigma fx}{N} \right]^2} = \sqrt{\frac{236700}{100} - \left[\frac{4500}{100} \right]^2}$$

$$= \sqrt{2367 - (45)^2} = \sqrt{2367 - 2025} = \sqrt{342} = 18.49$$

Calculation of S.D under deviation from Assumed Mean.

CI	f	x	$[A = 35] d = X - A$	d^2	fd	fd^2
0-10	5	5	-30	900	-150	4500
10-20	8	15	-20	400	-160	3200
20-30	7	25	-10	100	-70	700
30-40	12	35	0	0	0	0
40-50	28	45	10	100	280	2800
50-60	20	55	20	400	400	8000
60-70	10	65	30	900	300	9000
70-80	10	75	40	1600	400	16000
	N = 100				$\Sigma fd = 1000$	$\Sigma fd^2 = 44200$

$$\sigma = \sqrt{\frac{\Sigma fd^2}{N} - \left[\frac{\Sigma fd}{N} \right]^2} = \sqrt{\frac{44200}{100} - \left[\frac{1000}{100} \right]^2}$$

$$= \sqrt{442 - (10)^2} = \sqrt{442 - 100} = \sqrt{342} = 18.49$$

i. Calculation of S.D under deviation from Actual Mean.

C.I	f	x	fx	(x - $\bar{x}$)	(x - $\bar{x}$) ²	f(x - $\bar{x}$) ²
0 - 10	5	5	25	-40	1600	8000
10 - 20	8	15	120	-30	900	7200
20 - 30	7	25	175	-20	400	2800
30 - 40	12	35	420	-10	100	1200
40 - 50	28	45	1260	0	0	0
50 - 60	20	55	1100	10	100	2000
60 - 70	10	65	650	20	400	4000
70 - 80	10	75	750	30	900	9000
	N = 100		$\Sigma fx = 4500$			$\Sigma f(x - \bar{x})^2 = 34200$

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{4500}{100} = 45$$

$$\sigma = \sqrt{\frac{\Sigma f(x - \bar{x})^2}{N}} = \sqrt{\frac{34200}{100}} = \sqrt{342} = 18.49$$

j. Calculation on co-efficient of variation.

$$C.V = \frac{\sigma}{\bar{x}} \times 100 = \frac{18.49}{45} \times 100$$

$$C.V = 41.08\%$$

Problems on Standard Deviation

Illustration 1

Find standard deviation of 3, 5, 7, 9, 11.

Solution

x	(x - $\bar{x}$)	(x - $\bar{x}$) ²
3	3 - 7 = -4	16
5	5 - 7 = -2	4
7	7 - 7 = 0	0
9	9 - 7 = 2	4
11	11 - 7 = 4	16
$\Sigma x = 35$		$\Sigma (x - \bar{x})^2 = 40$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{35}{5} = 7$$

$$\sigma = \sqrt{\frac{\Sigma (x - \bar{x})^2}{n}} = \sqrt{\frac{40}{5}} = \sqrt{8} = 2.828$$

Illustration

Find the standard deviation of 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50.

Solution

x

8

9

15

23

5

11

19

8

10

12

$$\Sigma x = 120$$

Illustration

The yield of rice in different areas is given below. Find the standard deviation.

Distrib

Ric

Calculat

Solut

Convert

x (Ric

22

29

12

23

18

15

12

34

18

17

$$\Sigma x =$$

Illustration 2

Find the standard deviation for the following data:
8, 9, 15, 23, 5, 11, 19, 8, 10, 12.

Solution

x	(x - $\bar{x}$)	(x - $\bar{x}$) ²
8	8 - 12 = -4	16
9	9 - 12 = -3	9
15	15 - 12 = 3	9
23	23 - 12 = 11	121
5	5 - 12 = -7	49
11	11 - 12 = -1	1
19	19 - 12 = 7	49
8	8 - 12 = -4	16
10	10 - 12 = -2	4
12	12 - 12 = 0	0
$\Sigma x = 120$		$\Sigma (x - \bar{x})^2 = 274$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{120}{10}$$

$$\bar{x} = 12$$

$$\sigma = \sqrt{\frac{\Sigma (x - \bar{x})^2}{N}}$$

$$= \sqrt{\frac{274}{10}}$$

$$= \sqrt{27.4}$$

$$SD = 5.23$$

Problems on Standard Deviation For Discrete Data**Illustration 1**

The yield of rice per acre for 10 districts of a state is as under:

District	1	2	3	4	5	6	7	8	9	10
Rice	22	29	12	23	18	15	12	34	18	12

Calculate Standard Deviation and Coefficient of variation.

Solution

Convert the inclusive class interval into exclusive.

x (Rice)	(x - $\bar{x}$)	(x - $\bar{x}$) ²
22	22 - 19.5 = 2.5	6.25
29	29 - 19.5 = 9.5	90.25
12	12 - 19.5 = -7.5	56.25
23	23 - 19.5 = 3.5	12.25
18	18 - 19.5 = -1.5	2.25
15	15 - 19.5 = -4.5	20.25
12	12 - 19.5 = -7.5	56.25
34	34 - 19.5 = 14.5	210.25
18	18 - 19.5 = -1.5	2.25
12	12 - 19.5 = -7.5	56.25
$\Sigma x = 195$		$\Sigma (x - \bar{x})^2 = 512.5$

$$\bar{x} = \frac{\Sigma x}{N} = \frac{195}{10} = 19.5$$

$$\sigma = \sqrt{\frac{\Sigma (x - \bar{x})^2}{N}} = \sqrt{\frac{512.5}{10}}$$

$$= \sqrt{51.25}$$

$$\sigma = 7.16$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100 = \frac{7.16}{19.5} \times 100$$

$$C.V = 36.71\%$$

C.V for rice is 36.71.

Illustration 2

Calculate standard deviation and variance for the following distribution by direct method

x	4	10	15	20	23	28	32
f	1	6	3	4	8	2	3

Solution

x	f	fx	x ²	fx ²
4	1	4	16	16
10	6	60	100	600
15	3	45	225	675
20	4	80	400	1600
23	8	184	529	4232
28	2	56	784	1568
32	3	96	1024	3072
N = 27		Σfx = 525		Σfx ² = 11763

$$\begin{aligned}\sigma &= \sqrt{\frac{\Sigma fx^2}{N} - \left[\frac{\Sigma fx}{N}\right]^2} \\ &= \sqrt{\left[\frac{11763}{27}\right] - \left[\frac{525}{27}\right]^2} = \sqrt{435.67 - 190.9} \\ &= \sqrt{435.67 - 377.9} = \sqrt{57.76} \\ \sigma &= 7.6\end{aligned}$$

Problems of Standard Deviation for Continuous Data

Illustration 1

Calculate standard deviation variance and coefficient of variation deviation by direct method

Value	90 - 99	80 - 89	70 - 79	60 - 69	50 - 59	40 - 49	30 - 39
Frequency	3	11	25	21	16	3	2

Solution

Direct Method

C.I	Mid value (x)	f	fx	x ²	fx ²
89.5 - 99.5	94.5	3	283.5	8930.25	26790.75
79.5 - 89.5	84.5	11	929.5	7140.25	78542.75
69.5 - 79.5	74.5	25	1862.5	5550.25	138756.25
59.5 - 69.5	64.5	21	1354.5	4160.25	87365.25
49.5 - 59.5	54.5	16	872	2970.25	47524
39.5 - 49.5	44.5	3	133.5	1980.25	5940.75
29.5 - 39.5	34.5	2	69	1190.25	2380.5
		N = 81	Σfx = 5504.5		Σfx ² = 387300.25

$$\sigma = \sqrt{\frac{\Sigma fx^2}{N} - \left[\frac{\Sigma fx}{N}\right]^2} = \sqrt{\left[\frac{387300.25}{81}\right] - \left[\frac{5504.5}{81}\right]^2} = 12.77$$

$$\text{Variance} = \sigma^2 = (12.77)^2 = 163.0729$$

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{5504.5}{81} = 67.96$$

$$\therefore \text{Coefficient of Variation} = \text{SD/Mean} \times 100 = 12.77/67.96 \times 100 = 18.79\%$$

Problem

Illustration

Following data is given

C
D

Solution

Student C

x
70
35
50
83
90
50

$$\Sigma x = 378$$

Student D

x
60
90
80
62
53
45

$$\Sigma x = 390$$

Conclusion

Co-variation

Problems on Coefficient of Variation

Illustration 1

Following are the marks scored by two students named C and D. Find who is better scorer and who is more consistent.

C	70	35	50	83	90	50
D	60	90	80	62	53	45

Solution

Student C

x	$(x - \bar{x}) = x - 63$	$(x - \bar{x})^2$
70	7	49
35	-28	784
50	-13	169
83	20	400
90	27	729
50	-13	169
$\Sigma x = 378$		$\Sigma (x - \bar{x})^2 = 2300$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{378}{6} = 63$$

$$\sigma = \sqrt{\frac{\Sigma (x - \bar{x})^2}{n}} = \sqrt{\frac{2300}{6}} = 19.579$$

$$CV = \frac{\sigma}{\bar{x}} \times 100 = \frac{19.579}{63} \times 100 = 31.078\%$$

Student D

x	$(x - \bar{x}) = x - 65$	$(x - \bar{x})^2$
60	-5	25
90	25	625
80	15	225
62	-3	9
53	-12	144
45	-20	400
$\Sigma x = 390$		$\Sigma (x - \bar{x})^2 = 1428$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{390}{6} = 65$$

$$\sigma = \sqrt{\frac{\Sigma (x - \bar{x})^2}{n}} = \sqrt{\frac{1428}{6}} = 15.43$$

$$CV = \frac{\sigma}{\bar{x}} \times 100 = \frac{15.43}{65} \times 100 = 23.73\%$$

Conclusion : Mean of student D is more than student C, so student D is better scorer.
 Co-variance of student D is less than student C, so student D is more consistent.

Illustration 2

Following are the runs scored by two Batsman A and B.

A	60	50	40	80	90	30	70
B	60	50	40	30	80	70	20

- Find:
- Who scored good runs?
 - Which batsman is more consistent?

Solution

Batsman A	
x	x ²
60	3600
50	2500
40	1600
80	6400
90	8100
30	900
70	4900
Σx = 420	Σx² = 28000

$$\bar{x} = \frac{\Sigma x}{n} = \frac{420}{7} = 60$$

$$\begin{aligned}\sigma &= \sqrt{\frac{\Sigma x^2}{n} - \left[\frac{\Sigma x}{n}\right]^2} \\ &= \sqrt{\frac{28000}{7} - \left(\frac{420}{7}\right)^2} = \sqrt{4000 - [60]^2} \\ &= \sqrt{4000 - 3600} = \sqrt{400} \\ \sigma &= 20\end{aligned}$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100 = \frac{20}{60} \times 100 = 33.33\%$$

Batsman B	
x	x ²
60	3600
50	2500
40	1600
30	900
80	6400
70	4900
20	400
Σx = 350	Σfx² = 20300

$$\bar{x} = \frac{\Sigma x}{n} = \frac{350}{7} = 50$$

$$\begin{aligned}\sigma &= \sqrt{\frac{\Sigma x^2}{n} - \left[\frac{\Sigma x}{n}\right]^2} = \sqrt{\frac{20300}{7} - \left[\frac{350}{7}\right]^2} \\ &= \sqrt{2900 - 2500} = \sqrt{400} \\ \sigma &= 20\end{aligned}$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100 = \frac{20}{50} \times 100 = 40\%$$

Conclusion : The mean of A is greater than that of B. So, A has scored more runs than B. The CV of A is less than the CV of B. So Batsman A is more consistent.

Illustration

Following

Marks
Obtained

Find who
get the pr

Solution

Test

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10

Test

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10

Conclus
is less th

Illustration 3

Following are the marks obtained by two students Suraj and Dheeraj in tests of 100 marks each.

Test	1	2	3	4	5	6	7	8	9	10
Suraj	40	80	76	48	52	72	68	56	60	56
Dheeraj	48	75	54	60	63	69	72	51	72	60

Find who is the better scorer and if consistency is the criterion for awarding prize who should get the prize?

Solution

Test	Suraj (x)	$(x - \bar{x})$	$(x - \bar{x})^2$
1	40	-20.8	432.64
2	80	19.2	368.64
3	76	15.2	231.04
4	48	-12.8	163.84
5	52	-8.8	77.44
6	72	11.2	125.44
7	68	7.2	51.84
8	56	-4.8	23.04
9	60	-0.8	0.64
10	56	-4.8	23.04
$\Sigma x = 608$			$\Sigma(x - \bar{x})^2 = 1497.6$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{608}{10} = 60.8$$

$$\sigma = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n}} = \sqrt{\frac{1497.6}{10}} = 12.238$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100 = \frac{12.238}{60.8} \times 100 = 20.128\%$$

Test	Dheeraj (x)	$(x - \bar{x})$	$(x - \bar{x})^2$
1	48	-14.4	207.36
2	75	12.6	158.76
3	54	-8.4	70.56
4	60	-2.4	5.76
5	63	0.6	0.36
6	69	6.6	43.56
7	72	9.6	92.16
8	51	-11.4	129.96
9	72	9.6	92.16
10	60	-2.4	5.76
$\Sigma x = 624$			$\Sigma(x - \bar{x})^2 = 806.4$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{624}{10} = 62.4$$

$$\sigma = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n}} = \sqrt{\frac{806.4}{10}} = 8.98$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100 = \frac{8.98}{62.4} \times 100 = 14.39\%$$

Conclusion: The mean of Dheeraj is more than Suraj so Dheeraj is a better scorer. The CV of Dheeraj is less than Suraj, so Dheeraj is consistent.

Illustration 4

Find from the following the most consistent Batsman and better run getter.

Batsman A:	5	7	16	27	39	53	56	61	80	101
Batsman B:	0	4	16	21	41	43	57	78	83	95

Solution

Batsman A		
x	(x - $\bar{x}$)	(x - $\bar{x}$) ²
5	-45	2025
7	-43	1849
16	-34	1156
27	-23	529
39	-11	121
53	+3	9
56	+6	36
61	+11	121
80	+30	900
101	+51	2601
105	+55	3025
$\Sigma x = 550$		$\Sigma(x - \bar{x})^2 = 12372$

Batsman B		
x	(x - $\bar{x}$)	(x - $\bar{x}$) ²
0	-48	2304
4	-44	1936
16	-32	1024
21	-27	729
41	-7	49
43	-5	25
57	9	81
78	30	900
83	35	1225
90	42	1764
95	47	2209
$\Sigma x = 528$		$\Sigma(x - \bar{x})^2 = 12246$

Conclusion :

- Batsman 'A' is more consistent.
- 'A' is a better run getter also as his average is more.

$$\bar{x} = \frac{\Sigma x}{n} = \frac{550}{11} = 50$$

$$\sigma = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n}} = \sqrt{\frac{12372}{11}} = 33.537$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

$$= \frac{33.537}{50} \times 100 = 67.0736\%$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{528}{11} = 48$$

$$\sigma = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n}} = \sqrt{\frac{12246}{11}} = 33.365741$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

$$= \frac{33.3657}{48} \times 100 = 69.5119\%$$

Illustration 5

Life of two models of greater uniformity

Life (No. of years)	Frequency
0 - 2	5
2 - 4	16
4 - 6	13
6 - 8	7
8 - 10	5
10 - 12	4
Total	N = 50

Solution

CI	f
0 - 2	5
2 - 4	16
4 - 6	13
6 - 8	7
8 - 10	5
10 - 12	4
Total	N = 50

Conclusion : T

Illustration 5

Life of two models of refrigerators obtained in a survey is given below. Find which model has greater uniformity of life.

Life (No. of years)	No. of Refrigerators	
	Model 'A'	Model 'B'
0-2	5	2
2-4	16	17
4-6	13	12
6-8	7	19
8-10	5	9
10-12	4	1

Solution

Model 'A'					Model 'B'				
CI	f	M.V(x)	fx	fx ²	CI	f	M.V(x)	fx	fx ²
0-2	5	1	5	5x1=5	0-2	2	1	2	2x1=2
2-4	16	3	48	48x3=144	2-4	17	3	51	51x3=153
4-6	13	5	65	65x5=325	4-6	12	5	60	60x5=300
6-8	7	7	49	49x7=343	6-8	19	7	133	133x7=931
8-10	5	9	45	45x9=405	8-10	9	9	81	81x9=729
10-12	4	11	44	44x11=484	10-12	1	11	11	11x11=121
	N = 50		Σfx = 256	Σfx ² = 1706		N = 60		Σfx = 338	Σfx ² = 2236

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{256}{50} = 5.12$$

$$\sigma_x = \sqrt{\frac{\Sigma fx^2}{N} - \left(\frac{\Sigma fx}{N}\right)^2}$$

$$= \sqrt{\frac{1706}{50} - \left(\frac{256}{50}\right)^2}$$

$$= \sqrt{34.12 - 26.21}$$

$$= \sqrt{7.91} = 2.812$$

$$CV(A) = \frac{\sigma}{\bar{x}} \times 100$$

$$= \frac{2.812}{5.12} \times 100 = 54.93\%$$

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{338}{60} = 5.633$$

$$\sigma_x = \sqrt{\frac{\Sigma fx^2}{N} - \left(\frac{\Sigma fx}{N}\right)^2}$$

$$= \sqrt{\frac{2236}{60} - \left(\frac{338}{60}\right)^2}$$

$$= \sqrt{37.266 - 31.734}$$

$$= \sqrt{5.532} = 2.352$$

$$CV(B) = \frac{\sigma}{\bar{x}} \times 100$$

$$= \frac{2.352}{5.633} \times 100 = 41.754\%$$

Conclusion : The CV of B is less than A. So, B has greater uniformity of life than A.

Illustration 6

The goals scored by two teams A and B in the football matches were as follows

Goals	0	1	2	3
Team A	10	9	6	4
Team B	9	12	10	8

Find the team which is more consistent

Solution

For Team A :

Goals (x)	f	fx	fx ²
0	10	0	0x0=0
1	9	9	9x1=9
2	6	12	12x2=24
3	4	12	12x3=36
4	5	20	20x4=80
	N = 34	Σfx = 53	Σfx² = 149

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{53}{34} = 1.558$$

$$\sigma = \sqrt{\frac{\Sigma fx^2}{N} - \left(\frac{\Sigma fx}{N}\right)^2}$$

$$= \sqrt{\left(\frac{149}{34}\right) - \left(\frac{53}{34}\right)^2}$$

$$= \sqrt{4.382 - 2.429} = 1.397$$

$$CV = \frac{\sigma}{\bar{x}} \times 100 = \frac{1.397}{1.558} \times 100 = 89.66\%$$

For Team B :

No. of Goals (x)	f	fx	fx ²
0	9	0	0x0=0
1	12	12	12x1=12
2	10	20	20x2=40
3	8	24	24x3=72
4	3	12	12x4=48
	N = 42	Σfx = 68	Σfx² = 172

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{68}{42} = 1.619$$

$$\sigma = \sqrt{\frac{\Sigma fx^2}{N} - \left(\frac{\Sigma fx}{N}\right)^2}$$

$$= \sqrt{\frac{172}{42} - \left(\frac{68}{42}\right)^2}$$

$$= \sqrt{4.095 - (1.619)^2}$$

$$= \sqrt{1.4738} = 1.214$$

$$CV = \frac{\sigma}{\bar{x}} \times 100 = \frac{1.214}{1.619} \times 100 = 74.99\%$$

Conclusion : CV of team A is greater than CV of B. Team B is more consistent.

Illustration 7

In a data $n = 72$, $\Sigma x = 7$

Solution

$$\bar{x} = \frac{\Sigma x}{n}$$

$$\sigma = \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2}$$

$$CV = \frac{\sigma}{\bar{x}} \times 100$$

Thus, the coefficient

Illustration 8

Which company has

Number of employees
Standard deviation
Average monthly salary

Solution

Company X:

$$C.V =$$

Conclusion : As CV

Illustration 9

An analysis of mo
industry gives the

No. of workers
Average monthly wa
Variance of distribut

a. Which firm

Illustration 7

In a data $n = 72$, $\Sigma x = 713.4$ and $\Sigma x^2 = 9047.34$. Find the coefficient of variation.

Solution

$$\bar{x} = \frac{\Sigma x}{n} = \frac{713.4}{72} = 9.9083$$

$$\sigma = \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2} = \sqrt{\frac{9047.34}{72} - \left(\frac{713.4}{72}\right)^2}$$

$$= \sqrt{125.6575 - 98.1751} = \sqrt{27.4824} = 5.2424$$

$$C.V. = \frac{\sigma}{\bar{x}} \times 100 = \frac{5.2424}{9.9083} \times 100 = 52.91\%$$

Thus, the coefficient of variation is 52.91%.

Illustration 8

Which company has greater variability of salary?

	Company 'X'	Company 'Y'
Number of employees	250	200
Standard deviation	500	600
Average monthly salary (₹)	20,000	25,000

Solution

Company X:

$$C.V. = \frac{\sigma}{\bar{X}} \times 100$$

$$= \frac{500}{20000} \times 100 = 2.5\%$$

Company Y:

$$C.V. = \frac{\sigma}{\bar{X}} \times 100$$

$$= \frac{600}{25000} \times 100 = 2.4\%$$

Conclusion: As $CV (\text{Firm Y}) < CV (\text{Firm X})$, Company X has greater variability than company Y.

Illustration 9

An analysis of monthly wages paid to workers in the firms A and B belonging to the same industry gives the following results.

	Firm 'A'	Firm 'B'
No. of workers	500	600
Average monthly wages (₹)	480	475
Variance of distribution of wages (₹)	400	625

- Which firm pays a larger wage bill.

- b. In which firm is there greater variability in individual wages.
c. Find out the combined average wages of the two firms taken together.

Solution

Particulars	Firm A	Firm B
Total wage bill		
Average wages $\times$ No. of workers	$500 \times 480 = 2,40,000$	$600 \times 475 = 2,85,000$
$S.D = \sqrt{\text{Variance}}$	$\sqrt{400} = 20$	$\sqrt{625} = 25$
$C.V = \frac{\sigma}{\bar{x}} \times 100$	$\frac{20}{480} \times 100 = 4.167\%$	$\frac{25}{475} \times 100 = 5.26\%$

Conclusion :

- a. Firm B pays a larger wage bill.
b. As $C.V (\text{Firm A}) < C.V (\text{Firm B})$. So firm B has greater variability.

Illustration 10

- a. If $C.V = 25\%$ and if $S.D = 4$, find the mean.
b. If $C.V = 53\%$ and if $\bar{x} = 26$, find the standard deviation.

Solution

a. We have, $C.V = \frac{\sigma}{\bar{x}} \times 100$

$$25 = \frac{4}{\bar{x}} \times 100$$

$$\therefore 25\bar{x} = 400$$

$$\bar{x} = \frac{400}{25} = 16$$

b. We have, $C.V = \frac{\sigma}{\bar{x}} \times 100$

$$53 = \frac{\sigma}{26} \times 100$$

$$\sigma = 53 \times 26 = 1378$$

$$\sigma = \frac{1378}{100}$$

$$S.D = 13.78$$

Illustration 11

Mean and variance of 100 items are found to be 40 and 121. What is its C.V?

Solution

Given:

$$\bar{x} = 40, \sigma = \sqrt{121} = 11 \quad [S.D = \sqrt{\text{variance}}]$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100 = \frac{11}{40} \times 100 = 27.5\%$$

Illustration 12
For a distribution, var**Solution**

$$\sigma = \sqrt{\text{var}}$$

$$\sigma = \sqrt{\text{var}}$$

Illustration 13

If variance = 36, $\Sigma x =$

Solution

$$\sigma = \sqrt{\text{variance}}$$

$$\sigma = \sqrt{36} = 6$$

$$\bar{x} = \frac{\Sigma x}{n}$$

Illustration 14

If co efficient of var

Solution

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

$$\bar{x} = \frac{\Sigma x}{n}$$

Illustration 15

The arithmetic m
20 respectively. T
scorer?

Solution

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

$$= \frac{15}{50} \times 100$$

Student C is a cons

Illustration 12

For a distribution, variance is 121 and its C.V is 23.15%. Determine its mean.

Solution

$$\sigma = \sqrt{\text{variance}}$$

$$\sigma = \sqrt{121} = 11$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

$$23.15 = \frac{11}{\bar{x}} \times 100; \quad 23.15 \bar{x} = 1100$$

$$\bar{x} = \frac{1100}{23.15}$$

$$\bar{x} = 47.52$$

Illustration 13

If variance = 36, $\sum x = 150$ and $n = 10$ find CV.

Solution

$$\sigma = \sqrt{\text{variance}}$$

$$\sigma = \sqrt{36} = 6$$

$$\bar{x} = \frac{\sum x}{n} = \frac{150}{10} = 15$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

$$CV = \frac{6}{15} \times 100 \Rightarrow \frac{6}{15} \times 100 = 40\%$$

Illustration 14

If coefficient of variation and S.D of a distribution are 75 and 15 respectively, find its mean

Solution

$$C.V = 75, \quad S.D = 15, \quad \bar{x} = ?$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100 \Rightarrow 75 = \frac{15}{\bar{x}} \times 100 \Rightarrow 75\bar{x} = 1500$$

$$\bar{x} = \frac{1500}{75} = 20; \quad \bar{x} = 20$$

Illustration 15

The arithmetic mean of marks scored by 3 students A, B & C in a examination are 50, 44, 20 respectively. The S.Ds of marks are respectively 15, 11 and 3. Who is the most consistent scorer?

Solution

A	B	C
$C.V = \frac{\sigma}{\bar{x}} \times 100$ $= \frac{15}{50} \times 100 = 30\%$	$C.V = \frac{\sigma}{\bar{x}} \times 100$ $= \frac{11}{44} \times 100 = 25\%$	$C.V = \frac{\sigma}{\bar{x}} \times 100$ $= \frac{3}{20} \times 100 = 15\%$

Student C is a consistent scorer.

3.5 Review Questions

Short Answer Type Questions

1. Define dispersion.
2. What is Range and Coefficient of Range?
3. Write the formula for Range and for the Coefficient of Range.
4. What are Q1, Q2 and Q3?
5. Define Interquartile Range (IQR) and Quartile Deviation (QD).
6. State any two limitations of Range.
7. Define Mean Deviation.
8. What is Variance? How is it related to Standard Deviation?
9. What is Standard Deviation?
10. Define Coefficient of Variation (CV).
11. What is Absolute Measure of Dispersion?
12. Mention the Types of Absolute Measures of Dispersion.
13. What is Relative Measure of Dispersion?
14. Mention the Types of Relative Measures of Variation.
15. What is Inter Quartile Range?
16. What is Quartile Deviation or Semi Inter Quartile Range?

Long Answer Type Questions

1. What is Dispersion? Write the Objectives or Significance of the Measures of Dispersion.
2. Explain the Characteristics for an Ideal Measures of Dispersion.
3. Explain the Types of Measures of Dispersion.
4. What is Standard Deviation? Write merits and demerits of Standard Deviation.
5. For the raw data: 15, 22, 28, 33, 40, 45, 50, find:
 - a) Range
 - b) Coefficient of Range
 - c) Interquartile Range
 - d) Quartile Deviation and its Coefficient
 - e) Mean Deviation from mean and its Coefficient
 - f) Mean Deviation from median
 - g) Standard Deviation (direct method)
 - h) Standard Deviation (assumed mean method)
 - i) Standard Deviation (from actual mean)
 - j) Coefficient of Variation
6. For the raw data: 32, 36, 41, 44, 48, 52, 55, 60, calculate:
 - a) Range
 - b) Coefficient of Range

- c) Interquartile Range
 - d) Quartile Deviation
 - e) Mean Deviation from mean
 - f) Mean Deviation from median
 - g) Standard Deviation (direct method)
 - h) Standard Deviation (assumed mean method)
 - i) Standard Deviation (from actual mean)
 - j) Coefficient of Variation
7. Find the Standard Deviation.
 8. Find the Standard Deviation.
 9. The monthly wages (₹) of 10 workers are: 1350, 1550. Calculate Standard Deviation.
 10. For the following distribution:

x:	5, 12, 18, 22, 28,
f:	2, 5, 6, 10, 8, 4, 3
 11. For the following grouped data:

Class Interval:	10-15, 15-20, 20-25, 25-30, 30-35, 35-40, 40-45, 45-50, 50-55, 55-60
Frequency:	4, 8, 12, 16, 20, 18, 14, 10, 6, 4
 12. The marks scored by two students in a test are:

Student A:	55, 48, 60, 52, 45, 40, 35, 30, 25, 20, 15, 10, 5
Student B:	60, 70, 55, 45, 35, 25, 15, 10, 5, 40, 30, 20, 10

 Who is the better student?
 13. The runs scored by two batsmen in a match are:

Batsman A:	30, 45, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150
Batsman B:	20, 65, 35, 55, 40, 50, 60, 70, 80, 90, 100, 110, 120

 Find who is the better batsman.
 14. The marks of two students in a test are:

Ravi:	40, 45, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100
Raj:	42, 38, 55, 58, 60, 62, 65, 68, 70, 72, 75, 78

 Who is the better student?
 15. The life span (in years) of 100 persons is given below:

Life (years)	Me
0-2	
2-4	
4-6	
6-8	
8-10	
10-12	

- c) Interquartile Range
 d) Quartile Deviation and its Coefficient
 e) Mean Deviation from mean and its Coefficient
 f) Mean Deviation from median
 g) Standard Deviation (direct method)
 h) Standard Deviation (assumed mean method)
 i) Standard Deviation (from actual mean)
 j) Coefficient of Variation
7. Find the Standard Deviation of the raw data: 4, 6, 8, 10, 12.
 8. Find the Standard Deviation for the data: 7, 9, 12, 14, 17, 19, 20, 21, 25, 28.
 9. The monthly wages (₹) of 10 workers are: 1200, 1500, 1300, 1700, 1400, 1600, 1800, 1250, 1350, 1550. Calculate Standard Deviation and Coefficient of Variation.
 10. For the following distribution, calculate Standard Deviation and Variance:
 x : 5, 12, 18, 22, 28, 32, 38
 f : 2, 5, 6, 10, 8, 4, 3
 11. For the following grouped data, calculate Standard Deviation, Variance and Coefficient of Variation:
 Class Interval: 10-19, 20-29, 30-39, 40-49, 50-59, 60-69, 70-79
 Frequency: 4, 8, 12, 20, 15, 7, 4
 12. The marks scored by two students in 6 subjects are:
 Student A: 55, 48, 62, 71, 66, 59
 Student B: 60, 70, 50, 45, 85, 40
 Who is the better scorer? Who is more consistent? (Use mean and CV).
 13. The runs scored by two batsmen in 7 innings are:
 Batsman A: 30, 45, 60, 55, 40, 70, 50
 Batsman B: 20, 65, 35, 80, 25, 75, 30
 Find who is better and who is more consistent.
 14. The marks of two students in 10 tests (out of 100) are:
 Ravi: 40, 45, 55, 60, 65, 70, 50, 55, 60, 50
 Raj: 42, 38, 55, 58, 65, 62, 45, 50, 55, 58
 Who is the better scorer? If consistency is the criterion, who deserves a prize?
 15. The life span (in years) of refrigerators of two models is given:

Life (years)	Model A	Model B
0-2	4	3
2-4	15	16
4-6	12	11
6-8	8	18
8-10	6	8
10-12	5	2

16. Which model shows greater uniformity in life span?
17. The goals scored by two teams in a football season are:
Goals: 0, 1, 2, 3, 4
Team X: 8, 10, 12, 6, 4
Team Y: 7, 14, 11, 5, 3
Which team is more consistent?
18. In a data set $n = 60$, $\Sigma x = 450$ and $\Sigma x^2 = 4020$. Find Coefficient of Variation.
19. Two companies pay the following salaries:
Company X: Mean = ₹18,000, SD = ₹450, Employees = 200
Company Y: Mean = ₹22,000, SD = ₹600, Employees = 250
Which company shows greater variability of salaries?
20. The mean wage of 400 workers is ₹500 and the variance is 625. Find the Coefficient of Variation.
21. In a distribution, if $CV = 20\%$ and mean = 25, find the Standard Deviation.
22. The Mean and Standard Deviation of two brands of Bulbs are given below:

Brand	A	B
Mean Life	1000 Hrs	820 Hrs
S.D	100 Hrs	65 Hrs

Which category of bulb has more consistency in its life?

23. Which company has greater variability of salary?

	Company X	Company Y
No. of employees	250	200
Standard deviation	500	600
Average monthly salary(₹)	20,000	25,000

24. Calculate SD from the following:

Size	25	35	45	55	65
Frequency	28	38	50	45	40

25. Find which of the following two classes is more consistent in scoring marks from the table.

Marks:	20-30	30-40	40-50	50-60	60-70
Class A	7	10	20	18	7
Class B	5	9	21	15	6

□□□□

4.1 Introduction

After understanding the and present data in a me like mean, median, and in this chapter, we move — Correlation and Re are related and how one

? What is Correlation

The term correlation is means a relationship or changes, we say there is

Everyday Examples of

1. Study and Exam

- When parent
- More study
- This means correlation

2. Rainfall and Crop

- When we s between ra
- More rainfa
- This again

3. Watching TV and

- A student v
- More TV ti
- This is an decreases.